



**POLITECNICO**  
MILANO 1863



## Contents

|          |                                                 |           |
|----------|-------------------------------------------------|-----------|
| <b>1</b> | <b>Project Management</b>                       | <b>2</b>  |
| 1.1      | Roles and Organizational Chart . . . . .        | 2         |
| 1.2      | Meetings and Communication Structure . . . . .  | 2         |
| 1.3      | Work and Time Schedule . . . . .                | 3         |
| 1.4      | Finances . . . . .                              | 3         |
| 1.5      | Partners and Social Media . . . . .             | 4         |
| 1.6      | Outreach . . . . .                              | 4         |
| <b>2</b> | <b>Flight Mechanics</b>                         | <b>4</b>  |
| 2.1      | Optimization Algorithm . . . . .                | 5         |
| 2.2      | Optimization Results . . . . .                  | 7         |
| 2.3      | Empennage Geometry . . . . .                    | 7         |
| 2.4      | Mass and CG Computation . . . . .               | 8         |
| 2.5      | Payload Prediction . . . . .                    | 9         |
| 2.6      | TO Prediction . . . . .                         | 9         |
| 2.7      | Propeller Wind Tunnel Testing Results . . . . . | 9         |
| 2.8      | Software Analysis . . . . .                     | 10        |
| <b>3</b> | <b>Structure</b>                                | <b>11</b> |
| 3.1      | Fuselage . . . . .                              | 12        |
| 3.2      | Wing . . . . .                                  | 14        |
| 3.3      | Tail . . . . .                                  | 16        |
| <b>4</b> | <b>Aerodynamics</b>                             | <b>17</b> |
| 4.1      | Airfoil . . . . .                               | 17        |
| 4.2      | Wing . . . . .                                  | 19        |
| 4.3      | Fuselage . . . . .                              | 19        |
| 4.4      | Tail . . . . .                                  | 22        |
| 4.5      | Landing Gear . . . . .                          | 22        |
| 4.6      | Polar Curves . . . . .                          | 23        |
| <b>5</b> | <b>Electronics and Propulsion</b>               | <b>23</b> |
| 5.1      | Flight Controller . . . . .                     | 23        |
| 5.2      | Radio Control Protocol . . . . .                | 24        |
| 5.3      | ESC . . . . .                                   | 25        |

|          |                                                          |           |
|----------|----------------------------------------------------------|-----------|
| 5.4      | Acquisition System for the Wind Tunnel Testing . . . . . | 27        |
| 5.5      | Static Tests Stand and Acquisition System . . . . .      | 27        |
| 5.6      | Battery Testing . . . . .                                | 29        |
| 5.7      | Servos . . . . .                                         | 30        |
| <b>A</b> | <b>Acronyms, Abbreviations and Nomenclature</b>          | <b>33</b> |
| <b>B</b> | <b>Flight Mechanics Data</b>                             | <b>34</b> |
| <b>C</b> | <b>Structures definitions</b>                            | <b>51</b> |
| <b>D</b> | <b>Aerodynamics</b>                                      | <b>55</b> |
| <b>E</b> | <b>Electronics &amp; Propulsion</b>                      | <b>56</b> |

# 1. Project Management

FlyMi can currently count on 75 active members working in 5 different departments. The association is based in Milan, and main part of the production is carried out in the laboratories of the Department of Aerospace Sciences and Technology (DAER) at Politecnico di Milano.

## 1.1. Roles and Organizational Chart

The team is split into a technical and a managerial division. The first accounts for the Aerodynamics, Electronics and Propulsion, Flight Mechanics and Structural departments directly dealing with the executive design and the manufacturing of the drone. The latter instead is the marketing segment, which divided into 3 subsections, events, sponsorship and media, takes care of the economical aspects of the project.

Each division is headed by a technical and an administrative responsible, who together with the President, Vice-President, Project Manager, Treasurer and Secretary, constitute the board of the association. Two leaders coordinated the executive design phase guaranteeing continuity and fluidity of the work. The lamination and assembly phases were instead guided by the three most expertise members of the association in the manufacturing process.

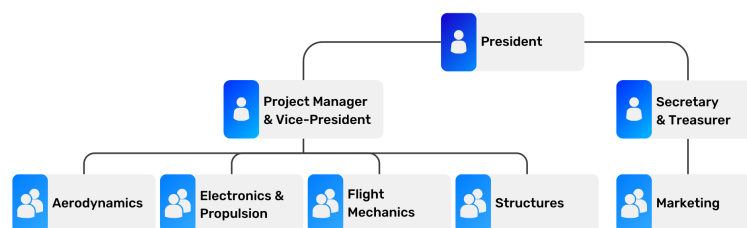


Figure 1: Organizational Chart

## 1.2. Meetings and Communication Structure

Each department holds weekly meetings, either online or in presence, attended if needed, also by one or both technical leaders. The same frequency is used to organize interdepartmental reunions to check on the status of the project and coordinate further advancements. Twice a month the whole board gathers together, to allow the executive part of the association to have a broad and always updated view on the project in its entirety. Plenary councils involving all associated members are held at least twice a semester.

### 1.3. Work and Time Schedule

Project Nyx was started at the end of September 2023 with a deep analysis of competitions' constraints and regulations. Several brainstorming sessions on both possible designs and project strategies were held, while a first recruitment to enlarge the association took place. During the conceptual design phase, many different configurations were analyzed so to find the optimal one. Once individuated this latter, the executive design was carried out. During it, every part of the aircraft was deemed and optimized to maximize performance minimizing the mass. To date, few elements are still undergoing an improvement process such as the horizontal tail and the landing gear fairings. Lamination procedure is almost done, and assembly one will begin shortly. Most of the experimental trials have been already performed while the flight testing campaign is expected to start in late May. Up to now, main issues were related to finding sponsors for moulds' production and optimal materials. If any better fabric, compared to the already owned one, will be found, some parts of the drone will undergo a further optimization of the lamination sequence resulting in lighter components.

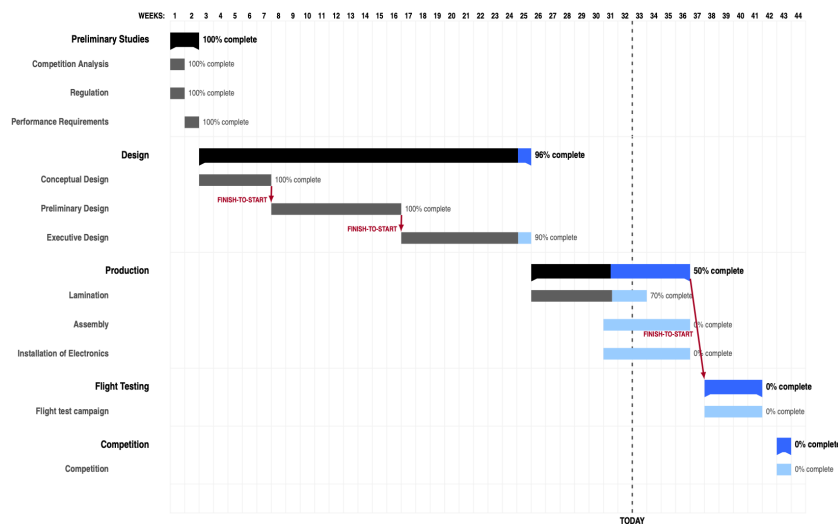


Figure 2: Gantt chart

### 1.4. Finances

During A.Y. 2023/2024 the team received a total amount of 10.500€ from Politecnico di Milano taking part to a competitive founding call hosted by the university itself and aimed at granting certain amount of money to all the associations responding.

Given the budget surplus from previous year, Nyx project started with a solid base of 15.075€. Prototype and competition expenses, such as material, tools, servos, participation fee etc., amount to 7.547€. Thanks to the hard work of the MKGT team it has been possible to get sponsorship contracts with several companies which provided us with, both money and manufacturing tools. These contracts account for a total of 10.310€. An expenditure forecast for the upcoming months including merchandise, travel expenses and participation fee, Table 4 was drawn up. A total of 3.328€ is used as emergency fund to cover any unexpected problem or need. Any further surplus will be used for future projects.

| Starting funds               |          |
|------------------------------|----------|
| Remaining from previous year | 4.575 €  |
| Politecnico di Milano        | 10.500 € |
| TOTAL:                       | 15.075 € |

Table 1: Starting funds

| Project expenses      |         |
|-----------------------|---------|
| Tools                 | 1.500 € |
| Electronic components | 2.070 € |
| Fiber Materials       | 367 €   |
| 3D print Materials    | 200 €   |
| Batteries             | 750 €   |
| Other Materials       | 1.900 € |
| Soldering Materials   | 560 €   |
| TOTAL:                | 7.347 € |

Table 2: Project expenses

| Sponsors                                            |           |
|-----------------------------------------------------|-----------|
| MKS                                                 | 1.503,5 € |
| Makita                                              | 1.463 €   |
| Bercella, Re Fraschini (lamination, autoclave etc.) | 2.500 €   |
| Others: materials                                   | 3.443,5 € |
| Others: software                                    | 1.400 €   |
| TOTAL:                                              | 10.310 €  |

Table 3: Sponsors

| Competition costs |         |
|-------------------|---------|
| Merch             | 500 €   |
| Competition fee   | 2.400 € |
| Transportation    | 1.500 € |
| TOTAL:            | 4.400 € |

Table 4: Competition costs

## 1.5. Partners and Social Media

Since its birth, FlyMi has been affiliated with the DAER, which ensures both economic support and working spaces within the university environment. Moreover the association managed to establish connections with several companies which started sponsoring the project both technically and financially. Main supporter is "MeglioBanca", while among big international names it is possible to find "MKS" and "MAKITA" which provided crucial help for the technical components. "Bercella" has been another key sponsor for the development of the project, their team provided the association with a series of moulds produced through 3D printing which using a brand new technology could sustain up to 3 cycles of autoclave. Many local realities are also supporting FlyMi supplying any kind of object instrument and material that could be needed. Social media channels are crucial to the team for reaching out individuals and companies unfamiliar with this reality, interacting with students and giving sponsors a media boost. Main drivers are *LinkedIn* and *Instagram*, the social content is constantly improved by Marketing Department members through the development of graphics and materials enabling the association to expand its network and keep people interested in the project.

## 1.6. Outreach

Despite the technical commitment of the association, several social activities and events related to engineering and aviation are often organized and attended by FlyMi members and not. During the first year of life of the association (founded in 2022), a bunch of workshops on aerodynamics and structures utility software (Ansys, SolidWorks) were held and welcomed with enthusiasm by many students. A special conference was also organized to interview Paolo Nespoli, an Italian former astronaut. FlyMi is also the Milan headquarter of EUROAVIA the European Association of Aerospace Students. As such, several international events were attended by the associated people, and one will be organized in Milan in September together with the IAC.

# 2. Flight Mechanics

## 2.1. Optimization Algorithm

For the conceptual and preliminary design of Nyx, an optimization algorithm has been developed by the Flight Mechanics Dept.: starting from a generic configuration input as a guess, the algorithm finds the design maximizing the competition's score. This code requires initialization data from: Sizing Matrix Plot (SMP), aerodynamic polar, and propeller data.

### 2.1.1. Aerodynamic Polar Guess

In order to build a SMP, a basic aerodynamic input is required, even just a  $C_L^{\max}$  achievable in different configurations or  $C_L^{\text{cruise}}$ . Instead of relying solely on statistical data from other aircraft (e.g. other universities models), FlyMi's first model (Aethra) has been used.

### 2.1.2. Sizing Matrix Plot

To construct a Sizing Matrix Plot, requirements have to be imposed, and for Nyx they are:

- Stalling speed (Team proposed)
- Maneuvering Speed (Team proposed)
- TO ground roll (Competition imposed)
- Rate of Climb (Team proposed)
- Angle of Climb (Team proposed)
- Cruising Speed (Team proposed)

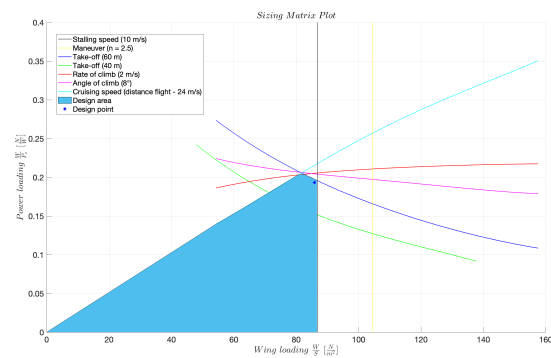


Figure 3: Sizing Matrix Plot for Nyx

From Figure 3 it can be seen that the design point chosen is:  $\frac{W}{S} = 85.88 \frac{N}{m^2}$  ;  $\frac{W}{P_S} = 0.1933 \frac{N}{W}$ . This configuration presents the smallest wing possible and the maximum achievable weight considering a fixed maximum motor power (as specified by the competition rulebook).

From Figure 3 another fact is visible: the chosen TO requirement has been 60m and not 40m, even though a 40m one ensures bonus points. This design choice has been taken considering different aspects: 5% bonus points given by a 40m TO would not have covered the weight penalty related to the design achieving it, or the current penalty imposed by the competition for exceeding a 30A supply.

### 2.1.3. Propeller Performance

The competition rules specify the *T-Motor AT2826 900KV* as the mandatory electric motor, to which 4 propellers have been coupled in its datasheet. The initial 4 selected for Nyx are 12x6, 13x6.5, 14x7, and 15x8. APC provides performance datasheets for each propeller that were used to compare them in terms of Thrust, Power, Torque, etc. (Figures 30 to 41). Based on this analysis and predicted airspeed for Nyx, together with the E&P Dept., the initial propeller chosen is 12x6, which guarantees higher Thrust and Power.

### 2.1.4. Algorithm Variables

After establishing the first-level inputs (SMP, Aerodynamic Polar, and Propeller), the optimization method can be initialized by defining its variables, which comprise the lifting surfaces' primary geometric features, such as MGC, TR, and  $\Lambda_{C/4}$ .

Each variable is determined by three parameters: initial guess, lower, and upper boundary, to ensure no unattainable geometries. Boundaries are determined based on design constraints (e.g. tail smaller than wing) and geometrical limitations (e.g. transport box limits).

The optimization algorithm only considers the main lifting surfaces, with the fuselage ana-

lyzed separately as its design does not significantly affect the flight mechanics results. This simplification is deemed suitable for conceptual and preliminary design, as the optimization results are not exact since the algorithm does not consider full non-linear dynamics.

Other inputs are configuration specifications, which have been defined during the design process in agreement between all departments, such as the High-Wing Configuration (enables a better cargo bay in the fuselage) and the T-Tail configuration (decided in accordance with all Departments, a detailed analysis is given in Section 2.3).

Wing and Horizontal Tail dihedrals have been imposed to  $0^\circ$  since no stability issues related to them are present. Furthermore, straight wings are easier to design and produce.

The code performing aerodynamic analyses must take into account also Wing and Tails airfoils, in particular for the wing an *S9000* has been used (afterward optimized, details in Section 4.1.2), while for both Tails, a *NACA 0012* first guess, based on Aethra, has been found to work well and therefore later confirmed.

The 2D airfoil aerodynamic polars have been computed using *XFOIL*, considering different Reynolds Number values, relevant to the specific flight conditions predicted for Nyx.

Lastly, an objective static margin ( $SM_{obj}$ ) has been imposed. A 14% SM has been deemed sufficient in the early stage of design, being the algorithm free to find solutions within a range of  $(0.8*SM_{obj} ; 1.3*SM_{obj})$ . The Static Margin Formula used is:

$$Static\ Margin = SM = \frac{X_{AC} - X_{CG}}{MAC_W} * 100 \quad (1)$$

### 2.1.5. Algorithm Optimization Process

The core of the Optimization Algorithm follows 9 main steps:

1. Generate aircraft geometry based on inputs
2. Generate new aerodynamic polar using a panel method implemented in Matlab
3. Compute aircraft Aerodynamic Centre (AC)
4. Estimate aircraft Center of Gravity (CG) (explained in more detail in Section 2.4)
5. Compute Static Margin (SM)
6. Find available Payload Mass ( $M_{PL}$ )
7. Compute Competition Score (Payload + Efficiency + Distance)
8. Compute eventual TO bonus
9. Compute Total Score

The basic starting geometry assumes rectangular wings and tails, however, the algorithm is free to find configurations with tapered and/or swept surfaces.

Knowing the aircraft's geometric and aerodynamic properties, the total AC can be computed together with the aircraft's mass and subsequent CG through a specific algorithm (detail in Section 2.4).

The  $M_{MTO}$  is known through SMP and motor-propeller coupling. Entering the coupled motor-propeller datasheet with the 30A current limit imposed by regulations, a maximum  $P_S$  of 296W has been found, leading to a  $M_{MTO}$  of 5.846kg. The available Payload Mass is individuated subtracting the  $M_{OE}$  from the  $M_{MTO}$ .

The competition score is computed with specific functions for each category (Payload, Efficiency, Distance), considering also different flight attitudes to gain more points.

Lastly, TO distance is evaluated, to check if the 5% bonus can be applied.

However, every configuration found has to satisfy constraints, such as:

- Geometric Considerations (for instance, surfaces relative positions)
- Aerodynamic Considerations (downwash angles and wake)
- Static Stability (unstable configurations are not allowed)
- Dynamic Stability (preliminary assessment, subject to post-processing)

Once a design has passed all the checks imposed by the above-illustrated bounds, the algorithm tries to modify it in a way that increases the total score without violating the re-

quirements.

The algorithm ultimately converges, when modifying each parameter, and does not result in a higher score compared to the previous configuration.

## 2.2. Optimization Results

The best configuration found through the algorithm is shown in Figure 4 using the software *XFLR5*, with its main parameters displayed in Table 5:

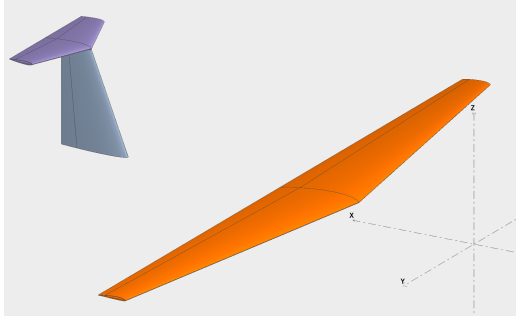


Figure 4: 1<sup>st</sup> Configuration in *XFLR5*

| Component | Parameter                       | Value                | Component       | Parameter            | Value                |
|-----------|---------------------------------|----------------------|-----------------|----------------------|----------------------|
| Wing      | b                               | 3.02 m               | Horizontal Tail | b                    | 0.39 m               |
|           | S                               | 0.612 m <sup>2</sup> |                 | S                    | 0.072 m <sup>2</sup> |
|           | AR                              | 14.92                |                 | AR                   | 2.13                 |
|           | TR                              | 0.45                 |                 | TR                   | 0.50                 |
|           | Root Chord                      | 0.28 m               |                 | Root Chord           | 0.24 m               |
|           | Tip Chord                       | 0.13 m               |                 | Tip Chord            | 0.12 m               |
|           | MAC                             | 0.212 m              |                 | MAC                  | 0.183 m              |
|           | X <sub>LE</sub> Root            | 0.50 m               |                 | X <sub>LE</sub> Root | 1.27 m               |
| Aircraft  | Λ <sub>C/4</sub>                | 4.40°                | Vertical Tail   | Z <sub>LE</sub> Root | 0.46 m               |
|           | Static Margin                   | 17%                  |                 | Λ <sub>C/4</sub>     | 24.98°               |
|           | Number of Payload Balls         | 12                   |                 | b                    | 0.39 m               |
|           | X <sub>AC</sub> of the Aircraft | 0.652                |                 | S                    | 0.072 m <sup>2</sup> |
|           | X <sub>CG</sub> of the Aircraft | 0.616 m              |                 | TR                   | 0.52                 |
|           |                                 |                      | Vertical Tail   | Root Chord           | 0.24 m               |
|           |                                 |                      |                 | Tip Chord            | 0.13 m               |
|           |                                 |                      |                 | MAC                  | 0.184 m              |
|           |                                 |                      |                 | X <sub>LE</sub> Root | 1.16 m               |
|           |                                 |                      |                 | Λ <sub>C/4</sub>     | 12.75°               |

Table 5: 1<sup>st</sup> Configuration Data from Algorithm

The Aerodynamics Dept. then further optimized the wing to minimize drag and, since in Figure 4 the Vertical Tail looks quite big, in post-processing it has been shrunk but still guarantees an adequate stabilizing effect (details in Section 2.8.1). The final Nyx configuration can be seen in Figure 5, with its main changed characteristics listed in Table 6:

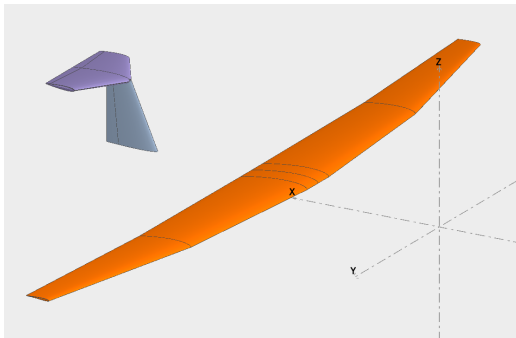


Figure 5: Final configuration in *XFLR5*

| Component | Parameter                        | Value                | Component     | Parameter            | Value               |
|-----------|----------------------------------|----------------------|---------------|----------------------|---------------------|
| Wing      | b                                | 3.08 m               | Vertical Tail | b                    | 0.25 m              |
|           | S                                | 0.617 m <sup>2</sup> |               | S                    | 0.04 m <sup>2</sup> |
|           | AR                               | 15.40                |               | AR                   | 6.45                |
|           | TR - Inboard                     | 0.78                 |               | TR                   | 0.48                |
|           | TR - Outboard                    | 0.44                 |               | Root Chord           | 0.21 m              |
|           | Root Chord                       | 0.27 m               |               | Tip Chord            | 0.1 m               |
|           | Inboard Section                  |                      |               | MAC                  | 0.16 m              |
|           | Sections Interface Chord         | 0.21 m               |               | X <sub>LE</sub> Root | 1.16 m              |
|           | Tip Chord                        |                      |               | Λ <sub>C/4</sub>     | 18.26°              |
|           | Outboard Section                 | 0.09 m               |               |                      |                     |
|           | MAC                              | 0.214 m              |               |                      |                     |
|           | X <sub>LE</sub> Root             | 0.50 m               |               |                      |                     |
|           | Λ <sub>LE</sub> Inboard Section  | 4.81°                |               |                      |                     |
|           | Λ <sub>LE</sub> Outboard Section | 12.14°               |               |                      |                     |

Table 6: Final Configuration Data

## 2.3. Empennage Geometry

### 2.3.1. Choice of Configuration

Empennage geometry has been defined after the decision to use a High-Wing configuration for Nyx. Initially, with a Low-Wing, the empennage was supposed Traditional; it represented the simplest choice still guaranteeing adequate stability and control.

The change to a High Wing configuration meant different wing wake interactions with the tail, and therefore a comparison between different configurations has been performed.

Three designs have been investigated: *Traditional Tail*, *T-Tail*, and *V-Tail*. For each configuration its pros and cons are shown in Table 10, however, the analysis is mostly made of qualitative considerations; quantitative investigations are based on Aethra's model in *XFLR5* since this study has been conducted before deciding the construction methods.

Modeling with *XFLR5* shows that to achieve the same aircraft's AC position with a V-Tail compared to a Traditional one, a 7% reduction in the wetted surface is achieved. As a consequence a similar decrement can be considered for the drag, being directly related to the interested area's dimension.

This 7% reduction of the V-Tail surface can be translated into a weight recovery. This latter will not be as near as 2/3 of the Traditional Tail case considering the same wing, materials, and construction methods. Moreover, mass reduction is even less when considering that the increment in needed tail lift requires bigger servos and potentially a bigger tail.

V-Tail's advantages have been deemed as too marginal if compared to the disadvantages identified in the analysis, therefore the T-Tail has been chosen as configuration for Nyx.

In particular, the main performance sought for the competition is drag minimization. From what can be seen by the analysis of the principal components in an aircraft, percentages as high as 50% of  $C_{D,0}$  of the model itself come from the wing and 30% of  $C_{D,0}$  comes from the fuselage. Only 3% and 5% come respectively from Vertical and Horizontal Tail, meaning that efforts should be aimed at optimizing the wing shape rather than the tail.

Last but not least in general, and especially in aeromodels, an ill-designed V-Tail usually makes the plane unusable, while its benefits might not be great even if realized with utmost detail. Analysis references: [15], [16], [17], [19], [20], [21], [23], [24], [26], [27]

### 2.3.2. Empennage Defining Equations

The empennage has been dimensioned in the algorithm considering a Tail Volume approach:

$$TVC_H = \frac{S_H * (X_{CG} - X_{AC_H})}{S * MAC_W} ; TVC_V = \frac{S_V * (X_{CG} - X_{AC_V})}{S * b_W} \quad (2)$$

Some parameters have been related together, especially the longitudinal position of both tails, while their Surface has been constrained by different conditions such as the AR. For the TVC there are general values derived from statistical data of other aircraft, therefore a boundary has been imposed to the allowable values in the algorithm.

In particular, it has been found that typical values for the Horizontal Tail TVC are between 0.4 and 0.5, as a consequence limits have been set between 0.35 and 0.55.

For the Vertical Tail, instead, in a T-Tail configuration the TVC is smaller, and typical values are between 0.02 and 0.04, therefore the boundaries have been set to 0.025 to 0.045.

## 2.4. Mass and CG Computation

A critical part of the design process, and in particular of the optimization algorithm, has been the definition of structural mass, and therefore the computation of the CG of the aircraft.

To achieve this feat, a function has been created together with the Structures Dept., which considers for each component (Wing, Horizontal and Vertical Tail, Fuselage) the different internal skeleton parts, in terms of both weight and arrangement. The E&P Dept. also provided information related to the masses of the electronic components needed to fly.

During the initial part of the design, no clear data was available, therefore the team had to rely only on the previous project's experience, considering similar materials and internal structure (scaled down) to have a rough estimate of both structural mass and CG location.

As the design progressed, the team agreed on a new internal structure, thus the CG estimation function could be completed. Given the team's little experience in building models, the estimations for components' weights were increased by a 10% safety factor due to construction contingencies, impacting the available payload mass ( $M_{PL}$ ).

## 2.5. Payload Prediction

Preliminary tests on the algorithm have been performed using Aethra's data to estimate the structural mass. Algorithm tuning revealed that a payload between 9 and 12 balls was achievable with Aethra's first guess. Consequently, the Structures Dept. began designing a payload support structure for up to 12 balls.

Going on with the project, and having defined the construction methods, the 12-ball payload became a clear feasible objective, confirming the first estimate.

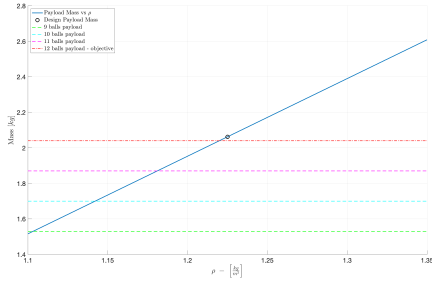


Figure 6:  $M_{MTO}$  vs  $\rho$

The most critical condition for payload has been found to be TO, therefore an analysis regarding the  $M_{MTO}$  at different air density  $\rho$  has been performed. Taking as reference air density  $\rho_0 = 1.225 \text{ kg/m}^3$ , a range of  $\pm 10\%$  has been considered:  $(1.100 ; 1.350) \text{ kg/m}^3$ . The relationship shown in Figure 6 is based on a standard TO at the Lift-Off speed determined in  $\rho_0$  conditions. In case of lower air density, maintaining the same payload requires a higher Lift-Off speed in TO conditions. Mass vs Air Density relationship is represented by:

$$M_{PL} = 4.3713 * \rho - 3.2926 \quad (3)$$

## 2.6. TO Prediction

TO analysis has been performed by integrating the general equation of motion:

$$a_{TO} = \dot{V}_{TO} = \frac{g}{W} [T - D - \mu(W - L)] \quad (4)$$

A very important parameter in the above equation is  $\mu$ , the dynamic friction coefficient which depends on the kind of take off surface.  $\mu$  has been assumed as 0.1, based on a range of 0.05-0.1 for grass runways [19].

A standard industry requirement for Lift-Off speed is  $V_{LOF}$  no less than  $1.1 * V_{stall}$ , for Nyx:

$$V_{stall} = \sqrt{\frac{2 * W_{MTO}}{\rho S C_{L,max}}} = 9.5 \frac{m}{s} \rightarrow V_{LO} = \sqrt{\frac{2 * W_{MTO}}{\rho S C_{L,TO}}} = 11.3 \frac{m}{s} > 1.1 * V_{stall} = 10.45 \frac{m}{s} \quad (5)$$

By integrating Equation 4, with the Lift-Off speed known, the TO run can be found.

During the first stages of design, having chosen the 12x6 propeller, a 40m TO was deemed not feasible. However, after the wind tunnel testing campaign (more on WTT in Section 2.7), it has been found that 40m is actually theoretically feasible.

## 2.7. Propeller Wind Tunnel Testing Results

WTT was extremely important, giving the possibility of testing a range of twelve propellers with different diameters and pitch, rather than just the initial four.

The main parameters measured in the Wind Tunnel were Thrust, Dynamic Pressure, and Airspeed. With the radio controller other parameters were obtained as RPMs, Voltage, Current, and Power. All the parameters made it possible to identify each propeller in terms of Advance Ratio, Power Coefficient, Thrust Coefficient, and Efficiency, all shown in Figures 42, 59. The identification of the best propeller (APC 12x8) in the set was a non-trivial problem, therefore using WT data, the Score for Efficiency and Distance Flights has been computed, determining the propeller guaranteeing the best Overall Score. The results are shown in Figures 44 and 45. Moreover, WTT enabled a better TO analysis, showing that a 40m TO could actually be feasible with the chosen propeller. Results are presented in Figures 46 and 47.

## 2.8. Software Analysis

### 2.8.1. Stability Analysis

The optimization algorithm is also able to compute some basic derivatives, in order to evaluate stability. In particular, static one is defined through three of those:  $C_{m_G/\alpha}$  (negative for longitudinal static stability),  $C_{l_G/\beta}$  (negative for lateral static stability), and  $C_{n_G/\beta}$  (positive for directional static stability).

The value of  $C_{m_G/\alpha}$  has been computed using a standard two-surface scheme in Matlab. The value of  $C_{l_G/\beta}$  has been instead calculated considering just the Vertical Tail contribution with an approximated formula [17] as well as that of  $C_{n_G/\beta}$ . Moreover, a High Wing configuration is in general stabilizing due to the fuselage contribution.

Implementing the algorithm's optimized configuration into *XFLR5* (Figure 4) it was possible to cross-check these derivatives, in order to effectively analyze the aircraft's stability. The results coming both from Matlab and *XFLR5* are summarized in Table 11. Derivatives from *XFLR5* are coherent with the expected ones in terms of sign, however it is more stable than forecasts in longitudinal and directional terms, while less stable than expected laterally. This is probably due to an over approximation used in the optimization algorithm.

Eigenvalues also are obtainable with *XFLR5*, representing the aircraft natural motion modes. In the MATLAB code the eigenvalues have been computed using simplified models which exploit specific aircraft characteristics. The comparison between simplified MATLAB models and *XFLR5* are shown in Figures 48 and 49 with Table 12 listing specific eigenvalues.

However, these results are related to the first configuration coming out of the optimization algorithm while, as stated above, Nyx's final configuration is different. The final shape (shown in Figure 5) has not been implemented in Matlab, therefore results are only available through *XFLR5*, and are shown in Figures 50 and 51 together with Tables 7 and 13:

| Derivative       | Software Results |                              |                    |
|------------------|------------------|------------------------------|--------------------|
|                  | Matlab           | XFLR5 1 <sup>st</sup> Config | XFLR5 Final Config |
| $C_{m_G/\alpha}$ | -0.9811          | -2.252                       | -0.7196            |
| $C_{l_G/\beta}$  | -0.03673         | -0.02909                     | -0.00950           |
| $C_{n_G/\beta}$  | 0.07164          | 0.1419                       | 0.03215            |

Table 7: Stability derivatives obtained with Matlab and *XFLR5*

It can be seen from Table 7 that the final configuration is actually less stable than the first optimization result. This is perfectly in line with the expectations, since the goal of the post-processing leading to the final design was drag minimization not stability increment. From Table 13 it can be seen that Spiral and Phugoid modes are less unstable if compared to the previous configuration; moreover with some further CG adjustment during the flight testing campaign the aircraft can achieve completely stable phugoid.

### 2.8.2. Control Surface Sizing

#### Flap Analysis

Flap requirements are based on TO constraints. To be more conservative, an initial guess for TO attitude has been of  $C_L=0.8$  and  $C_D=0.1$ , guaranteeing a TO in 60m. A less conservative guess for TO attitude considered a  $C_L=1.1$  and a  $C_D=0.12$ , which would guarantee a 40m TO with the new propeller chosen after wind tunnel testing.

Using the configuration of Figure 5, flaps have been introduced in the inboard section of the wing, covering its full span, being 6cm in chord (25% of mean chord, straight flaps lead to easier mechanism). The results are shown in Figures 52 and 53, and it is clear that a  $C_L$  of 1.1 is easily achievable by 25% chord flaps,  $C_D$  values shown in Figure 52 are particularly low as they just consider lifting surfaces in *XFLR5*, and not the whole aircraft.

## Aileron Analysis

A parametric analysis has been performed to describe the effect of aileron deflections on rolling moment. Based on Aethra, a value of  $-0.4$  for the control derivative  $C_{l/\delta_a}$  has been considered too high in module, therefore a target of  $-0.3$  has been chosen.

The study considered two different aileron definitions: the first, shown in Table 14, defines the aileron starting from the tip of the outboard section of the wing while the second, shown in Table 15, defines the aileron starting from the root.

Based on Tables 14 and 15, two possible configurations have been chosen:

- Configuration 1: full span (47.95%) and 15% chord
- Configuration 2: 40% span (from root of outboard section) and 20% chord

On these two designs, an analysis has been performed also concerning hinge moment ( $M_H$ ) at  $8^\circ$  and  $20^\circ$  deflections, leading to results shown in Table 16. Based on these numbers, Configuration 1 was initially chosen, as it guaranteed adequate control and lower hinge couples. Moreover, this latter seemed easier to produce being a full span surface.

After some structural considerations, Configuration 1 has been discarded due to attachment problems with the wing. Configuration 2 has therefore been chosen and produced for Nyx.

## Rudder and Elevator Analysis

Elevator and Rudder have been designed in parallel considering the tail assembly as a whole. In particular, both control surfaces are full span, therefore the parametric analysis is done only on the chord. Tables 17 and 18 show the previous study results.

Taking Aethra as reference, its control derivatives reflect statistical trends and can be assumed as targets ( $C_{m/\delta_e} = 1.2611$  and  $C_{n/\delta_r} = 0.04124$ ). Looking at Tables 17 and 18 Nyx's ones are quite low in comparison.

Rudder control has been deemed as not too critical, thus accepting lower values even if this implies less directional maneuverability during the TO run.

Elevator analysis from *XFLR5* has left the team puzzled, being its control derivatives very low if compared to other aircraft of similar size, which in general tend to have values around 1.2. Moreover,  $M_H$  seems off by an order of magnitude if compared to ailerons and rudder (Tables 16 and 18), which is quite odd considering their resembling dimensions.

Since longitudinal control is crucial also for TO performance, an all-moving Horizontal Tail has been hypothesized based on textbooks considerations regarding elevator chord percentages [19] and the flexibility allowed by 3D printing technology for the tail assembly.

However, another analysis on the elevator has been performed using OpenVSP, in order to assess if the problem was within *XFLR5* or if the tail was actually under-dimensioned.

The conducted analysis involving 40% chord elevator returned a control derivative  $C_{m/\delta_e} = -1.0398$ , which is not that high if compared to *XFLR5*, however it should be noted that, as seen in Figure 54, the elevator is not full span in OpenVSP as it is in *XFLR5*.

Considering all the aspects stated above, the Flight Mechanics Dept. assessed the elevator as being suitable for flight and, if during flight testing it proves to have little control authority, a new all-moving stabilizer will be investigated in greater detail.

## 3. Structure

As soon as Flight Mechanics department optimization script specifications came out the Structures department started working on a preliminary geometry of the drone. In the meantime, Aerodynamics department took care of the optimization of the wing's characteristics aided by CAD parametric modelling. The general sizing of the model was decided from the following criteria:

1. ensuring structural strength while minimizing weight wherever possible

2. satisfying the transportation box requirements (280 x 300 x 820 mm)
3. allowing manufacturing feasibility for inexperienced people

In order to make the work more efficient the structures team was divided into three groups: fuselage, wing and tail. From now on work is presented following this internal subdivision.

### 3.1. Fuselage

A versatile design was chosen for the fuselage, to encounter the necessity of having a CAD in a relative short period allowing for the milling of MDF moulds, guaranteeing at the same time enough space for the stowage of both electronics and payload whose configuration was yet undefined. This model included Flight Mechanics department indications concerning relative positioning of relevant aerodynamic points. The central part of the fuselage was obtained starting from a triangle adapted to follow the shape of the payload (three billiard balls in each section), while the front part was obtained lofting the cargo section to the circular motor mount.

This “conical” part and the payload bay sustain all the internal weights (electronics and payload) as well as those coming from wing and tail connections. As a result F.E. analyses were performed only on these parts.

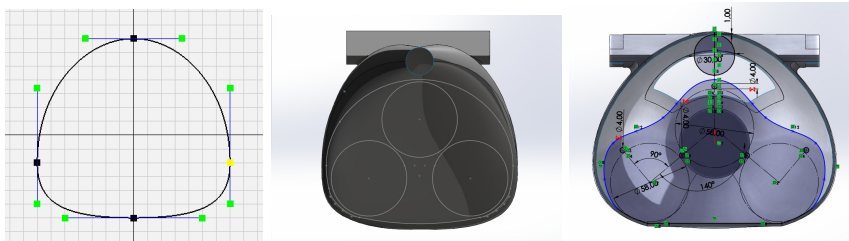


Figure 7: Stages of central fuselage design

#### 3.1.1. Payload Support

The design of this component went through several iterations to minimize weight and improve loading and unloading time. The aim was to have three billiard balls in different sections, each divided by bulkheads made in additive manufacturing (and then topologically optimized). Components and balls are tightened in place by a pass-through screw. The design also contemplates an external case to maintain the various pieces in place, made in additive manufacturing (vase mode) with low weight PLA. The case is then coated in resin to increase its strength (design 1). Currently another configuration is under evaluation in which external case is removed, first bulkhead is glued to the fuselage and the single central screw is replaced by 5 thin carbon fibre sticks that act as guides for the balls (design 2). The final solution will be chosen once tests will be performed on both designs.

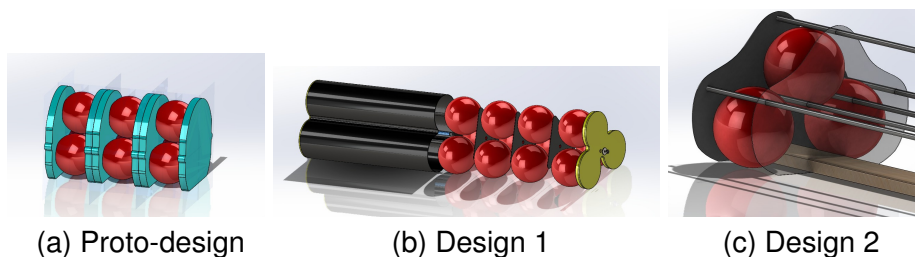


Figure 8: Different payload support designs

### 3.1.2. Central Fuselage

Given the structural requirements to sustain the loads coming from wing and tail, this part was made of two plies of lightweight glass fabric with one ply of carbon fabric in between (all plies are oriented at  $\pm 45^\circ$ ). This layup was verified using a FEA with seeds ranging from 0.5 to 5mm with the loads coming from the wing, tail and motor mount. For this model shell elements were used in a quad-dominated mesh (mainly S4R elements, with S3R elements used in case of high gradients of the geometry). The results showed an unsatisfying pitch deformation, resulting in the addition of Rohacell reinforcement ribs in the lamination sequence (1mm thick Rohacell core in between four plies of carbon fabric) along the longitudinal axis and at the motor mount. Below are shown the in-flight and the landing load conditions (both with load factor of 2.5 and safety factor of 2. Tsai-Wu failure index below 0.3 for both).

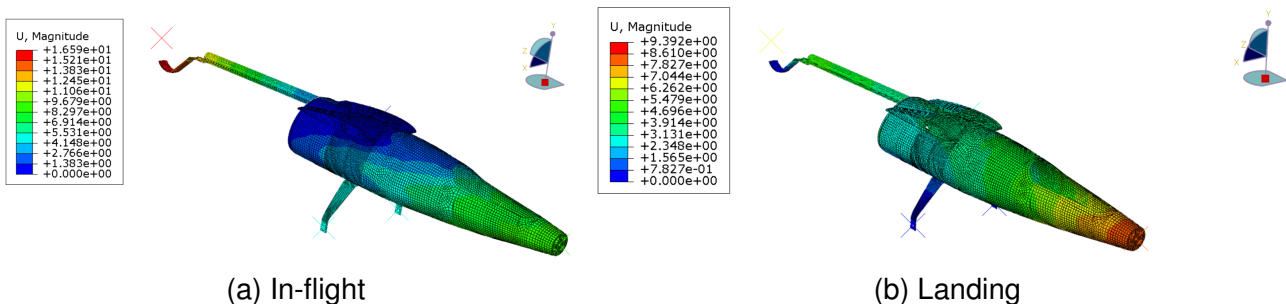


Figure 9: FEA results (U) - Fuselage

The load carrying carbon fiber part of the fuselage ends abruptly exposing the payload bay, then closed by a rotating capote, made of fiberglass as it just has an aerodynamic function. To minimize fuselage weight and ensure the distance between the aerodynamic centres of the wing and of the tail a carbon fabric tube was chosen for the tail connection. The tubular section was the best compromise between structural (need of sustaining torsional and bending loads coming from the tail minimizing the weight) and aerodynamic requirements. Moreover it is a slender component that can be easily put inside the transportation box. The front part of the fuselage ends with another capote, which ensure access to the electronics bay, and continuity of the external shape allowing in the meantime the “unobstructed vertical view of the sky” required by regulation. Since it has no significant structural role (FEA were performed without it), it was designed with the only goal of minimising weight. Several materials and building technologies were considered:

1. Glass fabric composite
2. Additive manufacturing
3. Thermoplastic moulding

The third solution was selected because it resulted to be the cheapest (does not need expensive moulds and autoclave cycles as 1) and lightest (especially with respect to 2:  $\rho_{PLA} = 1.25g/cm^3$ ,  $\rho_{plexiglass} = 1.18g/cm^3$ , and it allows to obtain a thickness lower than 0.4mm, the minimum wall thickness imposed by the team’s 3D printers, which is anyway a value not practically usable for a component). Ongoing tests will tell whether 3D printed ribs or carbon fibre sticks should be inserted to have lower deformations.

### 3.1.3. Landing Gear

A tail dragger solution was chosen because of its lower drag in flight, manufacturing simplicity and lightness. The design of the front landing gear was based on the indications on D. Raymer “Aircraft Design. A Conceptual Approach”, chapter 11, fig 11.4, *tail dragger* [16].

The height of the main landing gear was set to 120mm, based on propeller's diameter (12") while its width was set to obtain a satisfying triangle of stability. A C-shaped composite design was chosen to provide optimal energy dissipation thanks to its bending deformation allowing for the onset of dissipative visco-elastic effects in the matrix. As for the wheels, a commercially available 50mm diameter one seemed to be a good compromise between weight, width of the tire and general size. This way, the landing gear should be able to avoid the sinking of the wheel in the soft soil due to a too high contact pressure on the landing strip and dissipate the proper amount of kinetic energy during landing.

To enable an easier drilling of the four holes for screws and to properly redistribute concentrated loads, the component is attached to the fuselage in a skin reinforced section.

Two solutions were implemented to reduce its drag:

1. 3D printed PLA fairings for the wheels
2. Lamination of main landing gear with wider outer plies that, thanks to pressure granted by the vacuum bag during the autoclave cycle, allows for a more tapered shape of the whole component

For what concerns rear landing gear, the main constraint was a maximum clearance angle of  $15^\circ$  on the ground. The first design consisted of a 2mm wire-thick spring supporting a wheel, which was later discarded due to aerodynamic considerations, high weight (since it needed a PLA support) and difficulties in having such element custom made. Given these problems the design was changed to a simpler skid made either in aluminium alloy or carbon composite. Weight of both solutions was evaluated and FEA were carried out to ensure compliance with the maximum deformation and structural failure limits, resulting in the choice of carbon fabric. ( $m_{Al}=33g$ ,  $m_C=14g$ ). These analyses employed shell elements (S8R) with a seed refinement near the junctions, encastre boundary condition at the holes and with worst landing scenario loads (with 1.5 safety factor) distributed on 90% on the front landing gear and 10% on the skid. The resulting Tsai-Wu failure index for the main landing gear is 0.875 and for the skid 0.438. The lamination sequences chosen are (respectively for the front landing gear and the skid):

1.  $[(0 - 90)_3 \text{Fabric} / 90_{UD} / 0_{UD} / 0_{UD} / 90_{UD} / (0 - 90)_3 \text{Fabric}]_s$
2.  $[(0 - 90)_2 \text{Fabric} / (0)_2 \text{UD} / 90_{UD} / (0)_2 \text{UD} / (0 - 90)_2 \text{UD}]$

### 3.2. Wing

For the internal structure of the wing, taking inspiration from modern gliders, it was evaluated to have a load carrying skin made of carbon fabric (outer skin) and glass fabric (inner skin) with a Rohacell core. This solution allowed for the removal of all the internal ribs except for wing to wing and fuselage to wing interfacing ones and to have an overall lighter and less buckling prone structure.

These connection systems are made through carbon composite tubes that fit into round holes in the ribs and are then fixed with a pin.

Due to the limitations imposed by the transportation box, in order to guarantee the correct span, each half-wing has been divided into two parts.

Once the final design was set a FEA was performed to validate both strength and wingtip displacements during static test and flight. For these analyses shell elements (S4R) were used because the objective was to investigate the behaviour of the structure in normal operating conditions (requested outputs Tsai-Wu failure index and displacements). Since damages and failure modes such as delamination were not taken into account, the increase in computational cost coming from the adoption of 3D elements was not justified. Boundary conditions at the wing-fuselage connection were simulated by placing two reference points in the centres of the two holes in the wingroot rib. The first one blocks all degrees of freedom

except pitch rotation, which is constrained by the second reference point that acts as a pin blocking any translation in x and y directions.

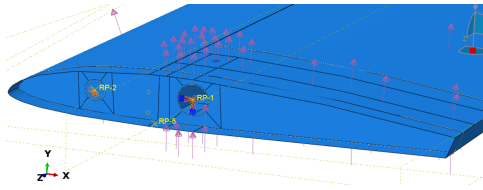


Figure 10: Wing root constraints

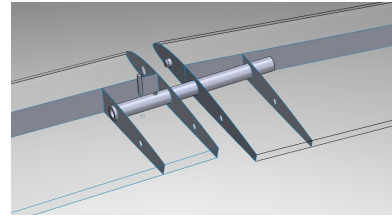


Figure 11: Wing connection system

The static test analysis was performed by imposing a vertical upwards load of 40N (corresponding to  $1.5 \cdot W_{MTO}/2$ ) at the wingtips.

For the in-flight simulation three different spanwise load distributions (triangular, elliptical and constant) based on the Aerodynamics department outputs were evaluated on a simplified structure. The results of the three different distributions were practically the same, thus the constant load distribution was applied on the real wing model since it converged faster and with fewer issues. It consisted in 100N ( $1.5 \cdot W_{MTO}/2$  with a load factor of  $n=2.5$ ) for each semi wing, applied for 2/3 on the upper skin and for the remaining 1/3 on the lower skin.

In the FEA two lamination sequences were tested, to verify the need of Rohacell core (lamination sequences listed below go from the outside to the inside):

1. 1 glass fabric ply ( $\pm 45^\circ$ ) 3 carbon fabric plies ( $\pm 45^\circ$ )
2. 1 glass fabric ply ( $\pm 45^\circ$ ), 2 carbon fabric plies ( $\pm 45^\circ$ ), Rohacell (1mm), 2 glass fabric plies ( $\pm 45^\circ$ )

The glass fabric used is very light ( $25g/m^2$ ) and has the role of increasing the surface finish of the carbon plies (avoiding any spots with lower resin content that may cause an exposure of the fibres, as happened in FlyMi's previous model Aethra) and restoring the resin absorbed by the Rohacell core.

The first option was from the static analysis slightly better, but with a higher overall weight and a higher carbon fabric consumption (which was available only in limited quantities), thus second lamination sequence was chosen.

Since FEA revealed an unsatisfying bending stiffness of the wing (tip displacement in the static test of 36cm), it was decided to add a UD reinforcement of two plies 2cm wide with the fibres along the spanwise direction to help the spar restrain the deflection due to the bending moment. Due to convergence problems the effectiveness of the UD reinforcement was tested only in a simplified model consisting of the external skin. The positive result of this analysis led to its adoption in the final design.

The Tsai-Wu index value exceeded 1.0 at the interface between the root ribs and the spar, problem that was solved with a mesh refinement in that area.

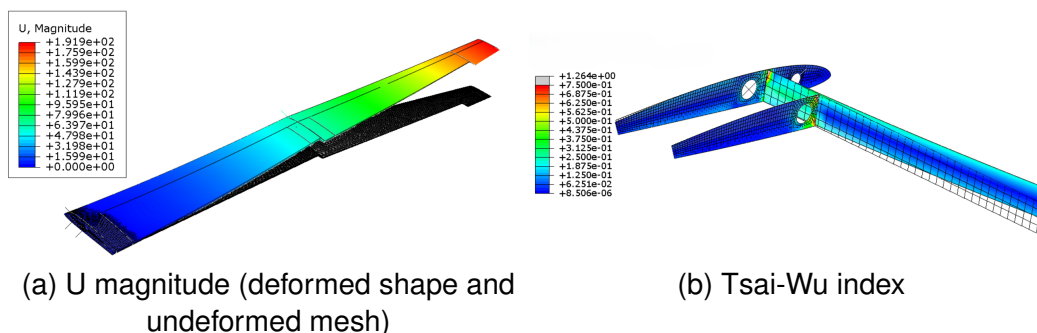


Figure 12: FEA results - wing

To obtain the servo-actuators carvings in the outer section of the semi wings it was needed to split the spar into two parts, thus compromising its structural role. To restore the bending stiffness of the section an external carbon fibre reinforcement rectangular piece was added (lamination sequence: UD  $0^\circ$ , fabric  $\pm 45^\circ$ , UD  $0^\circ$ , where  $0^\circ$  is the spanwise direction).

As for the movable surfaces, initially classic hinges were considered but given the reduced thickness of the airfoil, a more suitable solution made of aramid fibres stripes placed in most of the length of the control surfaces was implemented. These aramid fibres elements are compliant enough to be bent easily under the action of the servo-actuator, while ensuring the continuity of the airfoil.

Movable surfaces are manufactured in the same manner as the wing, with a thin skin and internal ribs obtained from a sandwich panel (skins of carbon fabric and honeycomb core) cut into shape by waterjet.

### 3.2.1. Wing Moulds

The wing moulds were produced using an innovative additive manufacturing technique specifically designed to use high melting point polymers (XECARB® 40-C20-3DP -PC-20%FC 3D printing modified-) capable of sustaining autoclave cycles up to  $120^\circ\text{C}$  and to grant a high surface finish. Moreover, they are realised in a completely recycled material, constituting a major step forward for the environmental sustainability of the project.

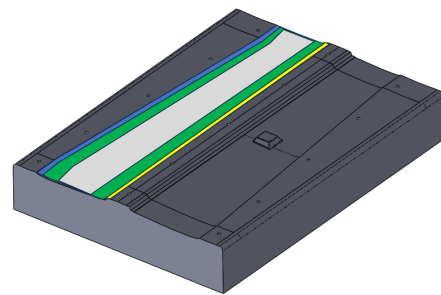


Figure 13: Wing mould

### 3.2.2. Preliminary Buckling Analysis

The choice between carbon skin and sandwich of carbon and Rohacell was based on a preliminary FEA of a plate ( $a=640\text{mm}$ ,  $b=220\text{mm}$ ) representative of a wing panel loaded in compression and shear. The analyses were performed with two load cases (shear equal to 20% and 40% of compression) and three possible layup sequences (carbon fabric  $0^\circ/0^\circ/0^\circ$ , carbon fabric  $0^\circ/\pm 45^\circ/0^\circ$ , carbon fabric  $0^\circ$  / Rohacell / carbon fabric  $0^\circ$ ). The sandwich turned out to be the lightest and the one with the highest buckling load (the eigenvalues obtained for the sandwich are between 18.7 to 19.2 times higher than those of the all carbon plates depending on the layup and load condition).

## 3.3. Tail

Initially the tail was meant to be constructed of composite materials, following the same design philosophy of the rest of the aircraft. This first design approach resulted in complex geometries in the attempt to create demountable joints. These joints could only be made in additive manufacturing, thus making the overall structure heavy due to their bulkiness. As a result, it was decided to directly make the whole tail in additive manufacturing in vase mode, reinforcing it with internal load carrying carbon fabric tubes, that could be used to lock the various parts together too, rear landing skid included.

This choice allowed the deepening of the possibilities of additive manufacturing and to take advantage of the technique's high design flexibility since it allows to have samples and specimens to test in a shorter time than that of a sponsor or of a classic construction method and at relatively lower cost. Compared to a more classic composite construction a structure made in additive manufacturing has lower specific mechanical properties which a priori implies a higher mass. But given the small dimension of the tail, the fact that it would not be

manufactured by skilled workers and that additive manufacturing allows for many tests, the design can be pushed to the limit of the structure's strength (reduction of safety factors due to a reduction of the uncertainties) resulting in a lower overall mass.

Once the T shape was set by Flight Mechanics the final design with both servo-actuators in the vertical tail was obtained from an iterative process.

Given the difficulty of performing FEA on elements made in vase mode (technique used for all the lifting components) it was decided to directly test the components to verify their mechanical properties, and possibly redesign them. Prior to the first test it was thought that only the external wall would not sustain the aerodynamic loads, thus the design implemented ribs and internal channels for carbon composite tubes to grant for the strength and stiffness of the component.

The results of the first test on vertical tail showed that carbon tubes were not needed since the component failed at 4.6kg of tip load with a tip deflection of 1.6cm. This was a consequence of the carbon tubes channels acting as spars. After this test the internal structure was lightened and the shape of the carbon tubes channels changed into that of spars, since the carbon tubes were not needed anymore. In this new design the role of the carbon tubes switched from structural to functional, allowing for simple connections between the parts, thanks to their high coefficient of friction. The elevator and the rudder too are made in additive manufacturing, and are connected to the main surface with simple hinges whose slots are directly created during the printing process.

Some full 3D printed components were employed to connect the various elements of the tail (tube, H tail, V tail). Tests of these components are now being held and a possible design improvement is to make some channels in them for carbon sticks that would increase the specific mechanical properties.

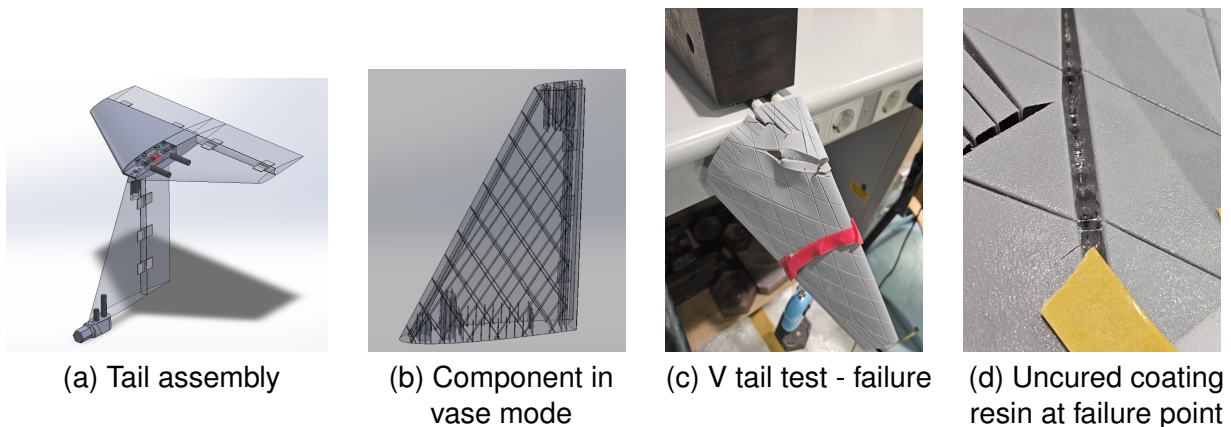


Figure 14

## 4. Aerodynamics

### 4.1. Airfoil

#### 4.1.1. Selection

Given the estimated Flight Mechanics performance requirements, the Aerodynamics department made a preliminary airfoil research using XFLR5 software to select the best-fitting one for the Nyx project. The aimed target was to find a shape guaranteeing a  $C_L^{max} \geq 1.3$  while maintaining a sufficiently high Lift-to-Drag ratio around  $\alpha = 0^\circ$ , which was the expected angle of attack (AoA) in cruise.

From a literature research involving various gliders datasheets, a total of 38 sailplane-inspired airfoils were selected and tested using the boundary layer equations solver of XFLR5 in 2D, within a range of low angles of attack to have more reliable results. Among all the analyzed geometries, the most suitable for the aimed performance was the S9000: in fact, despite not having the highest  $L/D$ , it emerged as the one with the lowest  $C_D$  in cruise condition, while not requesting for a negative pitching of the wing, being the  $C_L$  yet positive.

#### 4.1.2. Optimization

To further improve performances of the chosen airfoil, a 2D optimization was carried out through an XFOil-based code. Taking into account a particle-swarm type algorithm, a Hicks-Henne shape function was used to obtain a drag reduction in cruise configuration.

Given that at  $\alpha_{cruise} = 0^\circ$  the value of  $C_L$  for a standard S9000 is 0.31, trying to keep this property as fixed as possible (accepting eventually some small oscillations around the angle of such operating point), a  $C_D$  minimisation was imposed. From provided data by the Flight Mechanics codes, a cruise speed of 27 m/s was expected, therefore a  $Re = 350000$  was set for the analysis, together with  $N_{crit} = 9$  to better manage the laminar-to-turbulent transition over the airfoil in the studied case.

Further geometrical constraints were set to satisfy both structural and manufacturing needs: a sufficient thickness at  $\frac{3}{4}c$  was required to make clearance for the wing spar allocation, while other upper and lower bounds in camber, thickness and trailing edge angle were determined to make the output airfoil meet production necessities:

- thickness:  $0.07 \leq t \leq 0.10$ ;
- camber:  $0.02 \leq c \leq 0.10$ ;
- T.E. opening angle:  $\beta_{TE} \geq 4^\circ$

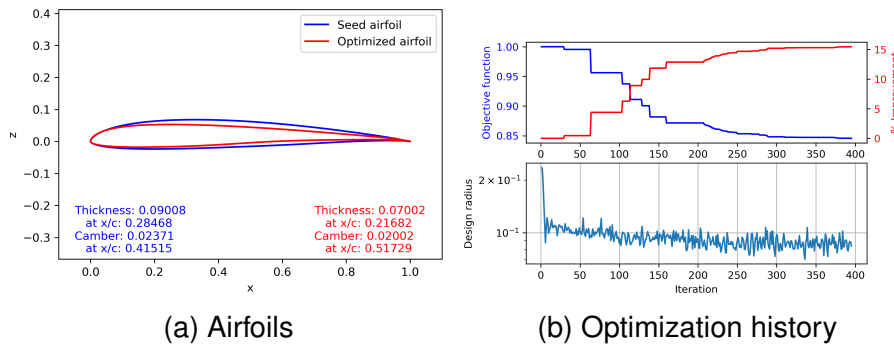


Figure 15: S9000 optimization for fixed  $\alpha = 0^\circ$  and fixed  $C_L = 0.32$

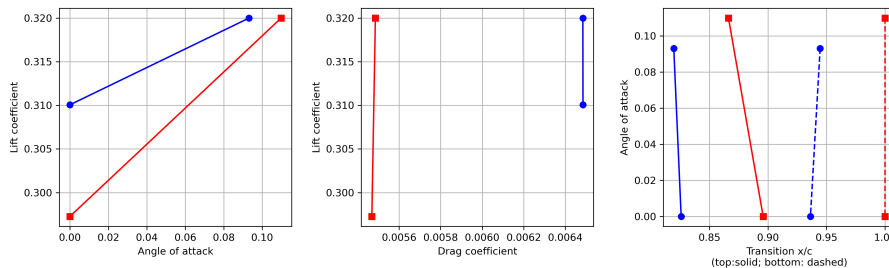


Figure 16: S9000 optimization for fixed  $\alpha = 0^\circ$  and fixed  $C_L = 0.32$

No moment constraints were imposed, other than the seed-airfoil values.

Despite the optimized shape lowers a bit the  $C_L$  ( $-4.19\%$ ) it positively acts on the  $C_D$  ( $-16.15\%$ ), further contributing to the delay of the flow transition, as shown from previous

graphs. The postponement of this phenomena on both back and belly of the airfoil made it suitable to be used at the wingtip. In fact, as indicated in Section 4.2, the final wing configuration presents a positive sweep, which could lead to a control loss if a stall condition occurs. The ailerons would be fully interested by flow detachment in this condition, thus using an airfoil with a late separation point towards the tip section can help maintaining control surfaces effectiveness if a stall circumstance.

## 4.2. Wing

A preliminary wing design was obtained in compliance with the competition dimensional limitations: with an estimated box length of 80cm, the best trade-off between aerodynamic efficiency and weight minimization was to limit the wing to two main sections (with a 3<sup>rd</sup> central one having the only function of constituting the structural attachment to the fuselage).

To maximize the wing  $L/D$ , a MATLAB code was created aiming at obtaining an ideal twist distribution along the span to achieve the highest Oswald factor possible with such geometry. Input parameters to the code were wingspan, root and tip chords (for a first estimation of a single-section wing), AoA of the root and wing surface, along with the S9000 zero-lift AoA obtained from the  $C_L - \alpha$  curve. With these entries, the script computes the prescribed AoA for each section to result in an elliptical lift distribution along the whole span. The corresponding twist is then computed as the sum between the latter and the zero-lift AoA. The latter result was then implemented in XFLR5 to test its accuracy; this process led to minor but necessary changes (such as value rounding) required to meet production' necessities and tolerances. A further 3D wing performance improvement in terms of  $L/D$  ratio was pursued testing various sweep and chord distributions along the span in XFLR5. During the optimization studies, particular focus was given to match requested values from Flight Mechanics department about wing area, MAC and  $x_{AC}$ , which were therefore fixed. Another important aspect was the capability of maintaining a  $\partial C_M / \partial \alpha$  as similar as possible to the preliminary design one.

The optimized airfoil S9000<sub>opt</sub> was chosen to better manage the lift distribution over the wingtip section, allowing the latter to better follow an ellipse-shaped distribution [Figure 60], along with the aforementioned advantages discussed in Section 4.1.2. Doing so, the outer section of the lifting surface had a morphing airfoil throughout the span. The resulting wing managed so to reach a better  $L/D$  at a lower angle of attack: near the expected operating condition, at  $\alpha_{cruise} = 0^\circ$ , a  $C_D$  reduction was obtained. Moreover, the pitching tendency of the whole wing remained unchanged during the optimization, with the new  $C_M - \alpha$  curve overlapping the previous one.

| Section | Airfoil (root/tip)         | $y$ [m] | $c$ [m]   | $\Gamma$ [°] | $\varphi$ [°] | $\Lambda _{c/4}$ [°] |
|---------|----------------------------|---------|-----------|--------------|---------------|----------------------|
| Inner   | S9000/S9000                | 0.077   | 0.27/0.27 | 0            | 0/0           | 0                    |
| Mid     | S9000/S9000                | 0.730   | 0.27/0.21 | 0            | 0/0           | 3.63                 |
| Outer   | S9000/S9000 <sub>opt</sub> | 0.740   | 0.21/0.09 | 0            | 0/-2          | 9.90                 |

Table 8: Wing data

## 4.3. Fuselage

### 4.3.1. Aircraft Configuration

Concerning the wing configuration choice, a double-stepped approach was followed. A raw study in XFLR5 was originally conducted followed then by an Ansys Fluent analysis to corroborate the initial results. This path allowed the team to obtain a general overview of the aerodynamic effects through the simple usage of a panel method, individuating in advance

critical regions where accurate refinements of the mesh and deeper analysis were required to completely capture the flow behaviour. The finer investigations, provided reliable results in relation to boundary layer and recirculation phenomena, which cannot be correctly captured if not solving the complete flow equations through dedicated CFD solvers.

In order to validate the outcomes a farfield independence study and a grid convergence one were carried out. To check that the dimension of the defined domain together with the one of its constituting cells were not influencing results, multiple simulations were run changing various geometrical parameters starting from a baseline mesh. A MATLAB code was created to automate the procedure: as input parameters, some describing the geometry of the problem were chosen as well as reference values at the operating conditions, receiving in output the variables  $D$  (characteristic length of the domain),  $H$  (length of the first cell, defining so number of points in which the body was segmented) and  $H/D$  which were consequentially changed in Ansys to assess farfield independence and grid convergence. The following tables show how despite the changes made on these variables no significant differences affect the results.

| D Farfield | H          | $C_L$    | $C_D$    |
|------------|------------|----------|----------|
| D_1        | $H_{cost}$ | 0.087201 | 0.012462 |
| D_2        | $H_{cost}$ | 0.088473 | 0.012402 |
| D_3        | $H_{cost}$ | 0.085638 | 0.012395 |

(a) Steady case farfield, D variation

| D Farfield | H   | $C_L$    | $C_D$    |
|------------|-----|----------|----------|
| $D_{cost}$ | H_1 | 0.088287 | 0.012404 |
| $D_{cost}$ | H_2 | 0.088473 | 0.012402 |
| $D_{cost}$ | H_3 | 0.087993 | 0.012385 |

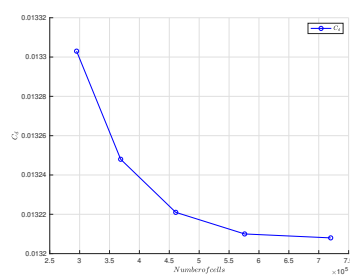
(b) Steady case farfield, H variation

| D Farfield | H    | $C_L$    | $C_D$    |
|------------|------|----------|----------|
| D'_1       | H'_1 | 0.087784 | 0.012521 |
| D'_2       | H'_2 | 0.088473 | 0.012402 |
| D'_3       | H'_3 | 0.090153 | 0.012403 |

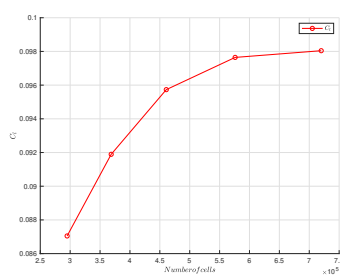
(c) Steady case farfield, H/D constant

Another point worth underlying is that different numerical methods and turbulent models combinations have been considered to find the best fitting for the case object of the study before defining the final mesh and the consequential outcome.

The case study created for the comparison of the two wings configurations will be used also at the end of the design to perform the simulations involving the whole aircraft, aimed at extracting the data to create the complete polar curve.



(a) Cd Grid Convergence



(b) Cl comparison

Figure 17: Grid Convergence

A CFD analysis to compare parasol high-wing configurations was done. During this study only the part of the aircraft mainly influencing the phenomena were taken into account. To date the back part of the fuselage was not adjusted yet and wing geometry was not optimized.

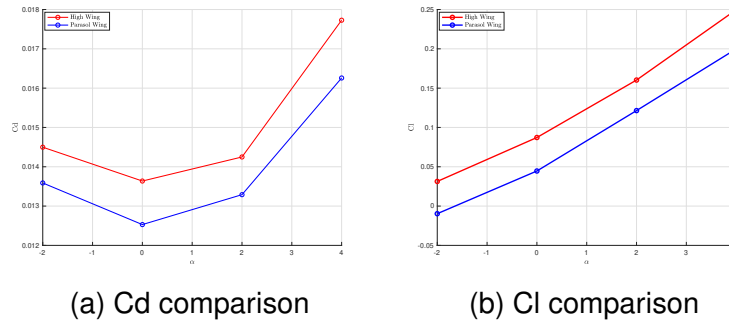


Figure 18: Cl comparison

From the shown results some considerations about the flow can be made.

The parasol configuration, due to the fuselage central shape (imposed by the payload cargo area) make the flow accelerate underneath the mid section of the wing, modifying the pressure gradient and resulting in a worse lift distribution, which couldn't compensate for the higher parasite drag given by the pylon and the higher wet area. So, even if this design allowed a more efficient droplet-like fuselage ending shape (not having to blend with the high tail cone) it had a globally lower lift-to-drag ratio which made the more conventional high wing shape the best choice.

To compensate the latter configuration main disadvantage of having a downforce-producing fuselage shape, a study on its ending part was conducted to make the flow follow a different path, thus preventing this limiting phenomena.

#### 4.3.2. Fuselage Ending

The target of this analysis was to match the structural constraints with an efficient aerodynamic shape. Being the fuselage ending the access way to the payload cargo area of the drone, it had to be detachable to allow a fast and easy loading procedure. With a maximum length constraint for this part imposed by the tail attachment on its cone, three different designs were tested:

- The first one tried to emulate the optimal fuselage ending that the parasol configuration would have suggested, namely a symmetrical one. To avoid uneven flow mixing in between the tail cone and the fuselage ending, a shark fin like panel was used.
- The second instead presented a simpler shape connecting the fuselage aft section, right behind the payload, to the tail cone with a smooth curvature.
- The last attempt was a slightly modified version of the second design, which added a hollow duct with the purpose of making the upper flow to merge with the lower one in a smoother way.

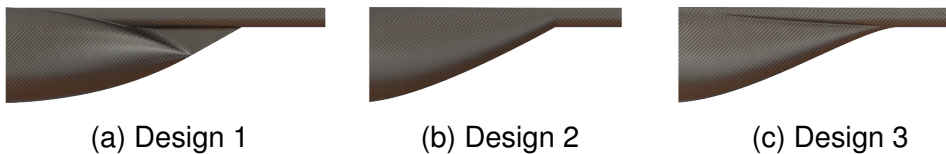


Figure 19: Fuselage ending tested designs

All three designs were forced to maintain the same frontal section, to be connected to the fuselage (which was not modified due to payload clearance). The total length instead was imposed short enough to allow for a back-slide and a rotation around the tail cone of this

part aimed at fully opening the rear cargo bay. Each design was tested with a CFD analysis conducted with Ansys Fluent. The fin of the first one didn't act on the flow as expected. The parasite drag increased significantly due to the larger wet area but no efficient flow redirection was obtained. The third one, instead of reenergizing the flow underneath the tail cone, created a chaotic mixing area between the upper and lower air, increasing the pressure drag. The second design resulted to be the best choice: it presented the least flow separation, minimizing also the recirculation of the created vortices. Moreover manufacturing speaking it was the simpler one to be laminated, not presenting surface discontinuities or acute curvatures.

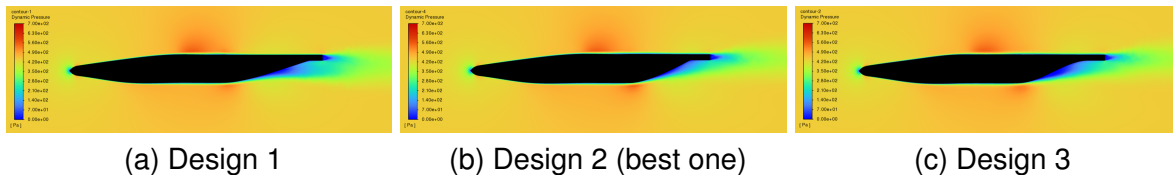


Figure 20: Fuselage ending CFD analysis

## 4.4. Tail

The tail configuration was individuated according to the necessities of the flight mechanics department, without performing any intense CFD investigation. Several studies were conducted through papers, books and manuals, from which it was evident that choosing a T-tail shape, would have brought the main aerodynamic advantages. In fact, in this configuration, the surfaces work in clean air, high above the disrupted flow coming from the propeller slipstream. Moreover, since the horizontal stabilizer acts as an end plate, for the vertical one a drag reduction occurs. Main disadvantages are related to the deep stall condition, and to the flutter instability. However as previously explained in the 2, a XFLR5 analysis has been conducted to make sure that Nyx's operating conditions were enough away from the one leading to these dangerous situations.

## 4.5. Landing Gear

Having the structure and flight mechanics departments opted for a taildragger configuration for the landing gear, the aerodynamic team took care of the careneeing of its constituting elements. Being the tail support very high, to guarantee among all, also sufficient clearance for the fuselage rear section, the tail gear, regardless of its type, should have been of non important dimensions, avoiding back part of the fuselage to scrape on the ground and having a wing stall right before the take off phase. As a consequence driving aspects for the study were its weight and related drag.

First evaluated option was a spring-shaped metal leg. This design choice would have requested also a long and wide fairing to limit the dirty air created by the presence of both the torsional spring and the wheel. The following weight estimation did not match requested targets, thus it was discarded.

Second option was that of a simple skid. Pros of this solutions could be found in the cons of the first one: it offered a narrower frontal section, helping reducing drag, while also being structurally simpler and therefore lighter. In particular, a spoon-shaped design was chosen to encounter the flight mechanics necessities of running the take off distance at specific angle of attack contemporary reducing friction between the grass surface and the skid itself. Therefore, this design option resulted to be the best choice.

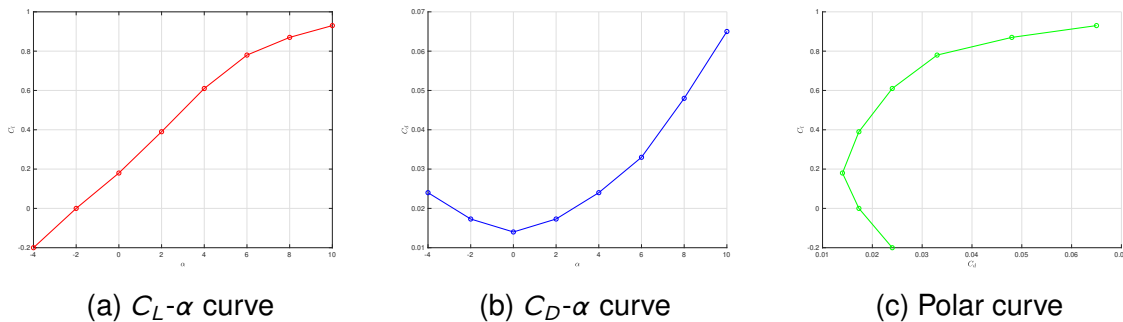
For what it concerns the main landing gear, given its dimensions defined by Structures department, a wheel fairing (*wheel pant*) design study was conducted. With the aid of the CFD

analysis, a comparison between the open wheel and two types of covers was done. The first wheel pant consisted in an airfoil-shaped fairing that incorporated also part of the gear leg. The second one was a slight evolution of the first option, having the inner face of the fairing planar and tangent to the gear leg outer side, leaving the leg itself fully uncovered. Each design had to respect a  $\sim 2\text{mm}$  margin around the wheel tyre and a ground clearance of  $\sim 1\text{cm}$ . To decide whether to use fairings or not, a trade-off between the weight added versus the  $C_D$  improvement (which, from CFD analyses displayed in Figure 61, resulted in a  $-50\%$  in respect to the uncovered wheel) has to be evaluated; for this purpose, various fairing 3D-printing solutions are undergoing a testing phase, and the final decision is expected to be made during the scheduled flight tests.

## 4.6. Polar Curves

The following polar curves were then obtained through CFD simulations involving the whole aircraft at different AoA. A range of simulations were performed from  $-4^\circ$  to  $10^\circ$  with designed wing, considering also an open flaps configuration both in flight and at ground level. The test case used for this simulations is the same of Section 4.3.1, but involving more refinement boxes to cover also the landing gear and the wheels.

No further grid convergence and farfield independence study will be presented being all simulations performed with the same criteria explained in Section 4.3.1 to validate result. To ensure adequate resolution of the analysis in the boundary layer region, the Spalart-Allmaras turbulence model was preferred to the SST  $k-\omega$  dealing the problem with a low Reynolds number. To save computational power and simulation time some gimmicks were used: for the geometry only half of the aircraft was simulated using the symmetry function, whereas for the numerical solver in the beginning of the solution a first order upwind scheme was selected before switching to a second order one. For the solutions methods a coupled scheme was selected. It is able to obtain a robust and efficient single phase implementation for steady-state flows, with superior performance compared to the segregated solution schemes. This pressure-based coupled algorithm offers an alternative to the density-based and pressure-based segregated algorithm with simple-type pressure-velocity coupling.



# 5. Electronics and Propulsion

## 5.1. Flight Controller

The flight controller was determined among the best full-featured ones available on the market for fixed-wing applications. A preliminary study involving all different options was carried out, individuating the Holybro Kakute H743-WING, the Matek H743 WING V3 and the Matek H743 W-LITE as the most suitable options for the department's goals.

A more comprehensive analysis of both the characteristics and the capabilities of these three flight controllers, hereafter illustrated, led to the choice of the **Matek H743 W-LITE**.

As shown in Figure 21, listing what are considered as the most important features, all three

|             | Holybro Kakute H743-WING | Matek H743 WING V3     | Matek H743-WLITE    |
|-------------|--------------------------|------------------------|---------------------|
| MCU         | STM32H743                | STM32H743VIT6          | STM32H743VIH6       |
| IMU         | ICM-42688P (1)           | MPU6000 & ICM42605 (2) | ICM42688 (1)        |
| Baro        | BMP280                   | Infineon DPS310        | DPS310              |
| PDB         | Yes                      | Yes                    | Yes                 |
| BEC outputs | Yes                      | Yes                    | Yes                 |
| PWM outputs | 14                       | 13                     | 13                  |
| UARTS       | 7                        | 7                      | 7                   |
| I2C         | 3                        | 2                      | 2                   |
| USB type C  | Yes                      | Yes                    | Yes                 |
| OSD         | Yes                      | Yes                    | Yes                 |
| CAN         | 1                        | 1                      | 1                   |
| ADC         | 5                        | 6                      | 6                   |
| Dimensions  | 45x 30 x 13.5 mm         | 54 x 36 x 13 mm        | 44 x 29 x 14.5 mm   |
| Weight      | 28g with USB extender    | 30g with USB extender  | 22g w/ USB extender |
| Firmware    | ArduPilot, INAV          | ArduPilot, INAV        | ArduPilot, INAV     |
| Price       | € 94,75                  | € 108,95               | € 105,90            |

Figure 21: Suitable Flight Controllers Comparison

options have similar characteristics and price ranges. In contrast to the other 2, the Matek H743 WING V3 presents 2 Inertial Measurement Units, used simultaneously to obtain a more accurate result if compared to the one of a single IMU. This ensures a performance improvement but it is unessential concerning safety and reliability.

The **Matek H743 W-LITE** stands out for its reduced weight, with just 22g it represents the best option, allowing mass minimization while still ensuring all the top-of-the-range features. Figure 22 shows the wiring diagram of the complete electronic system.

The **ArduPilot** firmware is flashed onto the Matek and is the one used in the competition: it supports LUA scripts, all telemetry protocols around, including CRSF (used by ExpressLRS), and guarantees a smooth and stable flight experience through its fly-by-wire modes.

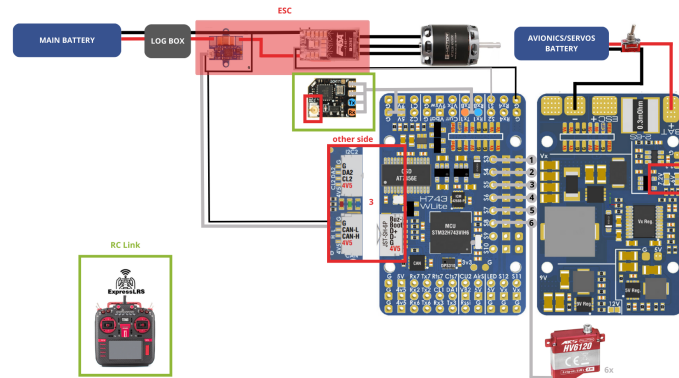


Figure 22: Wiring Diagram of the Complete Electronic System

## 5.2. Radio Control Protocol

**ExpressLRS** is the radio control protocol used to establish the RC link between the pilot's radio-transmitter and the on-board receiver. It is preferred over all the competitors for three main reasons: the outstanding ultra-low latency, the extended long-range capabilities, and the fact that it is an open-source RC system.

The latency is the lowest ever provided by a radio control link and it is available, not trivially, along with the widest range achievable so far, practically eliminating the risk of entering in

fail-safe mode. Being open-source, all ExpressLRS documentation is available online and innovative and increasingly user-friendly features are constantly implemented and released by the community. Finally, all this comes with significant cost-effective characteristics.

The leading control link alongside ELRS in terms of performance, stability and ease of use is TBS Crossfire. It has slightly inferior characteristics in comparison with the competitor and is only available with a frequency band of around 900 MHz.

In Figure 23<sup>1</sup>, the main differences between the ExpressLRS and the TBS Crossfire are presented, along with a display of FrSky ACCST and ACCESS, older and therefore robust protocols, but with less potential. They are enough to guarantee acceptable performance, especially in confined spaces, and present viable options for stand-alone receivers, but the similarity in price with the other alternatives on the market makes them unattractive. Moreover, the hardware is not developed to ensure and support compatibility with third-party products, making them less appealing.

| Radio Protocols   | TBS Crossfire                                           | ExpressLRS                                           | FrSky ACCST                     | FrSky ACCESS                                    |
|-------------------|---------------------------------------------------------|------------------------------------------------------|---------------------------------|-------------------------------------------------|
| Frequency bands   | 868 MHz (EU) / 915 MHz                                  | 900 MHz (915/868) / 2.4 GHz                          | 2.4 GHz                         | 900 MHz / 2.4 GHz                               |
| Latency           | Good latency<br>(Max update rate of 150 Hz)             | Ultra Low Latency<br>(Max update rate up to 1000 Hz) | High latency                    | Good latency                                    |
| Range             | LoRa in 50 Hz mode, switching to FSK at 150Hz 40/50 km* | LoRa modulation in all packet rates 100+ km*         | 1-2 km (open air)               | 2 km (open air)                                 |
| Compatibility     | Wide                                                    | Wide                                                 | Wide                            | Only supported in FrSky radios                  |
| Channels          | Up to 12                                                | Up to 14                                             | Up to 16                        | Up to 24                                        |
| Telemetry         | Extensive telemetry features                            | High amount of features                              | Requires a 2 <sup>nd</sup> UART | Supported                                       |
| Data encryption   | Yes                                                     | No                                                   | No                              | Yes                                             |
| Flashing firmware | Very easy to set up and manage                          | Easy, on everything                                  | Difficult <sup>1</sup>          | Automatic updates (wireless update and binding) |
| Opensource        | No but good community                                   | Yes                                                  | No                              | No                                              |
| Cost              | More Expensive but affordable                           | Affordable                                           | Inexpensive (least expensive)   | Inexpensive (least expensive)                   |

Figure 23: Radio Control Links Comparison

The hardware used to fly with ExpressLRS, at a frequency of 2.4 GHz, is composed of a **RadioMaster TX16S Mark II** and a **RadioMaster ExpressLRS nano receiver**.

### 5.3. ESC

The Electronic Speed Controller is composed of a **T-motor F45A 3-6S 32bit V2** (that runs BLHeli-32 and supports DShot up to 1200), that can hold a continuous current of 45A and a peak of 55A, plus a current sensor (**CAN-L4-BM**) controlling that delivered current is below the threshold.

The choice of having an ESC capable of standing more than 30A in continuous current was aimed at reducing the risk of overheating and overloading. Its mass is only 10g, making it one of the best options on the market from the perspective of mass minimization.

This F45A does not incorporate a BEC and is built for racing multi-rotors, thus for operating conditions characterized by a massive and constant airflow. Dealing with high current delivery, this component is exposed to the risk of overheating: ensuring that it remains within its operating temperature range is necessary to guarantee the mission's success and avoid inconveniences.

In Nyx, this component is placed on the carbon-fiber fuselage through a thermally conductive adhesive paste and a mechanical fastener: the temperature reached in the operating range and the heat flow generated, computed through experiment reported in the following section, suggested that no external cooling system was required. This solution also does not impact on fuselage aerodynamics.

<sup>1</sup> The available data concerns tests carried out by experts in the field

### 5.3.1. Electronic Speed Controller's Heat Flow Evaluation

Following tests are meant to measure the heat flow ( $W$ ) generated by the ESC in operating conditions. The assumption that the heat flow generated is equal to the power loss ( $P_{loss}$ ) of the system is made. Power is computed by measuring the values of voltage and current at a specific location.

$$ESC_{HeatFlow} = P_{loss} = P_{in} - P_{out}$$

Initially, a problem was faced: the available shunt resistor, supposed to be inserted on every cable connecting the motor to the ESC, had big resistance compared to the one of the motor windings, which would have brought certainly to a measurements' distortion.

New resistors were created from scratch, cutting 4 copper wires of the same length. Their resistance was then measured with a micro-Ohmmeter (accuracy to the 0.001 mOhm), and resulted  $R = 0.00522 \text{ Ohm}$ .

Through the voltage and resistance values in the various sections, it is possible to retrieve the current measurements by applying Ohm's first law  $V = R \cdot I$ . As shown in Figure 24, the input current is evaluated by acquiring the voltage across the shunt wire number 4 (soldered in series with the ground wire between the battery and the ESC) that is equal to the difference in voltage between the positive and negative pole of the battery. The same approach is used to retrieve the current in shunt wire number 1, 2 and 3, by measuring the voltage difference between the points preceding the shunts and the battery ground.

The differential measures were acquired through the DewesoftX software, by connecting all the cables into 8 differential inputs of an A/D converter.

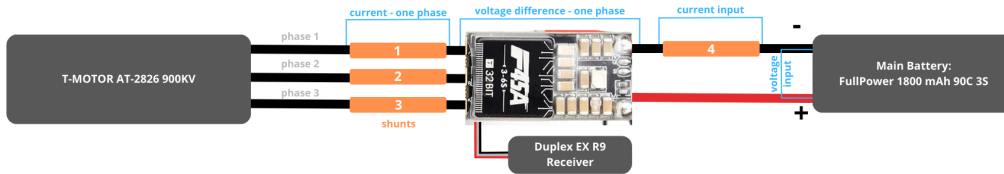


Figure 24: Wiring Diagram - Heat Flow Testing

The test is carried out for 180s in which the motor is powered at a voltage of 12.5V, this value decreases during the test due to battery discharge, and a current of 30A. The temperature reached by the ESC was also measured with a thermocouple.

The power entering and exiting the component, Figure 27, was computed through measured values, plotted over time and averaged on a window that starts 60s after the beginning of the experiment, to eliminate effects of initial transient and consider a fully operational condition.

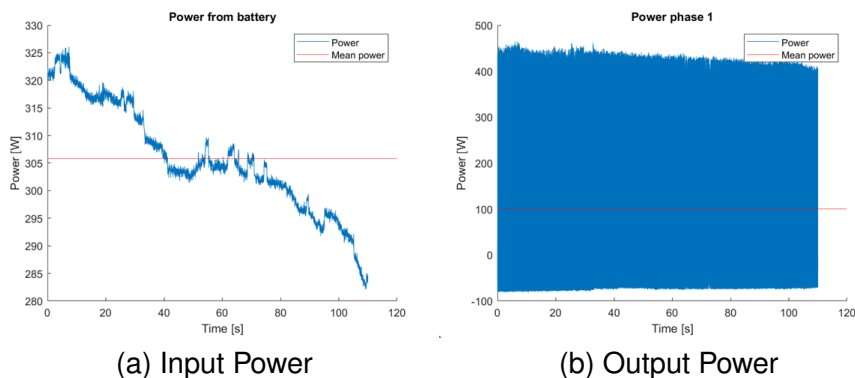


Figure 25: Power entering the ESC and exiting (one phase)

The difference between the entering power ( $P_{in} = 305.83 \text{ W}$ ), and the exiting one ( $P_{out} = 296.028 \text{ W}$  - considering all the three phases) is of 9.8 W approximately, data were smoothed to eliminate any spurious points present in the measurements.

The temperature reached in the operating condition with a current of 30 A is 75°C and is inside the operating range of commercial ESCs, estimated to be around 100°C, and far below the thermal shutdown protection value of 130°C suggested by the manufacturer.

#### 5.4. Acquisition System for the Wind Tunnel Testing

The electronic equipment consisted of the T-motor AT-2826 Long Shaft 900KV imposed by the competition, a T-motor AT55A as an ESC, deliberately oversized for this test campaign, to protect the electronics and avoid possible overheating, a Duplex EX R9 receiver, directly connected to the ESC, and a Jeti Duplex DS-12 Red as radio-transmitter, used to manually keep the current constant through feedback visualized on the on-screen telemetry, savable as a .log file to be later analysed.

A preliminary and rudimentary test stand was built to ensure the correct functioning of all components and validate the acquisition system, shown in Figure 26.

This latter, developed from scratch, is composed by a **UniSens-E**, which is a telemetry sensor, and a **GPS logger 3**.

The UniSens-E is placed between the battery and ESC and presents a cable connected to one of the three motor phases to acquire the rpm. It also measures voltage, current, power, capacity, altitude and vertical speed and is not able to generate a data file on its own. The GPS logger 3, when integrated with the UniSens-E, can generate data files, not only of the directly acquired quantities but also those caught by the UniSens-E.

To save these information on the SD card, the GPS must succeed in triangulating its position, which is not the case when using the equipment indoors. However, via reception of the telemetry system of the radio used, the values can be displayed and recorded. The power

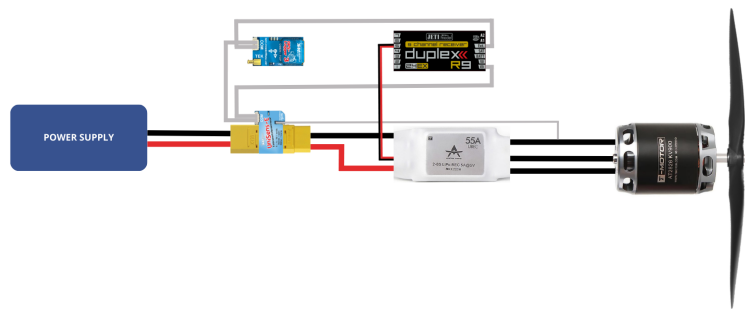


Figure 26: Acquisition System for Wind Tunnel Testing

supply was set to provide the engine with 321W (10.7V, 30A). The voltage value applied was defined through battery testing. Each acquisition was taken varying the wind tunnel speed from 0 to 30 m/s with 2.5 m/s steps.

#### 5.5. Static Tests Stand and Acquisition System

A functional engine test bench is designed and built to gather important results for the preliminary design phases: in particular, to estimate the maximum Take Off thrust and the electronic system efficiency and to complete the measurements of the wind tunnel testing of the propellant system, by better estimating the thrust at null velocity in an open environment. The development and realisation of the test bench had to be as economically feasible as

possible and built in a short period of time. These constraints dictated the project's evolution by utilizing a simple arrangement and cost-effective materials and components, still maximizing the precision and validity of the measurements.

Finally, the apparatus is conceived to be reusable with various kinds of similar engines, in either pusher or puller configuration, by simply swapping one 3D printed PLA piece.

The stand is composed of a framework in pine wood and 3D printed parts in PLA, designed using hand calculations and FEM analysis to be resistant to a various range of loads.

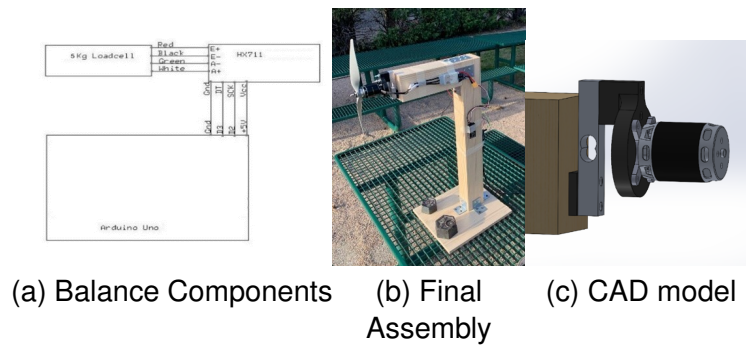


Figure 27: Static Test Stand

The acquisition system is composed by the electronic equipment previously explained in Figure 26 and powered by a FullPower Platinum 1800mAh 90-180C battery. In addition, a 5 kg load cell, with an operating voltage of 5-10V, is used, alongside with an Arduino UNO board and a HX711 module with A/D converter, with an operating voltage of 2.6 - 5.5V. The output data rate of the latter was set to 10Hz, and the resolution is  $\pm 0.1g$ . The Arduino UNO board is paired to a tailor-made program on the PC. The wiring diagram of the pairing of the balance components is shown in Figure 27.

The Arduino sketch loaded into the board allows the user to set the duration of the test, to carry out the initial calibration of the balance and to start a new test. The freeware serial port terminal application and software Coolterm is used to view the data and instructions to carry out the tests. In contrast to Arduino's serial monitor, this application enables the user to save the acquired data into a .txt file.

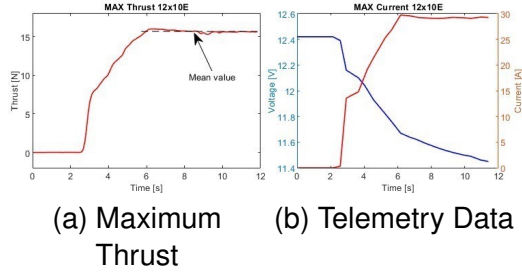
### 5.5.1. Maximum Take Off Thrust Evaluation

This first test is performed with the battery completely charged. The test methodology is the following:

1. Calibration of the balance and mounting of the engine-propeller assembly.
2. Starting of the 20s acquisition from the PC terminal and the radio-transmitter.
3. Gradual increase of the current input until 30A in a range of 3s.
4. Mean thrust values acquisition, together with the telemetry parameters for 5s once the current reaches 30A.

The data from the load cell and the telemetry were synchronized to the same time frame using MATLAB and the mean maximum thrust was computed. As an example, the results of the test performed with the APC 12x10E propeller is shown in Figure 28.

The results are shown in Table 9. The Thrust to Power ratio was additionally calculated to obtain an efficiency parameter of each electric system-propeller assembly.



| Propeller  | Mean MAX Thrust [N] | Thrust to Power ratio [ $\frac{N}{W}$ ] |
|------------|---------------------|-----------------------------------------|
| APC 12x8E  | 18.24               | 0.05171                                 |
| APC 12x10E | 15.69               | 0.04642                                 |
| APC 13x8E  | 19.39               | 0.05447                                 |

Table 9: Maximum Take Off Thrust Evaluation

Figure 28: Maximum Take Off Thrust Evaluation

### 5.5.2. Thrust at Null Velocity

The second test aimed to complete the characterization of the propulsive system by adding a point (in static and open environment conditions) to the curves found from the wind tunnel testing phase, being null airflow velocity a condition that could not be simulated.

The test replicated the same wind tunnel setup, especially the power of 321 W: the current is kept stable at 30A and the battery is discharged, the average voltage is 10.7V. In this case, the test methodology previously reported differs for the 50s of acquisition time, used to discharge the battery, from which a 10s period that met constraints was extracted.

### 5.5.3. Electronics Efficiency

To conclude, the efficiency of the electronics is computed as the ratio between the overall propulsive system and the propeller efficiencies:

$$\mu_{el} = \frac{\mu_{overall}}{\mu_{propeller}} \quad \mu_{overall} = \frac{T \cdot V}{P_{electric}}$$

The brushless engine's efficiency varies with several parameters as, for example, loading conditions (constant in this case) and angular velocity. For this reason, multiple flight conditions were investigated, exploiting the data gathered during WTT and static testing. The main focus was on cruising conditions, which showed a  $\mu_{el} = 80.29\%$ .

## 5.6. Battery Testing

The battery testing campaign aimed at studying both the battery discharge and the voltage sag phenomenon. The gained information are used to optimally power the propulsive system and to individuate the correct value of the voltage to be imposed to the power supply, used in the wind tunnel testing campaign to simulate the system performances as accurately as possible.

### 5.6.1. Preliminary Tests on Complete Discharge

In the preliminary testing phase, different batteries were completely discharged at 30A, without going beyond the safety threshold, to avoid permanent damage to the cells. The acquisition system described in Figure 26 was first validated with these experiments.

Figure 29(a) shows the graphs Voltage (V) - Capacity (mAh) of the FullPower Platinum 1800 mAh 90C 3S battery. Considering the voltage against the battery capacity instead of time frees from the test methodology.

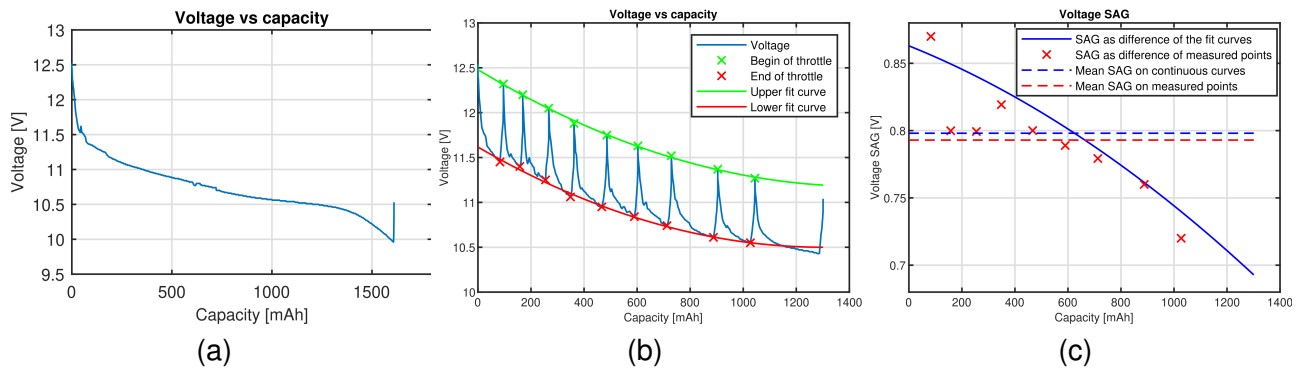


Figure 29: Battery and Voltage Sag Testing Results

The voltage decrement is related to the natural discharge of the battery over time and to a temporary reduction in the output which appears when a sudden increase in load or demand occurs. This phenomenon is known as voltage or battery sag (peukert effect in literature), and is dominant in rechargeable batteries.

The most suitable battery is clearly the one with the highest power density (the integral of the voltage discharge curve over the flight time normalized on the mass of the battery), which is strictly correlated to the lowest voltage sag. This represents the main parameter that will be used to compare batteries (single and in parallel configuration) in the next battery testing campaigns. The parallel configuration of more than one 3S battery will be further investigated since augmenting their number, the amount of current drawn by every battery is decreased. Since voltage sag is strictly correlated to current drawn, the previous configuration is supposed to be better in terms of power density.

Through the time derivative of the voltage, it is possible to understand when the effect of voltage sag stabilizes. This knowledge is used to develop an accurate test methodology.

### 5.6.2. Voltage Sag Testing

The Voltage Sag Testing is carried out in the following way:

1. The test starts with a battery charged to 12.45 V.
2. Throttle at 30A is imposed for 25s, and then it is cut to zero for 60s.
3. The cycles are repeated 12 times or until the battery reaches the limit of 10.2V.
4. Before unplugging the batteries, the system is left to rest for 120s.

The voltage sag–capacity graph, in Figure 29(b), is created and two best fit curves are created over peaks and troughs. The difference between them generates a curve (Figure 29(c)) from which to compute the average voltage sag for a FullPower 1800 mAh 90C 3S battery, which is equal to 0.7981V.

Finally, the value to be imposed through the power supply for the WTT campaign, considering a current of 30A, is chosen as 10.7V.

## 5.7. Servos

To meet the need for lightweight and compact servos with adequate speed and torque, MKS provided a set of **HV6120**. They are mini digital servo, optimally designed for this type of application, with an operating voltage that ranges from 3.7V to 8.2V, mass of 11g, and a torque and speed up to 5.4 kg-cm and 0.08s respectively. Moreover, the MKS HV6150, whether they are considered more suitable in the testing phases, can be used to support and control the flaps. Given the estimated current drawn of these servo motors, the FullPower 850 mAh 35C 2S is chosen as auxiliary battery. The first flight tests will reveal if it is possible to further reduce its capacity.

# Appendix

## References

- [1] 3D-printz. How to get strong lw-pla wings, 2023.
- [2] *Abaqus 6.11, Getting started with Abaqus: interactive edition.*
- [3] Michael F. Ashby. *Materials selection in mechanical design.* Butterworth-Heinemann, 2016.
- [4] Joshua Bardwell. Crossfire vs. tracer vs. ghost vs. expresslrs vs. r9 vs. access vs. accst — rc control link shootout, 2021.
- [5] K-J Bathe. *Finite Element Procedures.* Pearson Education.
- [6] bodge (MIT license). An arduino library to interface the avia semiconductor hx711 24-bit analog-to-digital converter (adc) for weight scales., 2021.
- [7] *Composite Materials Handbook Volume 3: Materials Usage Design and Analysis.*
- [8] ExpressLRS. Long range competition, 2024.
- [9] InternetArchive. Arduino model rocket motor test stand, 2015.
- [10] KremerFPV. Expresslrs vs. crossfire - 25mw 3km range test, 2021.
- [11] Oscar Liang. Why expresslrs is the best radio link in 2024: 2.4ghz vs. 900mhz, crossfire is dead?, 2021.
- [12] T.H.G. Megson. *Aircraft Structures for Engineering Students.* Elsevier, 2007.
- [13] Sala G.; Di Landro L.; Airoidi A.; Bettini P. Dispense del corso di tecnologie e materiali aerospaziali.
- [14] Painless360. Choosing between betafly. inav and ardupilot: A guide for new builders, 2023.
- [15] John P. Purser, Paul E.; Campbell. Experimental verification of a simplified vee-tail theory and analysis of available data on complete models with vee tails. Technical report, NASA, 1945.
- [16] Daniel P. Raymer. *Aircraft Design: A Conceptual Approach.* AIAA Education Series, 2018.
- [17] Jan Roskam. *Aircraft Design.* DAR Corporation, 2018.
- [18] Chris Rosser. 5 inch fpv battery testing: Round 1, 2023.
- [19] Mohamed H. Sadraey. *Aircraft Design - A Systems Engineering Approach.* Wiley, 2013.
- [20] Cristina Sanchez-Carmona, Alejandro; Cuerno-Rejado. Vee-tail conceptual design criteria for commercial transport aeroplanes. *Chinese Journal of Aeronautics*, 32(3):595–610, 2019.

- [21] Cristina; García-Hernández Luis Sanchez-Carmona, Alejandro; Cuerno. Unconventional tail configurations for transport aircraft. pages 127–148, 01 2017.
- [22] sebastien chedal bornu. pla vs lwpla vs polylyte foamy pla, 2021.
- [23] Jim Smith. T-time? why britain fell in love with the t-tailed aeroplane, 2020.
- [24] Andrea Spinelli. A study of unconventional wings and tails, AY 2019-2020.
- [25] Tim Station. How i designed a 3d printed wing, 2022.
- [26] Helmut Stettmaier. V-tails for aeromodels, 2001.
- [27] Singh Sumit. What are the advantages and disadvantages of t-tails?, 2022.
- [28] S.P. Timoshenko and J.M. Gere. *Theory of Elastic Stability*. Dover Publications, 2016.

## A. Acronyms, Abbreviations and Nomenclature

| Acronym               | Meaning                                                  |
|-----------------------|----------------------------------------------------------|
| AR                    | Aspect Ratio                                             |
| $b_W$                 | Wingspan                                                 |
| $c$                   | Chord                                                    |
| $C_{D,0}$             | Minimum Drag Coefficient                                 |
| $C_L^{\max}$          | Maximum Lift Coefficient                                 |
| $C_L^{\text{cruise}}$ | Cruise Lift Coefficient                                  |
| CFD                   | Computational Fluid Dynamics                             |
| CG                    | Center of Gravity                                        |
| DAER                  | Department of Aerospace Science and Technology           |
| FE                    | Finite Element                                           |
| FEA                   | Finite Element Analysis/Analyses                         |
| FEM                   | Finite Element Method/Model                              |
| $M_{\text{MTO}}$      | Maximum Take-Off Mass                                    |
| $M_{\text{OE}}$       | Empty Operative Mass                                     |
| MAC                   | Mean Aerodynamic Chord                                   |
| $\text{MAC}_W$        | Wing's Mean Aerodynamic Chord                            |
| MDF                   | Medium-density fibreboard                                |
| MGC                   | Mean Geometric Chord                                     |
| S                     | Wing Reference Surface                                   |
| $S_H$                 | Horizontal Tail Reference Surface                        |
| $S_V$                 | Vertical Tail Reference Surface                          |
| TO                    | Take-Off                                                 |
| TR                    | Taper Ratio                                              |
| TVC                   | Tail Volume Coefficient                                  |
| $\text{TVC}_H$        | Horizontal Tail Volume Coefficient                       |
| $\text{TVC}_V$        | Vertical Tail Volume Coefficient                         |
| $W_{\text{MTO}}$      | Maximum Take-Off Weight                                  |
| WTT                   | Wind tunnel Testing                                      |
| $X_{\text{AC}}$       | Aerodynamic Center Longitudinal Position                 |
| $X_{\text{AC}_H}$     | Horizontal Tail Aerodynamic Center Longitudinal Position |
| $X_{\text{AC}_V}$     | Vertical Tail Aerodynamic Center Longitudinal Position   |
| $X_{\text{CG}}$       | Center of Gravity Longitudinal Position                  |
| $y$                   | Spanwise Wing Coordinate                                 |
| $\Lambda_{C/4}$       | Quarter Chord Sweep Angle                                |
| $\Gamma$              | Dihedral Angle                                           |
| $\varphi$             | Twist Angle                                              |

## B. Flight Mechanics Data

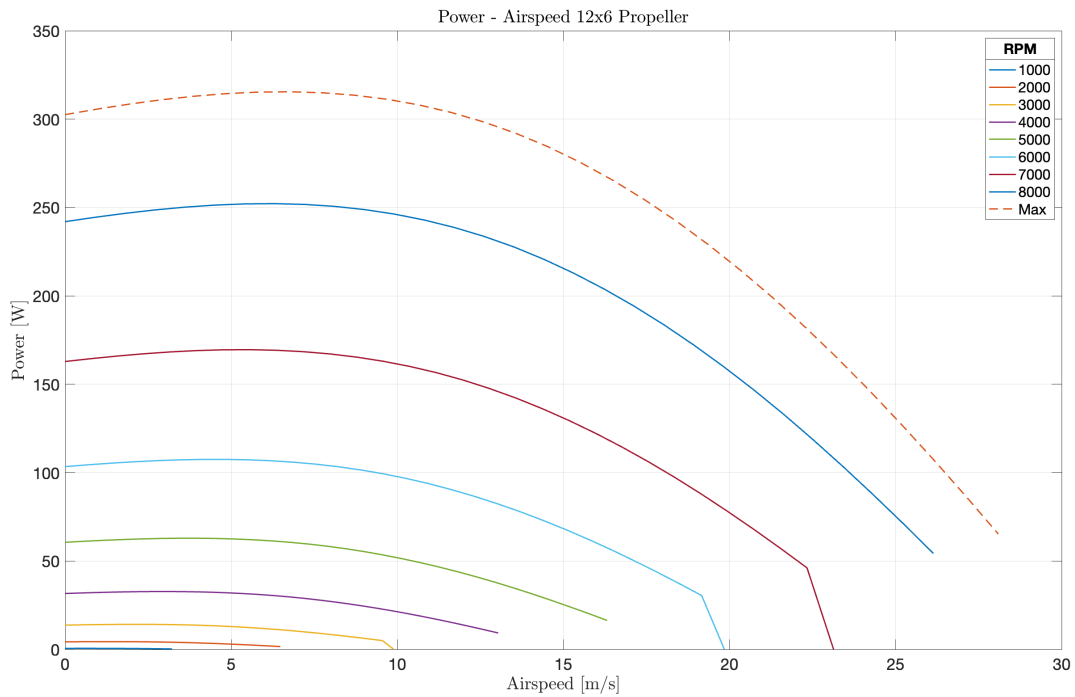


Figure 30: Power vs Airspeed for 12x6 Propeller from APC datasheet at different RPMs, with the maximum value of RPMs defined by the 30A limit imposed by the competition

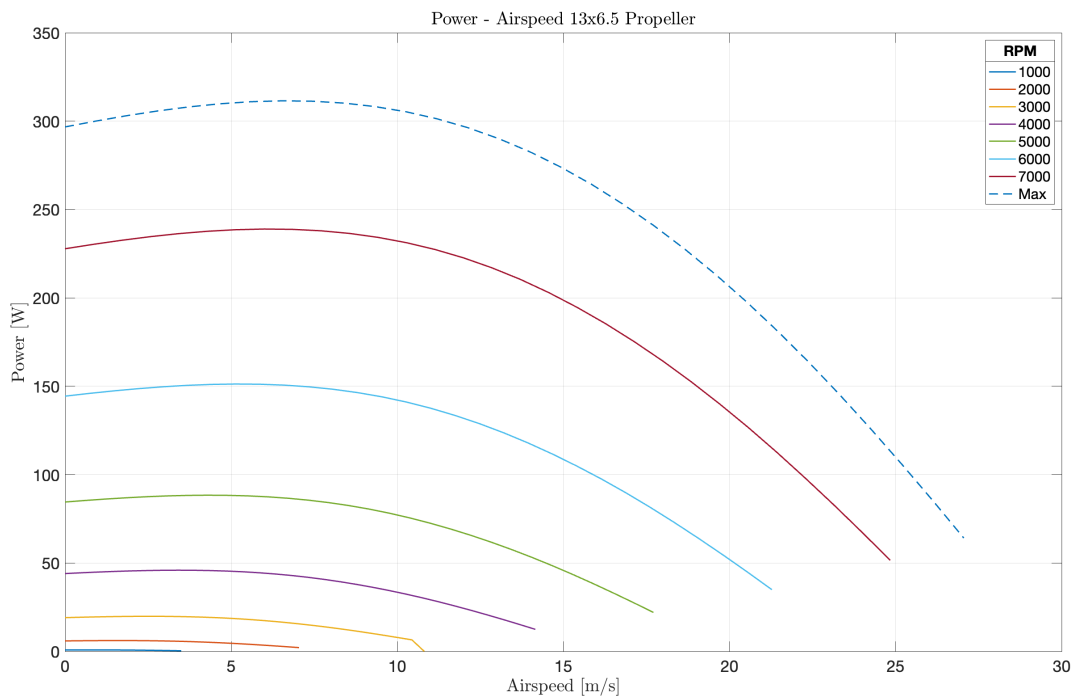


Figure 31: Power vs Airspeed for 13x6.5 Propeller from APC datasheet, at different RPMs, with the maximum value of RPMs defined by the 30A limit imposed by the competition

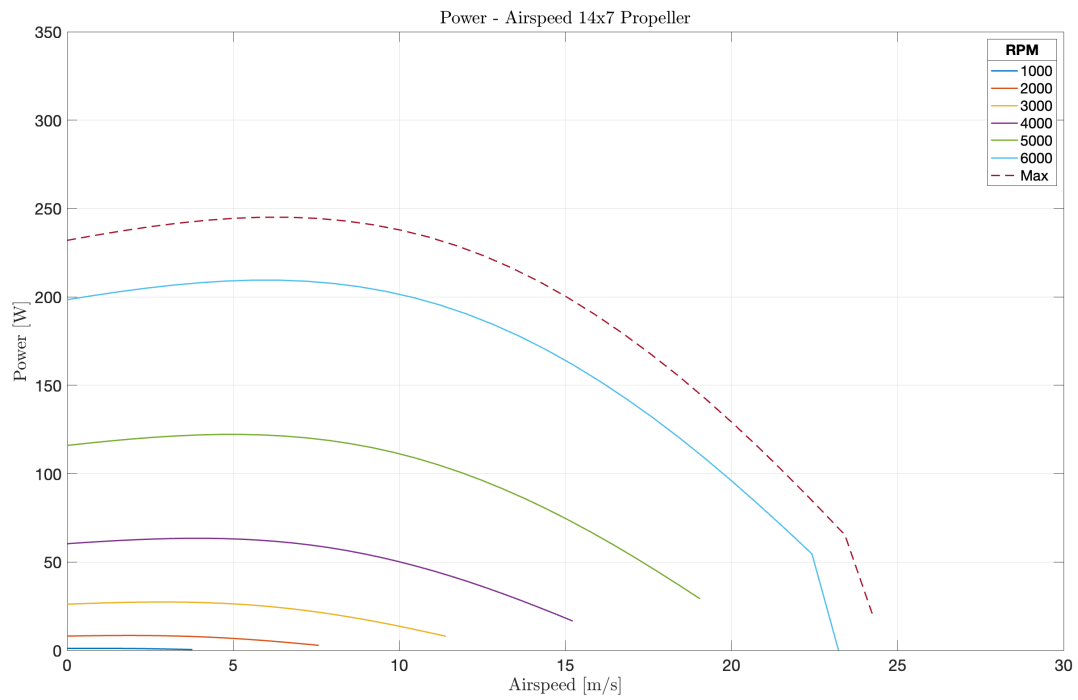


Figure 32: Power vs Airspeed for 14x7 Propeller from APC datasheet, at different RPMs, with the maximum value of RPMs defined by the 30A limit imposed by the competition

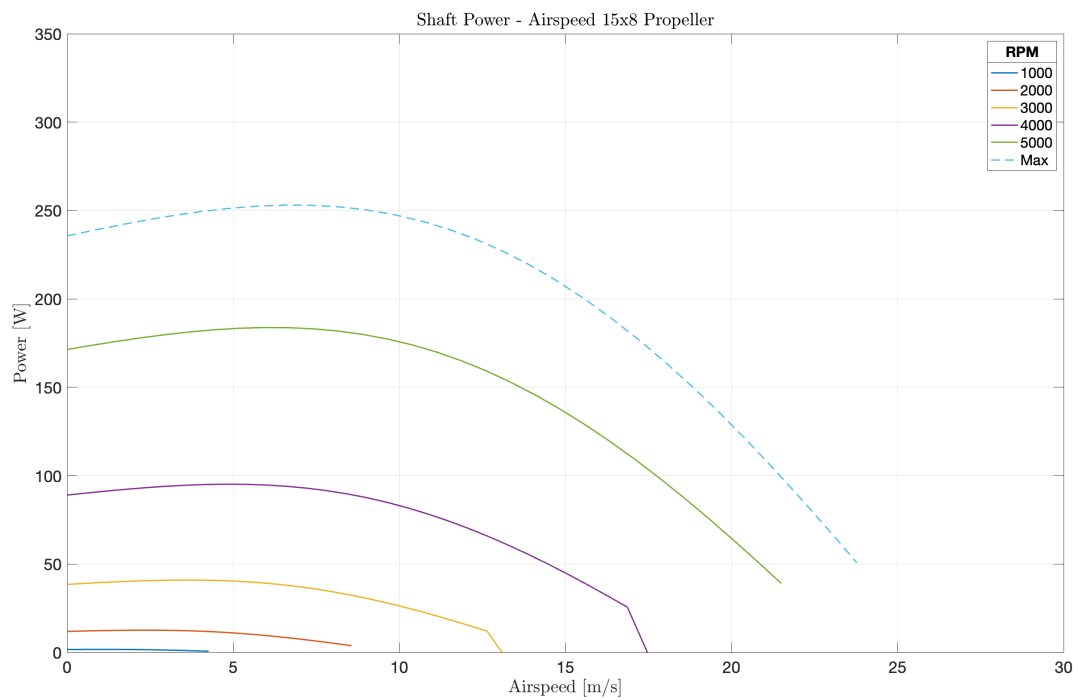


Figure 33: Power vs Airspeed for 15x8 Propeller from APC datasheet, at different RPMs, with the maximum value of RPMs defined by the 30A limit imposed by the competition

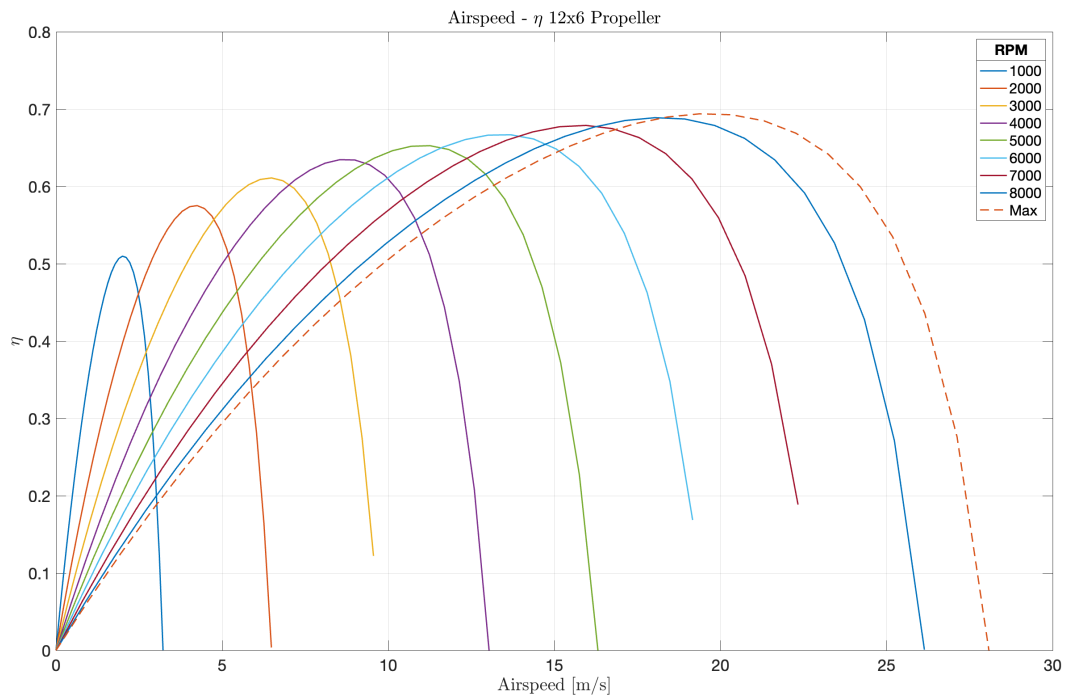


Figure 34:  $\eta$  vs Airspeed for 12x6 Propeller from APC datasheet, at different RPMs, with the maximum value of RPMs defined by the 30A limit imposed by the competition

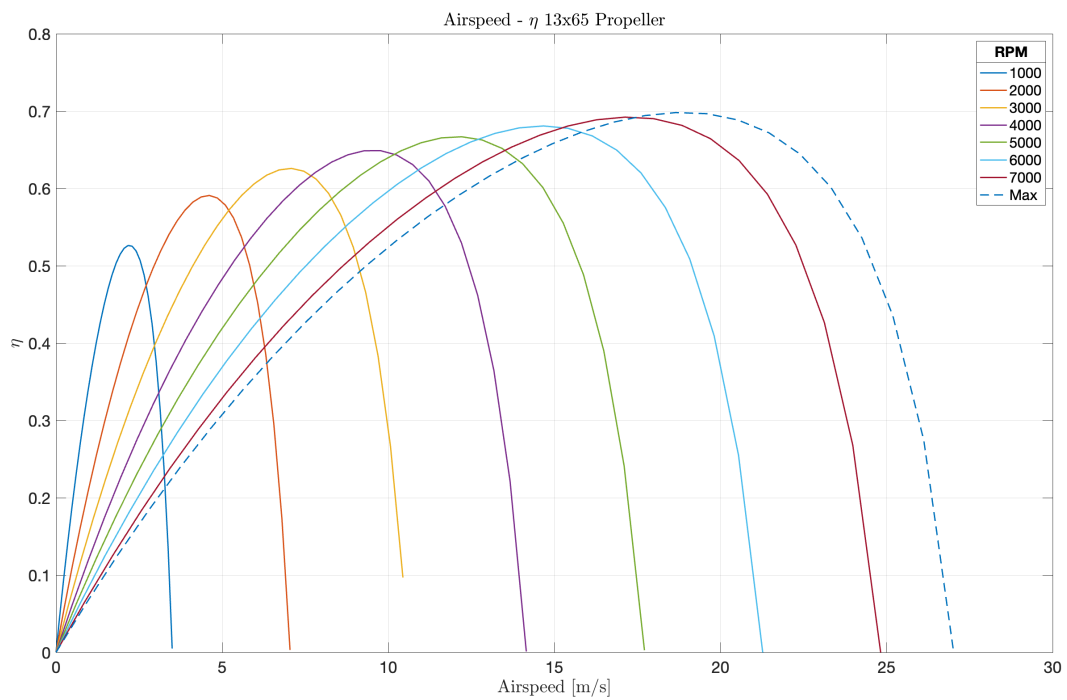


Figure 35:  $\eta$  vs Airspeed for 13x65 Propeller from APC datasheet, at different RPMs, with the maximum value of RPMs defined by the 30A limit imposed by the competition

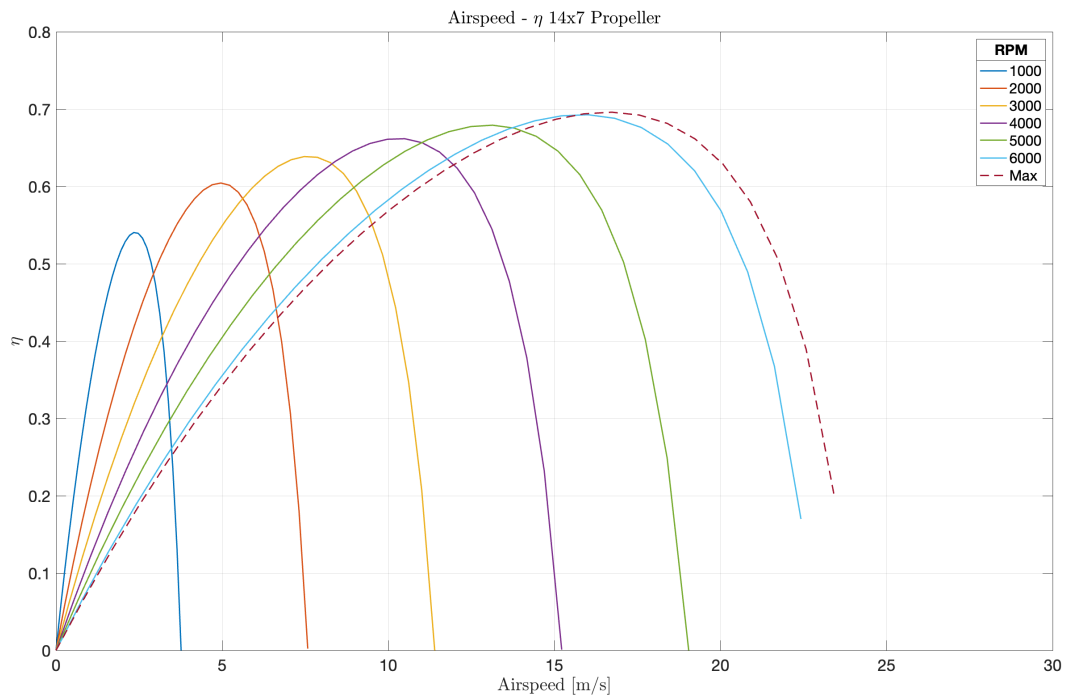


Figure 36:  $\eta$  vs Airspeed for 14x7 Propeller from APC datasheet, at different RPMs, with the maximum value of RPMs defined by the 30A limit imposed by the competition

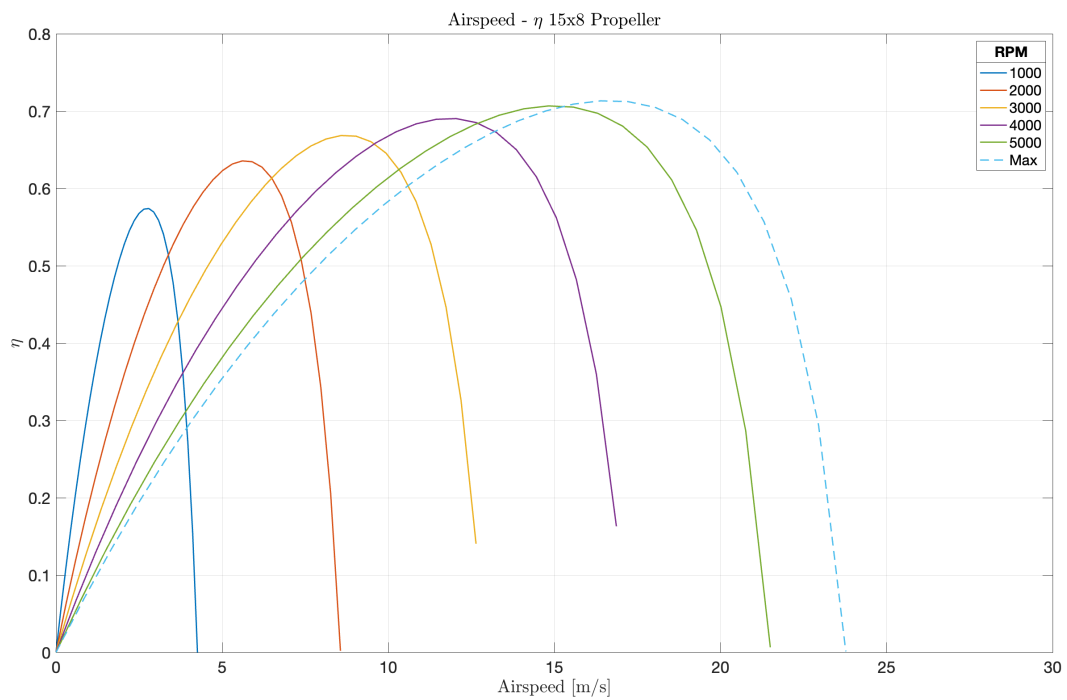


Figure 37:  $\eta$  vs Airspeed for 15x8 Propeller from APC datasheet, at different RPMs, with the maximum value of RPMs defined by the 30A limit imposed by the competition

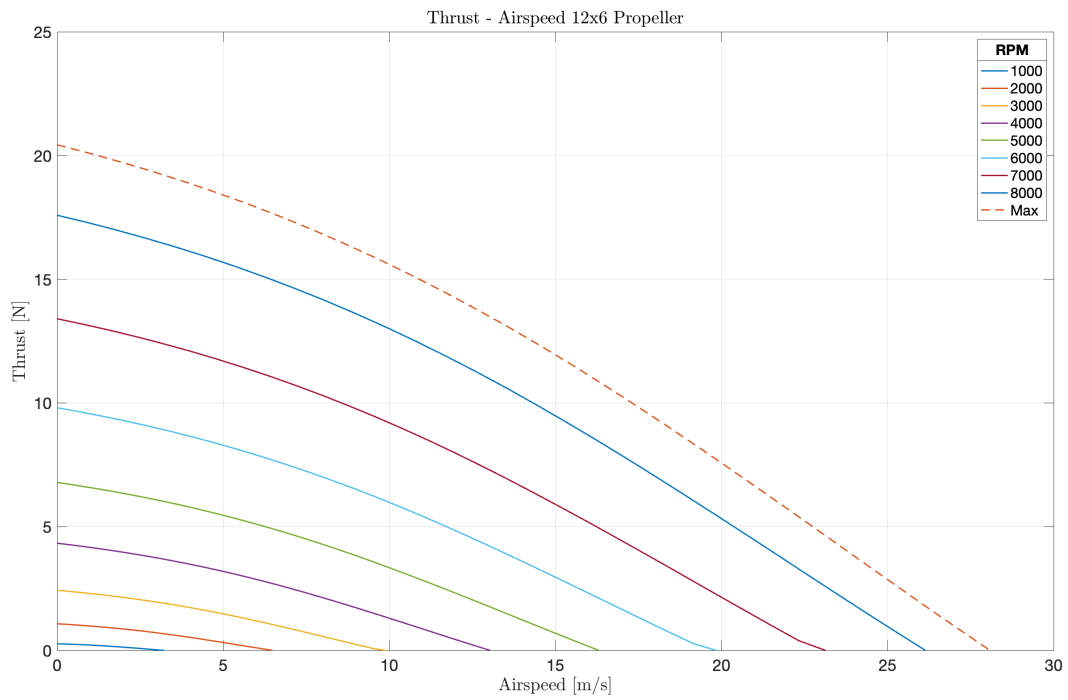


Figure 38: Thrust vs Airspeed for 12x6 Propeller from APC datasheet, at different RPMs, with the maximum value of RPMs defined by the 30A limit imposed by the competition

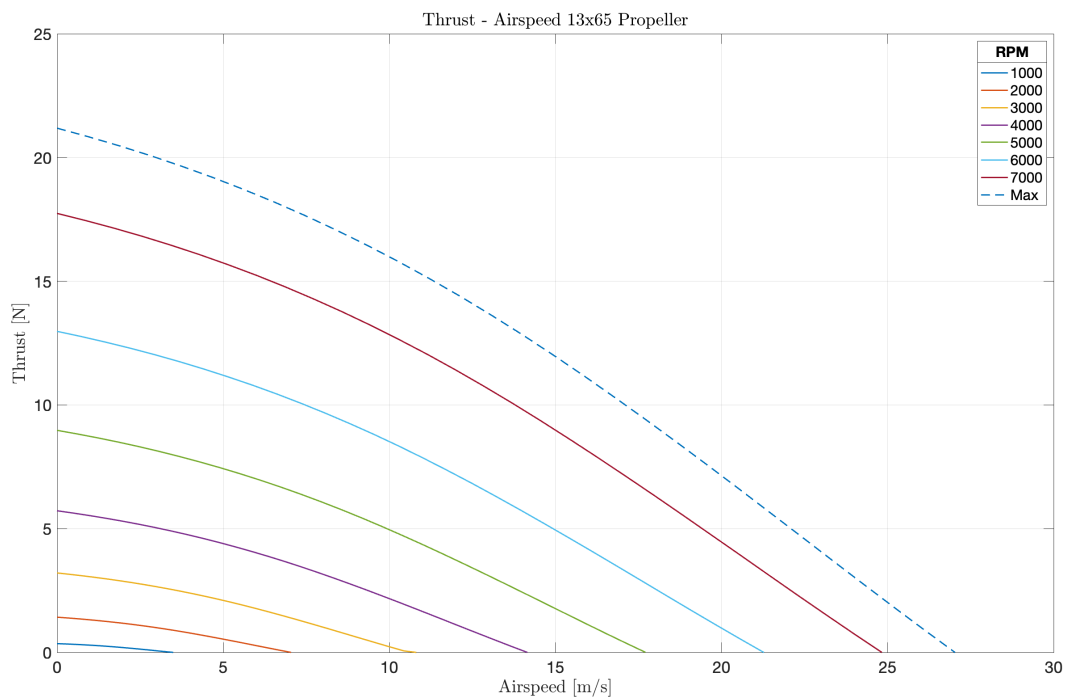


Figure 39: Thrust vs Airspeed for 13x65 Propeller from APC datasheet, at different RPMs, with the maximum value of RPMs defined by the 30A limit imposed by the competition

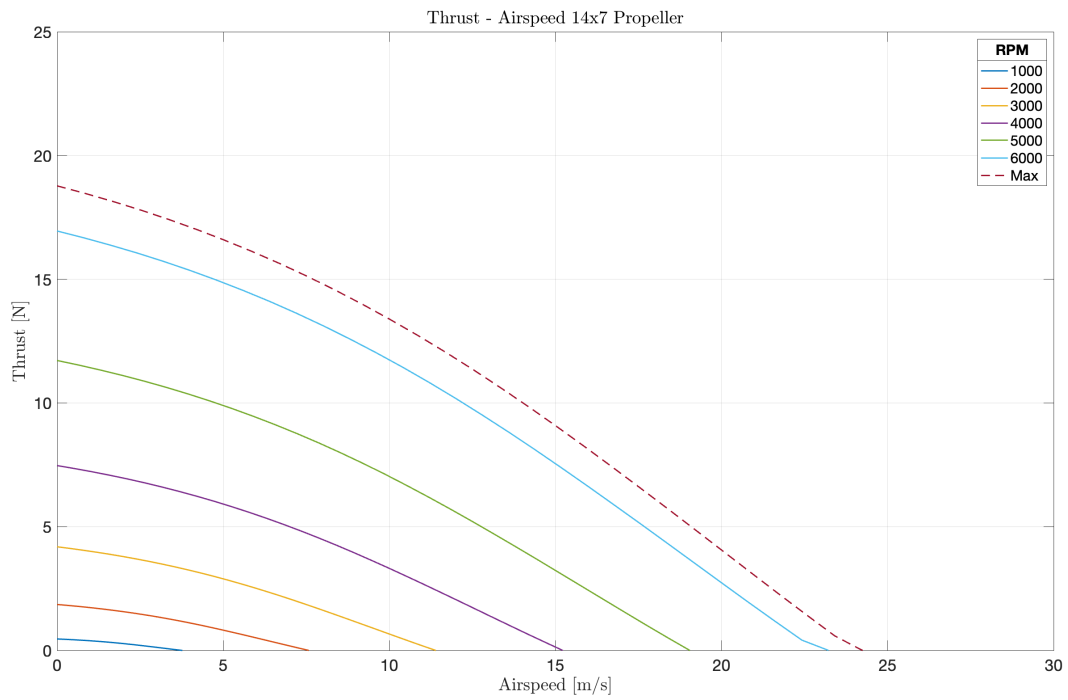


Figure 40: Thrust vs Airspeed for 14x7 Propeller from APC datasheet, at different RPMs, with the maximum value of RPMs defined by the 30A limit imposed by the competition

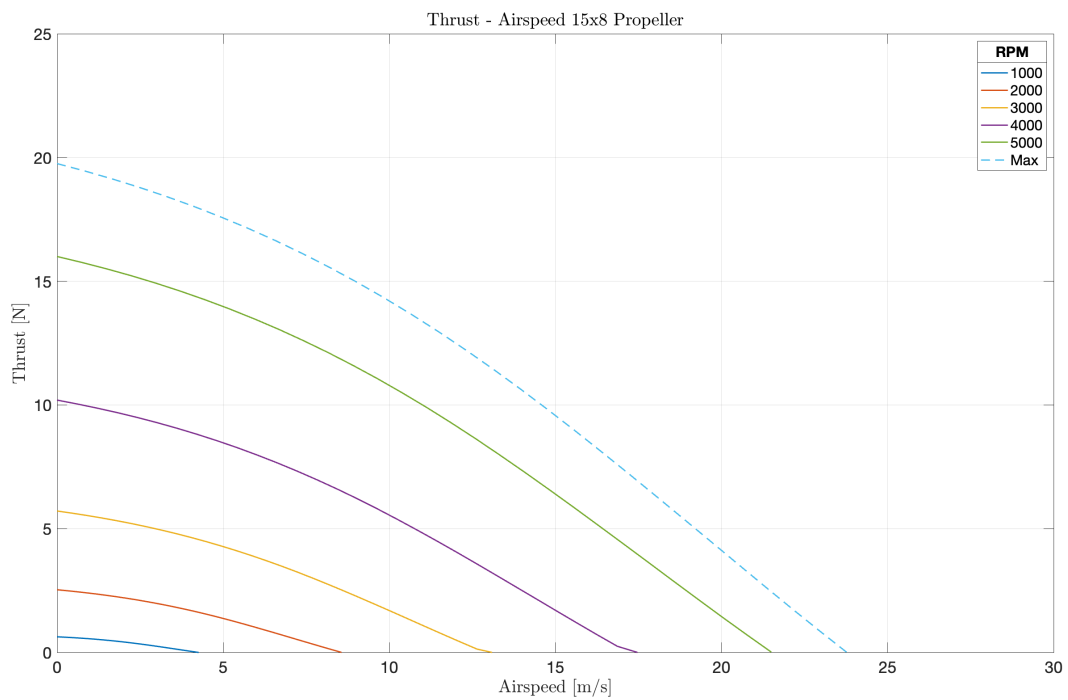


Figure 41: Thrust vs Airspeed for 15x8 Propeller from APC datasheet, at different RPMs, with the maximum value of RPMs defined by the 30A limit imposed by the competition

| <i>Configuration</i> | <i>Advantages</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <i>Disadvantages</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Traditional Tail     | <ul style="list-style-type: none"> <li>Standard design</li> <li>Simplest to design and validate</li> <li>Previous experience from Aethra</li> <li>Possibility of placing servos in fuselage using cables to move control surfaces (hence helping with weight adjustments for stability)</li> </ul>                                                                                                                                                                          | <ul style="list-style-type: none"> <li>With a High Wing configuration, the tail would lie in the wing's wake, causing the control surfaces to be ineffective, unacceptable condition</li> <li>Higher drag due to fuselage connections</li> </ul>                                                                                                                                                                                                                                                                                                                                                              |
| V-Tail               | <ul style="list-style-type: none"> <li>Marginal advantage in drag against Traditional Tail (quantitative analysis in Section ??)</li> <li>Weight reduction against both Traditional and T-Tail, however not as big as simply 2/3 of the weight (considering 2 surfaces against 3)</li> <li>Good clearance off the ground for grass landings</li> </ul>                                                                                                                      | <ul style="list-style-type: none"> <li>Mixed controls would nullify drag advantage when trimming with ruddervators in-flight</li> <li>Stability problems when the aircraft is slipping, not particularly suited for large gliders with high AR and with highly cambered airfoils for the main wing</li> <li>Servos inside the tail will be difficult to reach with cables</li> <li>No experience from our team with V-Tails, leading to difficult validation and problem-solving</li> </ul>                                                                                                                   |
| T-Tail               | <ul style="list-style-type: none"> <li>Horizontal tail lying above the main wing's wake, elevator keeps its effectiveness, which leads to better short field TO performance</li> <li>Swept Vertical tail enables smaller Horizontal tail as it increases its horizontal moment arm</li> <li>Drag reduction thanks to "End-Plate Effect" of the Horizontal tail on the Vertical Tail</li> <li>Similarity to Traditional Tail helps with understanding of problems</li> </ul> | <ul style="list-style-type: none"> <li>Heaviest configuration, Vertical tail needs to be reinforced to hold the weight of the Horizontal tail</li> <li>Structural vibration problems during landings, main fuselage structure (tube and front) has to be strong enough to sustain them, it is linked to aeroelastic problems that can be evaluated only with the aircraft completely designed</li> <li>Servos inside the tails will be difficult to reach with cables</li> <li>Deep stall problems, our flight condition, however, should not be anywhere near angles of attack causing deep stall</li> </ul> |

Table 10: Tail Configurations Comparison

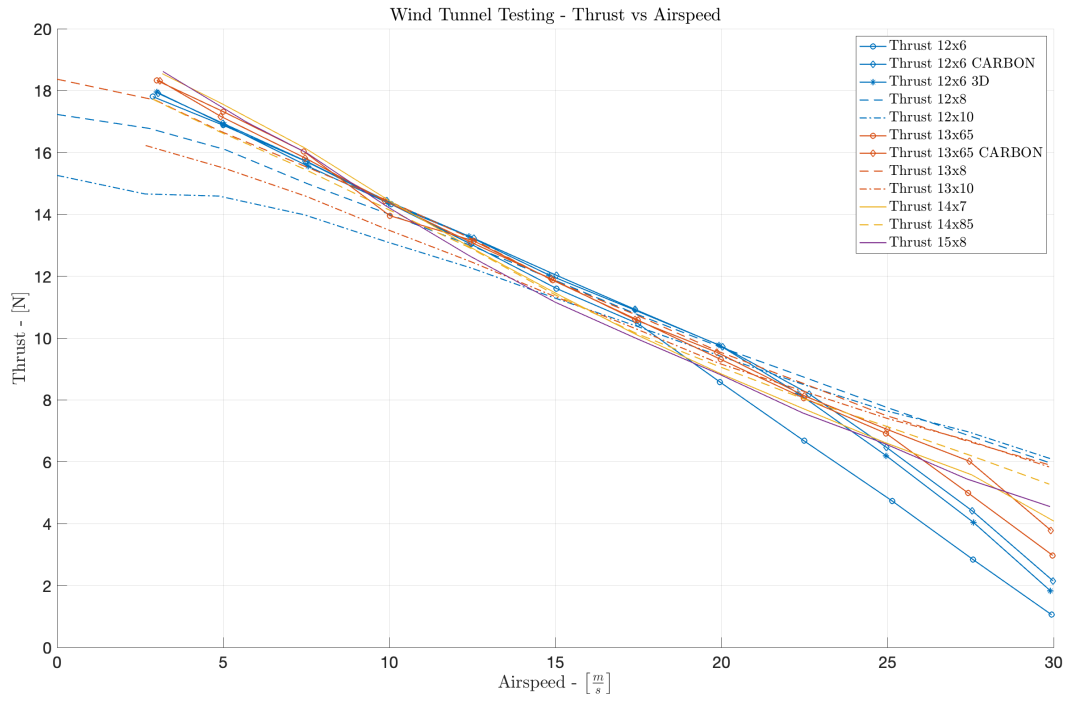


Figure 42: Thrust vs Airspeed in Wind Tunnel for each propeller

Values at 0  $\frac{m}{s}$  in Figure 42 were obtained through static testing, explained in Section 5.5

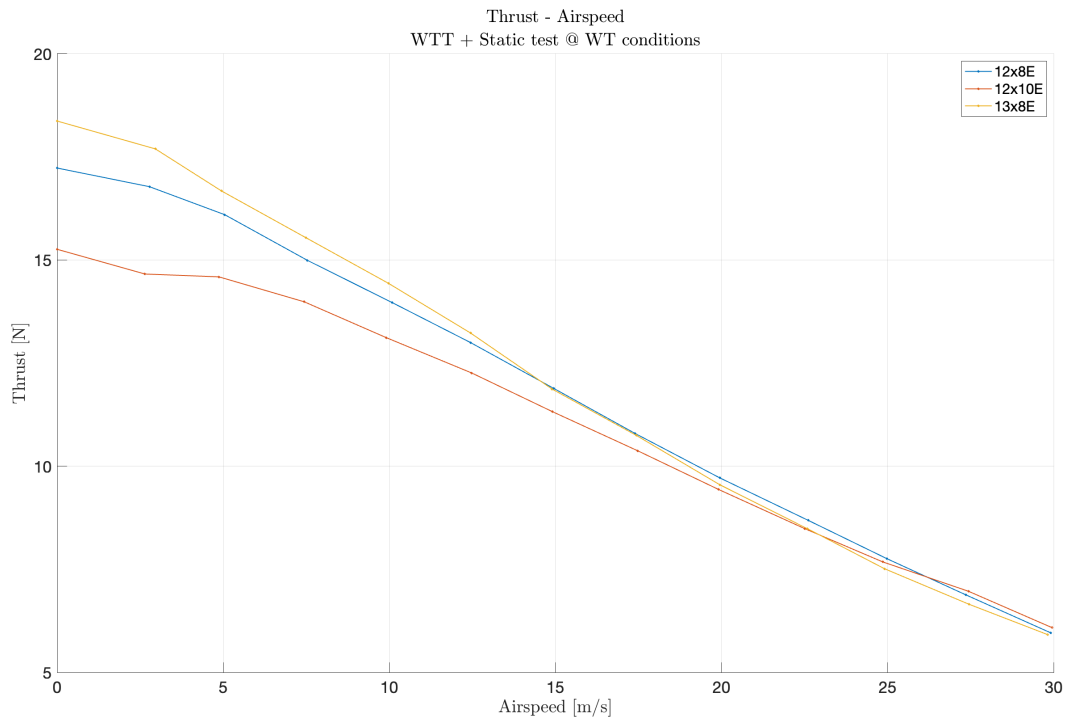


Figure 43: Detail of Thrust of 3 best propellers from WTT

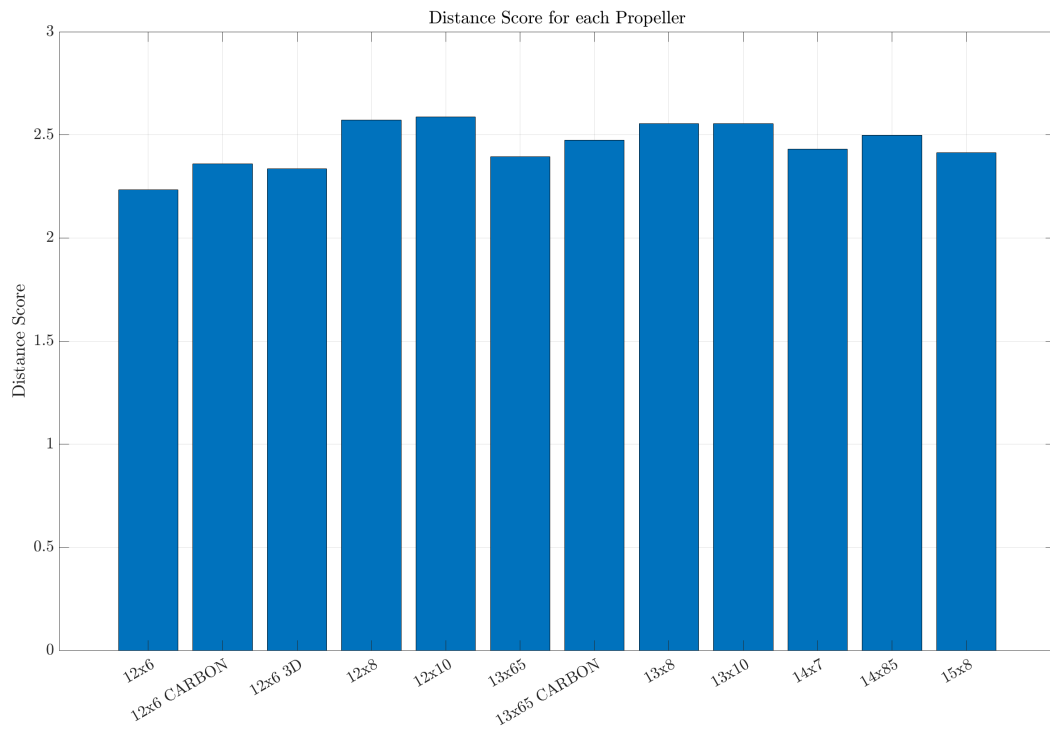


Figure 44: Distance Score achieved by each propeller

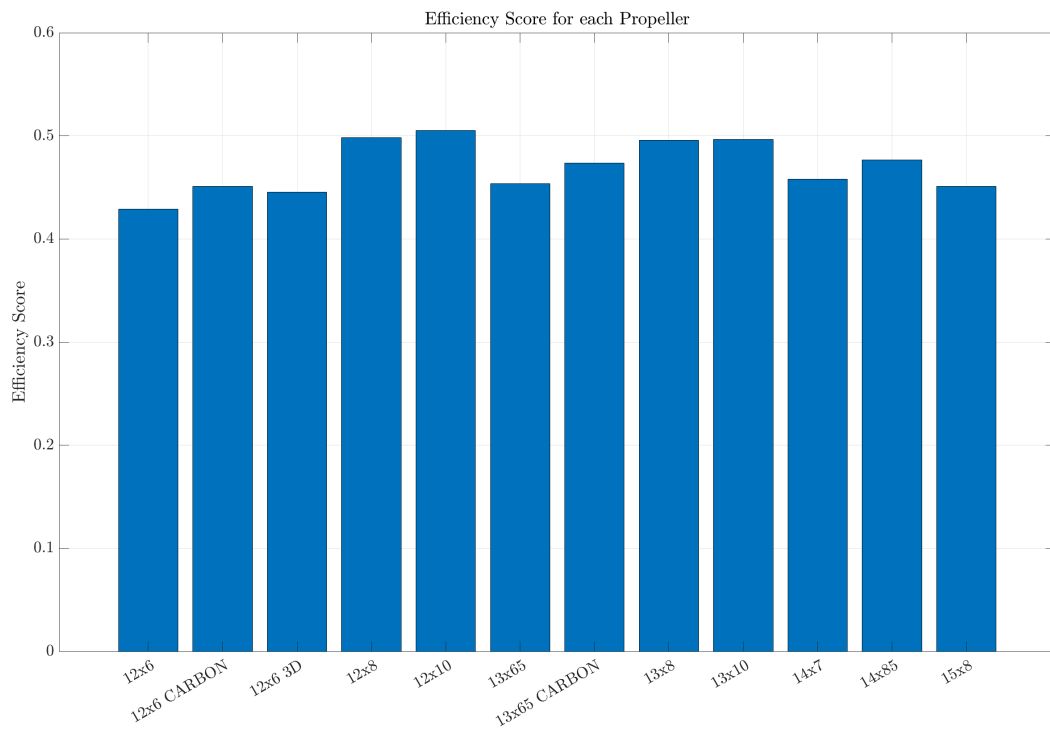


Figure 45: Efficiency Score achieved by each propeller

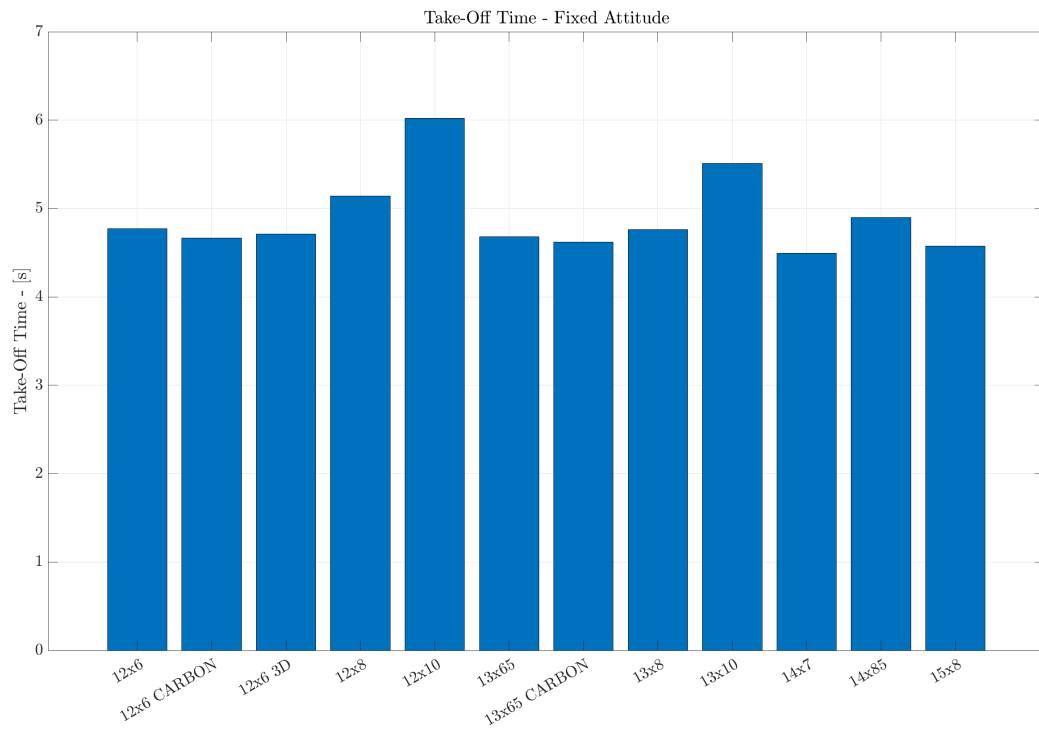


Figure 46: Theoretical TO Time achieved by different propellers

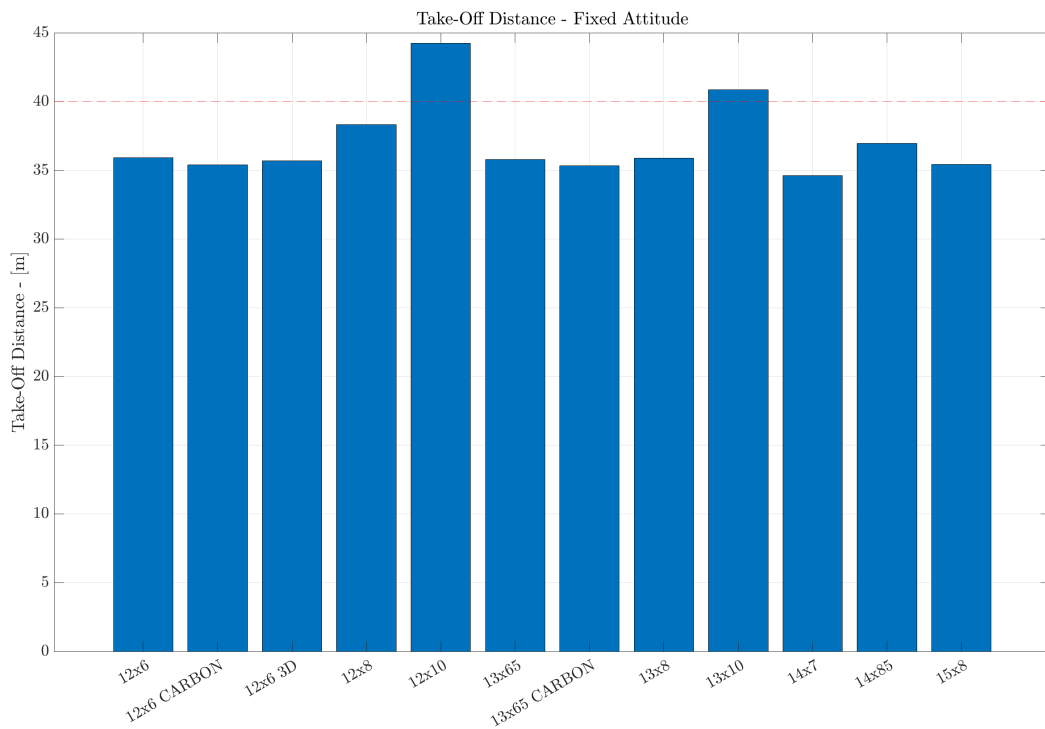


Figure 47: Theoretical TO Distance achieved by different propellers

| Derivative       | Software Results |          |
|------------------|------------------|----------|
|                  | Matlab           | XFLR5    |
| $C_{m_G/\alpha}$ | -0.9811          | -2.252   |
| $C_{l_G/\beta}$  | -0.03673         | -0.02909 |
| $C_{n_G/\beta}$  | 0.07164          | 0.1419   |

Table 11: Stability derivatives obtained with Matlab and XFLR5

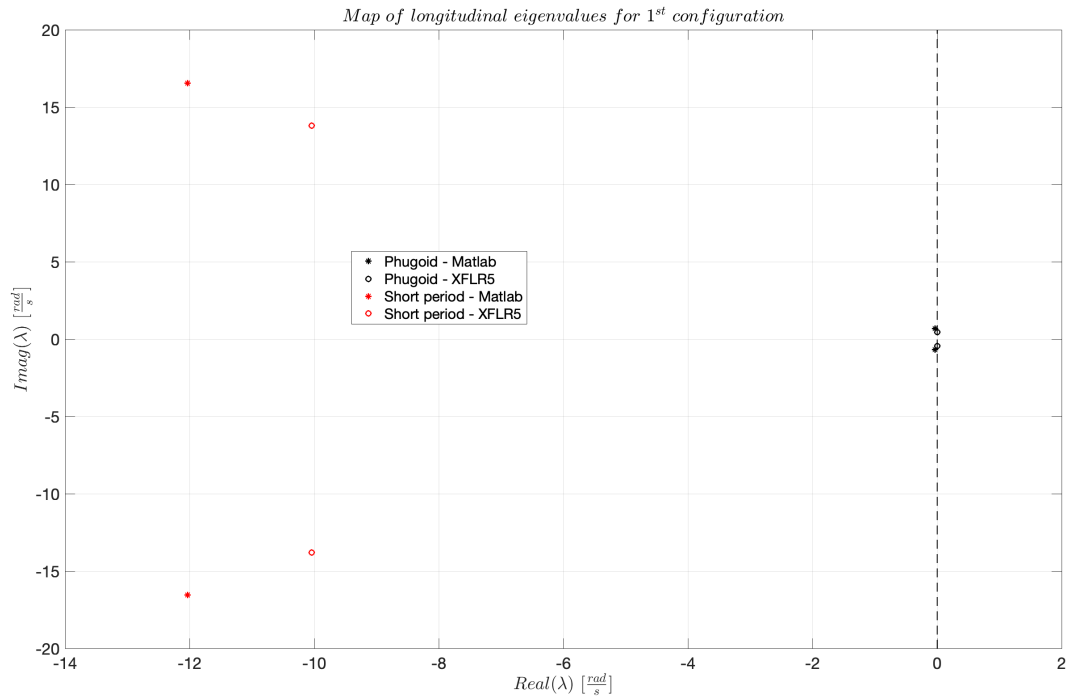


Figure 48: Map of Longitudinal Eigenvalues computed using Matlab and XFLR5, for the 1<sup>st</sup> Configuration

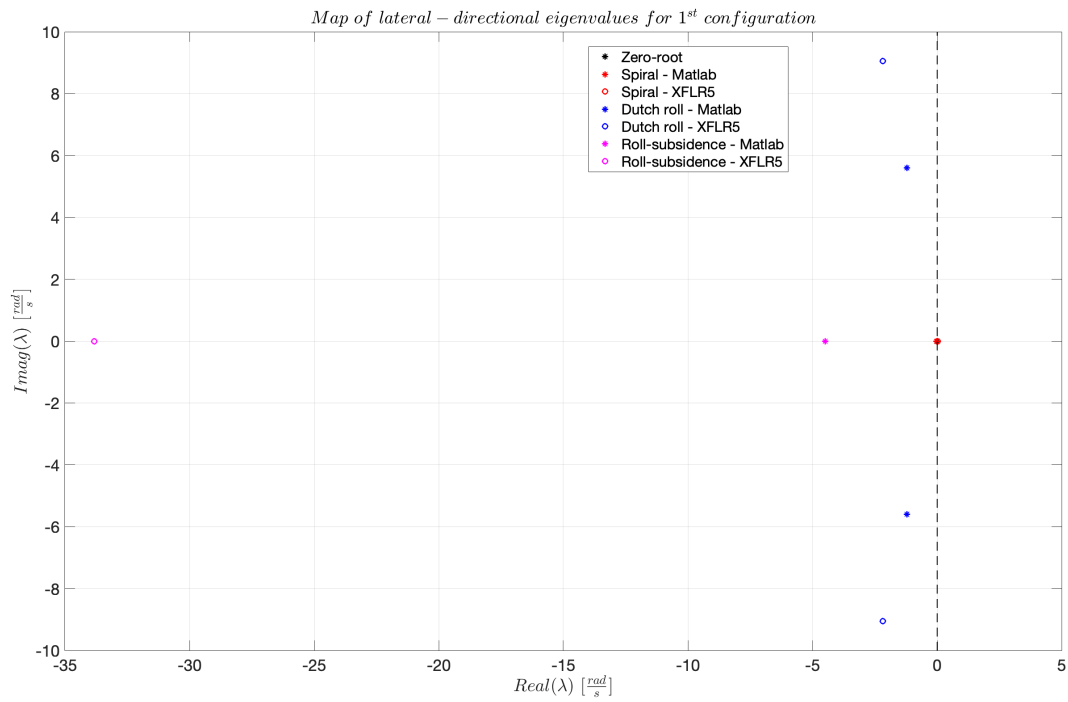


Figure 49: Map of Lateral-Directional Eigenvalues computed using Matlab and XFLR5, for the 1<sup>st</sup> Configuration

| Mode            | Eigenvalues             |                        |
|-----------------|-------------------------|------------------------|
|                 | Matlab                  | XFLR5                  |
| Phugoid         | $-0.0311 \pm 0.6939i$   | $0.0008 \pm 0.4478i$   |
| Short Period    | $-12.0310 \pm 16.5515i$ | $-10.042 \pm 13.7957i$ |
| Dutch Roll      | $-1.2132 \pm 5.6022i$   | $-2.1708 \pm 9.0605i$  |
| Spiral          | -0.03246                | 0.0333                 |
| Roll Subsidence | -4.488                  | -33.8233               |

Table 12: Eigenvalues of first optimized configuration from Matlab and XFLR5

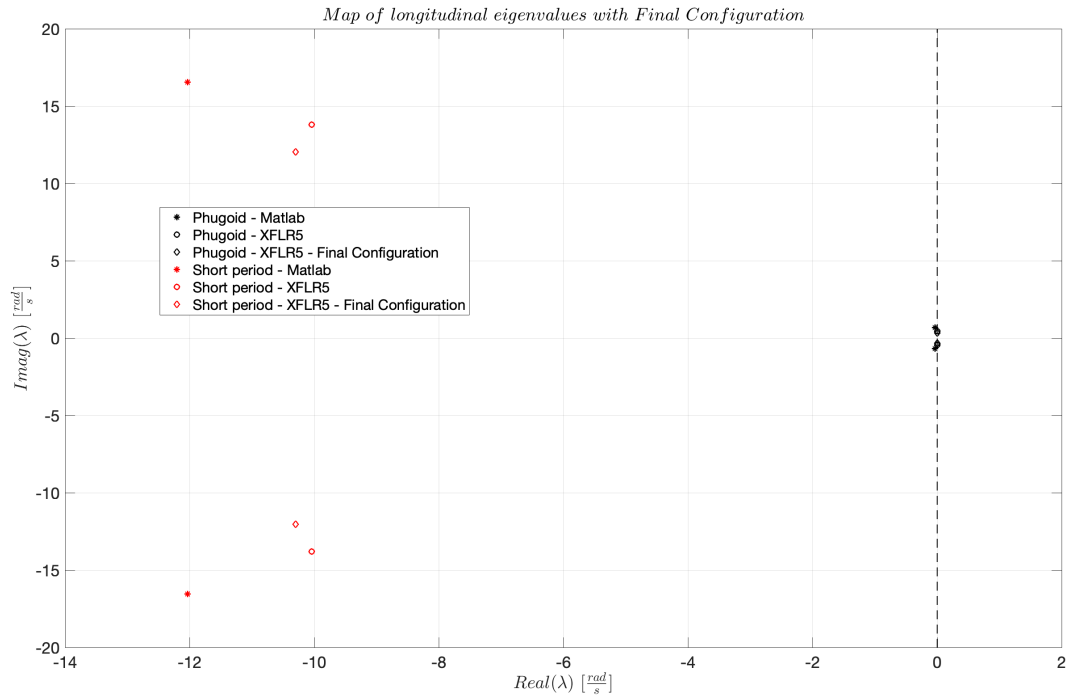


Figure 50: Map of Longitudinal Eigenvalues computed using Matlab and XFLR5, for the 1st and the Final Configurations

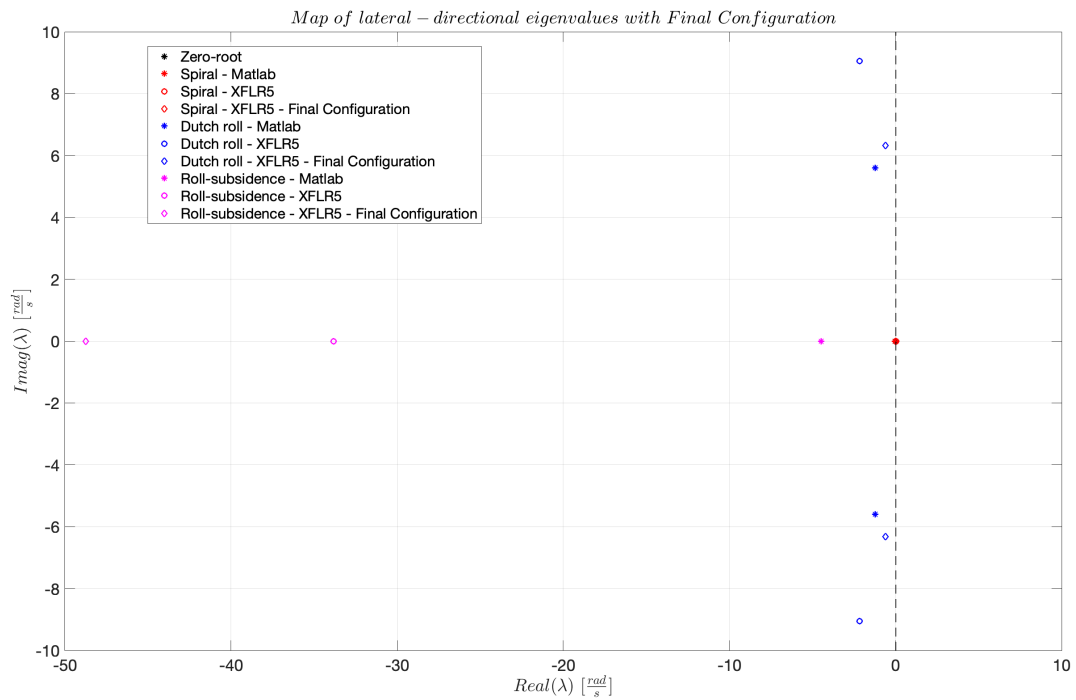


Figure 51: Map of Lateral-Directional Eigenvalues computed using Matlab and XFLR5, for the 1st and the Final Configurations

| Mode            | Eigenvalues             |                              |                         |
|-----------------|-------------------------|------------------------------|-------------------------|
|                 | Matlab                  | XFLR5 1 <sup>st</sup> Config | XFLR5 Final Config      |
| Phugoid         | $-0.0311 \pm 0.6939i$   | $0.0008 \pm 0.4478i$         | $0.0006 \pm 0.3330i$    |
| Short Period    | $-12.0310 \pm 16.5515i$ | $-10.042 \pm 13.7957i$       | $-10.2966 \pm 12.0476i$ |
| Dutch Roll      | $-1.2132 \pm 5.6022i$   | $-2.1708 \pm 9.0605i$        | $-0.6078 \pm 6.3264i$   |
| Spiral          | -0.03246                | 0.0333                       | 0.0112                  |
| Roll Subsidence | -4.488                  | -33.8233                     | -48.7594                |

Table 13: Eigenvalues of first optimized configuration from Matlab and XFLR5

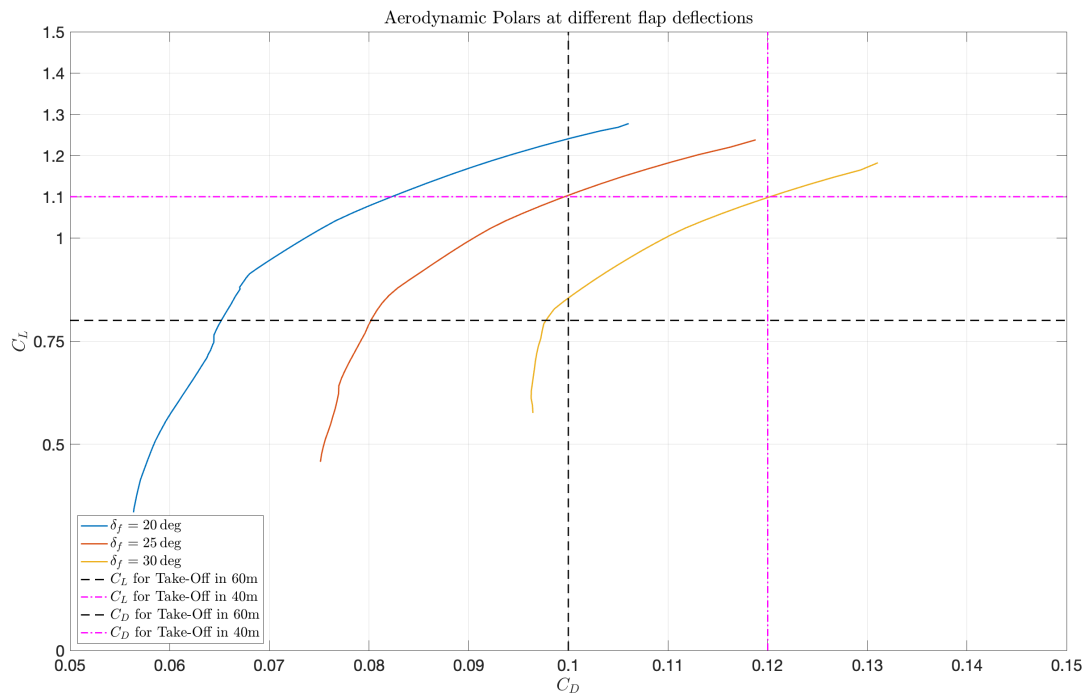


Figure 52: Aerodynamic Polars with different flap deflections

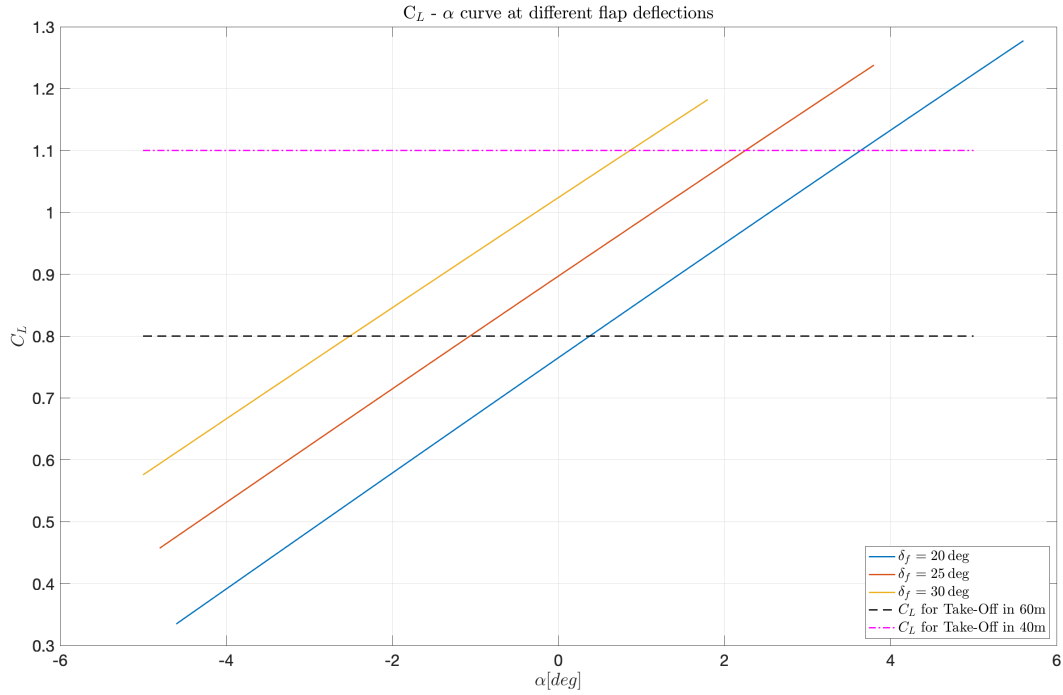


Figure 53: C<sub>L</sub> - α curve at different flap deflections

| Aileron span / wing span        | $C_{l/\delta_a}$ - 15% chord | $C_{l/\delta_a}$ - 20% chord | $C_{l/\delta_a}$ - 25% chord |
|---------------------------------|------------------------------|------------------------------|------------------------------|
| 20%                             | -0.1105                      | -0.1320                      | -0.1489                      |
| 30%                             | -0.1819                      | -0.2175                      | -0.2455                      |
| 40%                             | -0.2508                      | -0.2996                      | -0.3347                      |
| 47.95% (total outboard section) | -0.2966                      | -0.3321                      | -0.3787                      |

Table 14: Control derivative for different values of aileron chord and span, aileron starting from tip of outboard wing section

| Aileron span / wing span        | $C_{l/\delta_a}$ - 15% chord | $C_{l/\delta_a}$ - 20% chord | $C_{l/\delta_a}$ - 25% chord |
|---------------------------------|------------------------------|------------------------------|------------------------------|
| 30%                             | -0.1961                      | -0.2345                      | -0.3347                      |
| 40%                             | -0.2609                      | -0.3113                      | -0.3524                      |
| 47.95% (total outboard section) | -0.2966                      | -0.3321                      | -0.3787                      |

Table 15: Control derivative for different values of aileron chord and span, aileron starting from root of outboard wing section

| <i>Configuration</i> | $C_{l/\delta_a}$ | $M_H - \delta_a = 8^\circ [\text{N}^*\text{m}]$ | $M_H - \delta_a = 20^\circ [\text{N}^*\text{m}]$ |
|----------------------|------------------|-------------------------------------------------|--------------------------------------------------|
| 1                    | -0.2966          | 0.0369                                          | 0.0705                                           |
| 2                    | -0.3113          | 0.0773                                          | 0.1344                                           |

Table 16: Hinge moments for different aileron configurations

| Elevator Chord Percentage | $C_{m/\delta_e}$ | $M_H - \delta_e = 8^\circ [\text{N}^*\text{m}]$ | $M_H - \delta_e = 20^\circ [\text{N}^*\text{m}]$ |
|---------------------------|------------------|-------------------------------------------------|--------------------------------------------------|
| 15%                       | 0.60004          | 0.0041                                          | 0.0045                                           |
| 20%                       | 0.68722          | 0.0061                                          | 0.0068                                           |
| 25%                       | 0.75787          | 0.0081                                          | 0.0093                                           |
| 30%                       | 0.83045          | 0.0105                                          | 0.0119                                           |
| 35%                       | 0.87795          | 0.0131                                          | 0.0148                                           |
| 40%                       | 0.91702          | 0.0158                                          | 0.0180                                           |

Table 17: Elevator chord percentage analysis

| Rudder Chord Percentage | $C_{n/\delta_r}$ | $M_H - \delta_r = 8^\circ [\text{N}^*\text{m}]$ | $M_H - \delta_r = 20^\circ [\text{N}^*\text{m}]$ |
|-------------------------|------------------|-------------------------------------------------|--------------------------------------------------|
| 15%                     | 0.02154          | 0.0229                                          | 0.0568                                           |
| 20%                     | 0.02454          | 0.0389                                          | 0.0977                                           |
| 25%                     | 0.2693           | 0.0574                                          | 0.1495                                           |
| 30%                     | 0.02938          | 0.0791                                          | 0.2350                                           |
| 35%                     | 0.03112          | 0.1052                                          | 0.2136                                           |
| 40%                     | 0.03272          | 0.1352                                          | 0.3633                                           |

Table 18: Rudder chord percentage analysis

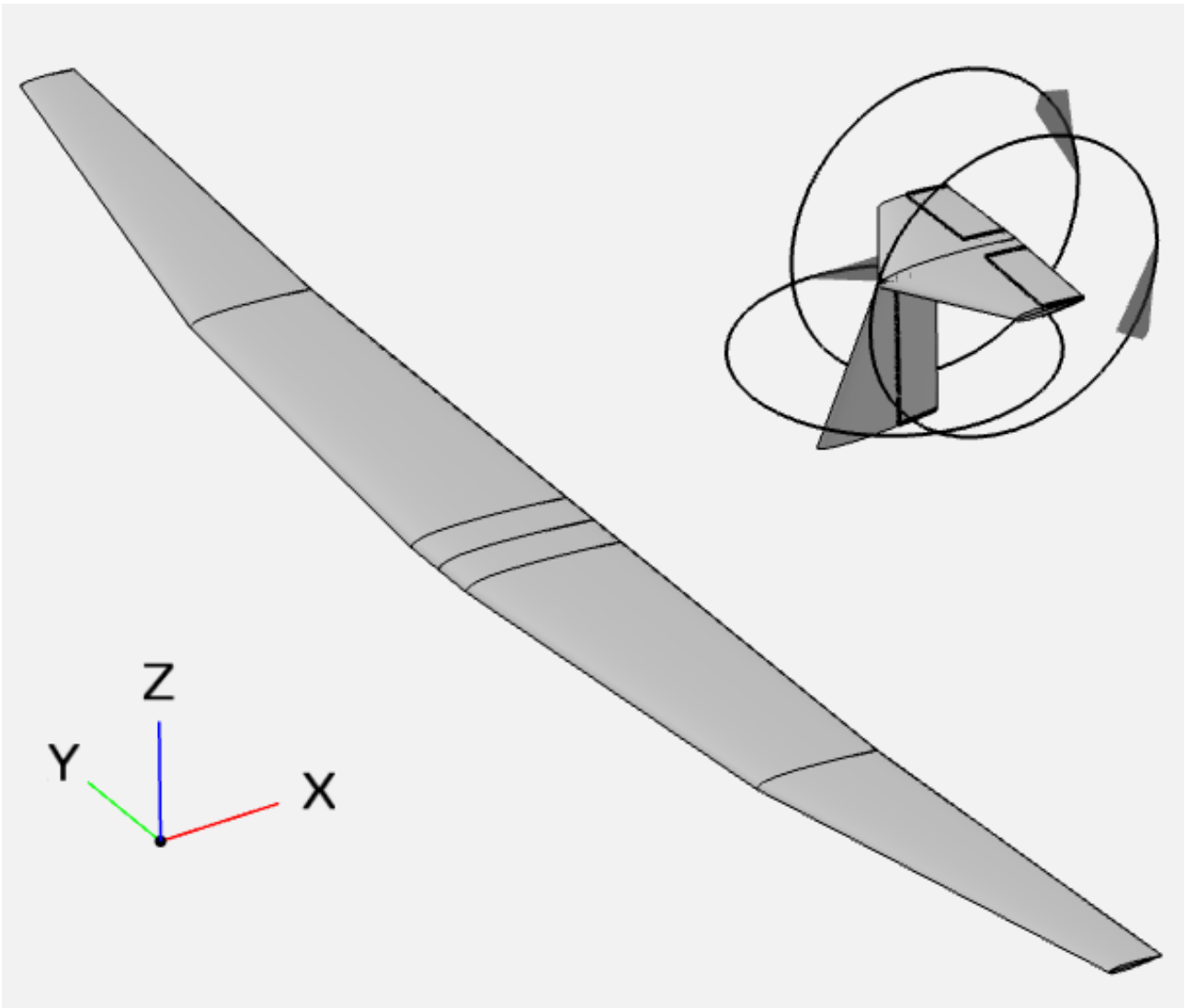


Figure 54: Final Configuration in OpenVSP

## C. Structures definitions

### Capote

Capotes in our aircraft are non-structural parts of the fuselage whose function is to obtain a closed and aerodynamic shape.

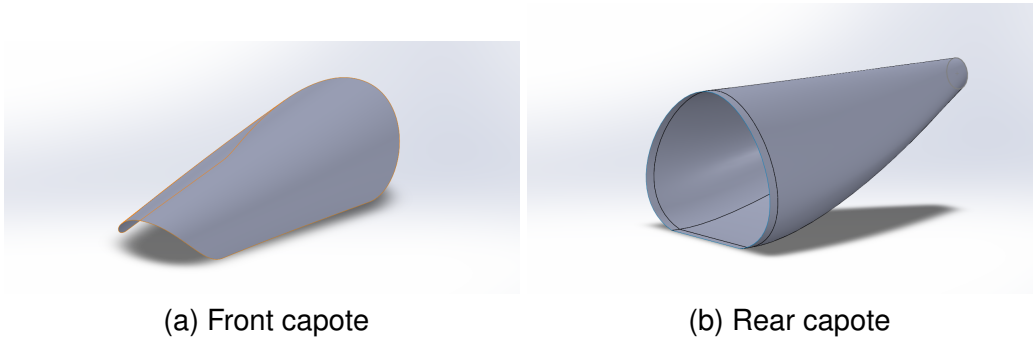


Figure 55: Capotes

### Clearance angle

Angle obtained by joining the contact point of the front landing gear with that of the rear landing gear

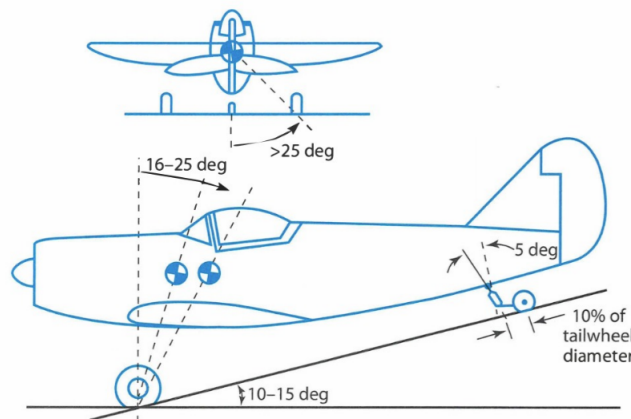


Figure 56: Tail dragger configuration; Image from [16]

### Parametric modelling

All the dimensions in the cad model are expressed with the help of functions that relate them with other significant dimensions, as in a master slave relation. This made possible to update quickly the cad model whenever new specifications were set by the flight mechanics or aerodynamics department.

### Vase mode

Classic 3D printed components are made by the machine in parallel layers laid down one by one. Each of them has an associated height and the printer changes z-height (being z the axis normal to the printing plate) only when the layer is completed. In this way the machine can create many different shapes, multiple wall layers and different infill thicknesses and shape. The normal 3D printing method is used in our drone in some parts

that need to be strong and stiff. The vase mode is a printing technique that is selectable from the slicer software (software that takes the cad model and converts it into a 3D path for the printer nozzle) that has the peculiarity of making an entire piece with a single line of filament, from the bottom to the top without any interruption in the nozzle flow rate. There are some consequences of that:

1. The layers are not parallel to the base plate but are tilted and positioned like a screw.
2. The geometry of the piece must be made in such a way that it is composed of two faces (bottom one and top one) that are connected by a single wall (it has to resemble a cylinder).
3. The piece must be done in a single rigid body otherwise it could end up with missing parts during the printing process.
4. The vase mode only permits a single print at a time, not multiple pieces in parallel like the normal layer by layer printing.

The cons of this technique are:

1. The wall thickness is only one and therefore the parts are very sensitive to forces perpendicular to the wall.
2. The parts if not printed at a sufficient temperature can suffer from delamination of the walls, a condition that can lead to a catastrophic failure of the part.
3. The top face cannot be created, thus the name vase mode.
4. There is no infill so when designing the part in the cad it is needed to account for internal structures that are obtained by cutting away material in such a way that in the middle the two cuts are so close one to the other to connect during the printing process but separated in the cad so the rigid body remains single.

If everything mentioned before is done successfully, the result is a stiff component whose weight is much lower than that of the same piece printed in normal mode. The weight saving depends on the shape of the object and there is no scientific formula that relates the weight of the same project done in the two different ways.

In the final cad model every feature inside is a cut from the outside and the body in the middle is full of material. Only during the slicing the software makes a path for the extruder that follows the object's perimeter in a spiral fashion in order to print in vase mode.

With the transparency turned off the section appears as:

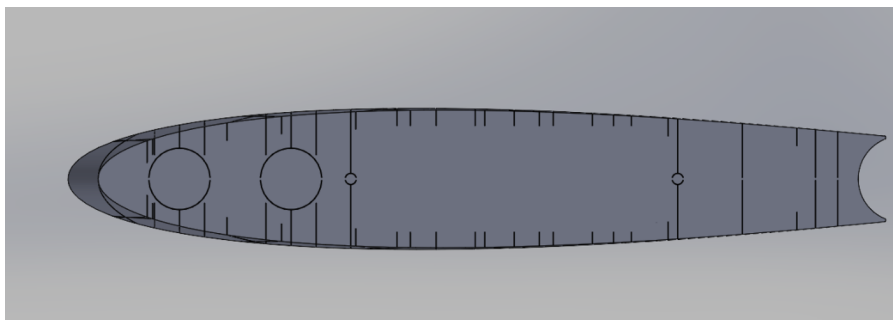


Figure 57: Section of vertical tail from CAD software prepare for the slicer

Every cut ends at 0.25mm from the centreline. This value has been tested in many previous prototypes and is the best distance that still manages to bond together the two sides during the vase mode print.

As it can be seen in the picture below the extruder head moves along the perimeter constantly extruding plastic material and when it encounters a cut it follows the line until it merges with the corresponding cut present on the other side. During the printing process the z values changes continuously so any point has a different z height from the others.

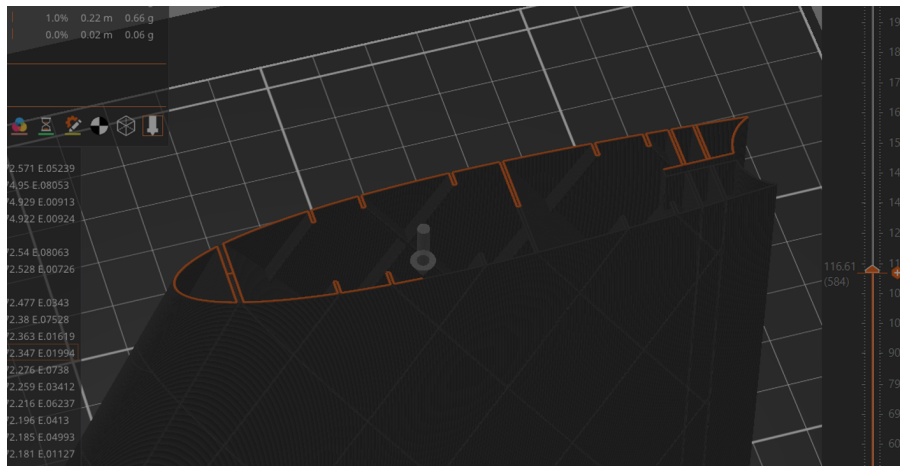


Figure 58: Extruder path from slicer software



Figure 59: Printing of a component in vase mode

An effective way to increase the strength and stiffness of a component manufactured in

vase mode is to coat it in a very thin layer of epoxy clear resin. Some water-based glues were tested but their performance was lower than that of a simple bi-component resin. The resin is mixed in a container and is applied on the component with a brush, having care of filling every gap on the surface.

Improvements due to the resin coating:

1. The layer adhesion is greatly improved thanks to the presence of a matrix of resin that goes transversal in direction of the layers and keeps them strongly bonded together.
2. The cuts made to make the structure more rigid sometimes act in the opposite way by stretching and creating weak points in their proximity (in the direction perpendicular to the ribs). The resin by going deep into the cut has the effect of improving the bonding force perpendicular to the ribs line.
3. 3D printed components in vase mode have low surface finish especially near the folding of the rib's lines. The resin smooths the surface reducing aerodynamics interference.

## D. Aerodynamics

### Wing

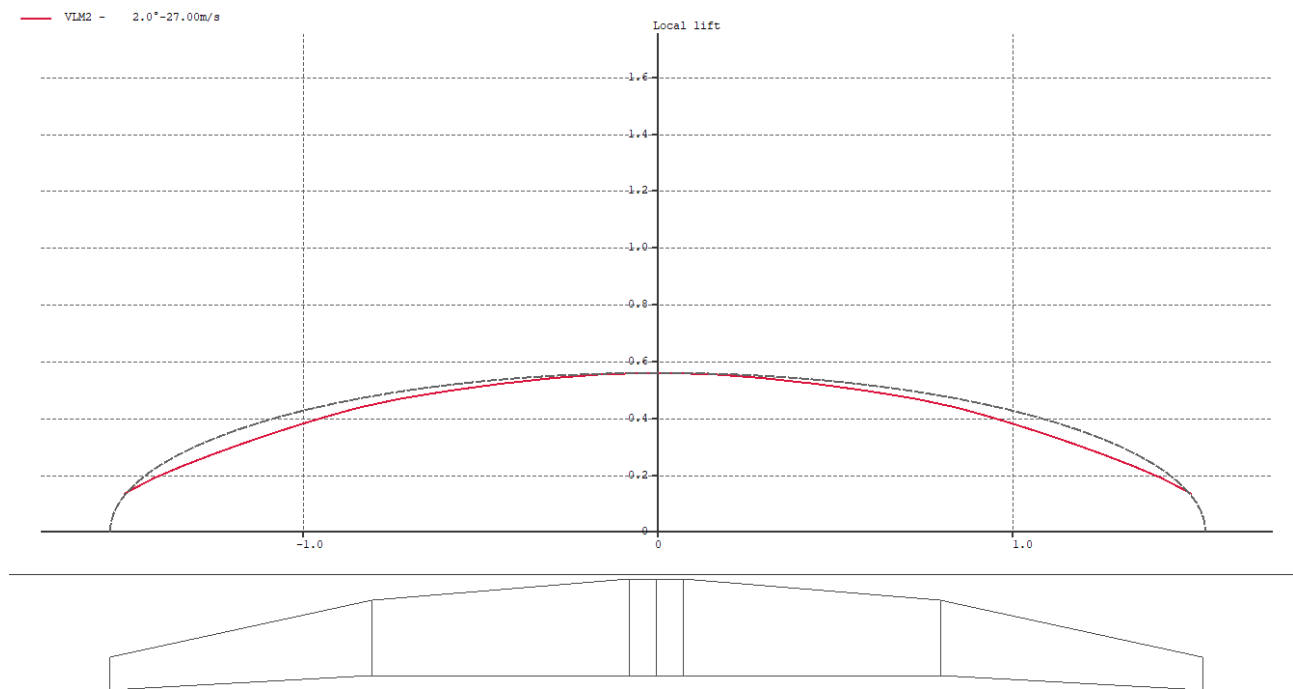


Figure 60: Lift distribution over wingspan vs elliptic

### Landing Gear

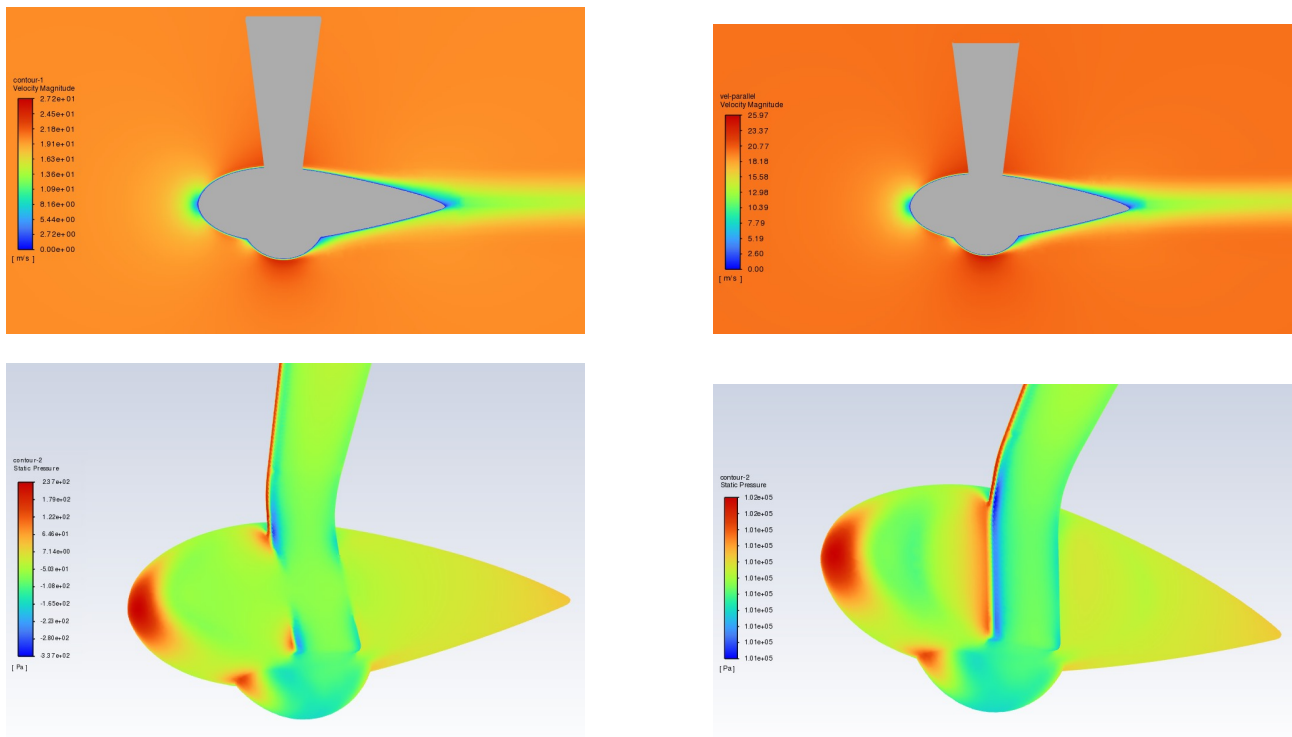


Figure 61: Main Landing Gear fairing CFD simulations

## E. Electronics & Propulsion

### Thrust at Null Velocity - Additional Table and Graphs

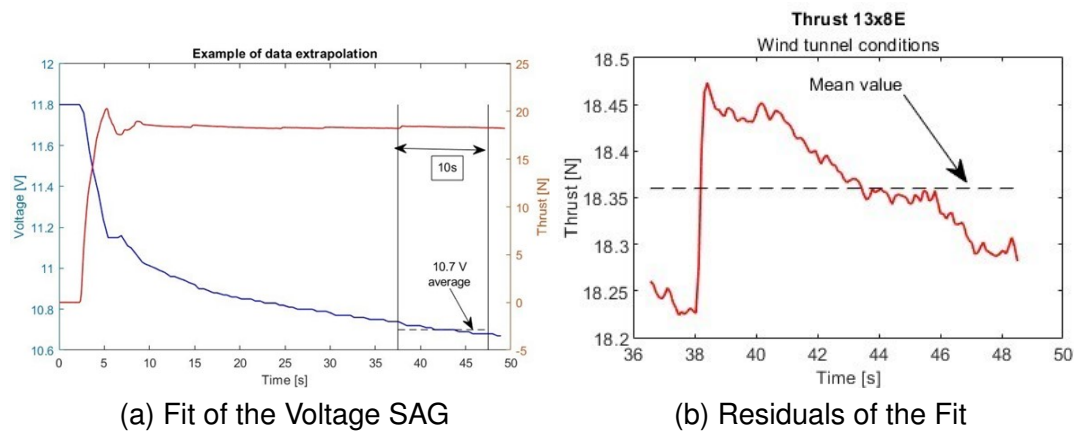


Figure 62: Extrapolated Data

| Propeller  | Mean Thrust [N] |
|------------|-----------------|
| APC 12x8E  | 17.23           |
| APC 12x10E | 15.26           |
| APC 13x8E  | 18.37           |

Figure 63: Mean Thrust Values

## Validation of Battery Testing Functions

The statistical analysis of the fit shows an accurate model. The fit in Figure 64a is done on the magnitude of the sag during the test with a parabolic fit (to the second degree). As seen in the figures, the values of the difference of the sag from the fit respect the main proprieties of a good model. The residues that can be seen in Figure 64b are not dispersed, or in other words, it respects the principle of the homoscedasticity. Also, the fitting instrument of MATLAB reports an elevate r-squared (for all the props is shown a  $> 90\%$  r-squared). After this test it is clear that the model provided is robust and accurate.

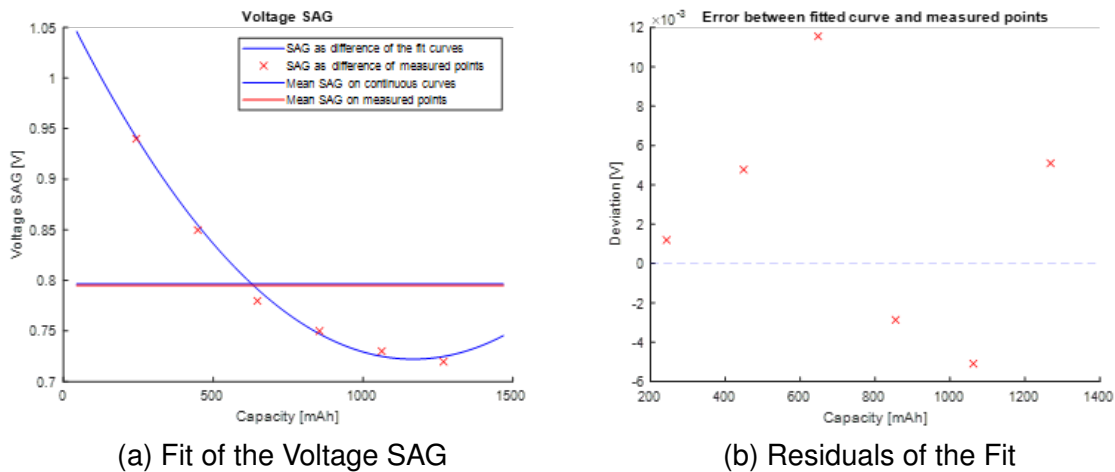


Figure 64: Voltage SAG

After proving the goodness of the sag-model, the single sag-pattern is analyzed. The presence of spikes requires the use of a Fourier fitting model. The MATLAB autofitter makes the creation of a 8 terms model very easy, as visualized in Figure 65.

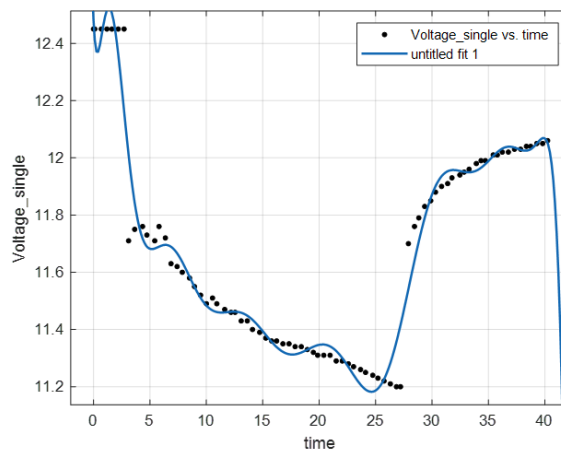


Figure 65: Fourier fit of the single SAG

This model gives back a very complex 19 terms series, but with high goodness values. In fact:

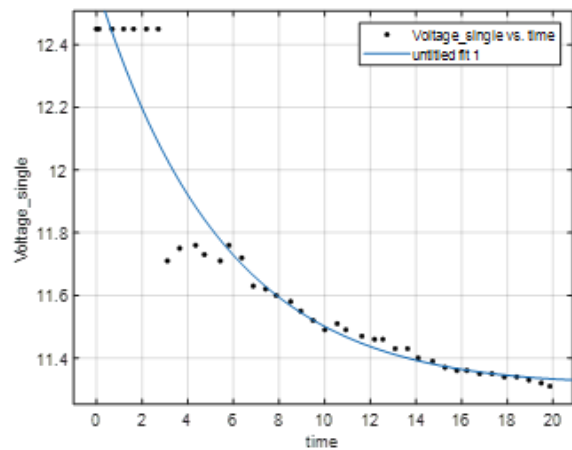
- *sse*: 0.3741
- *rsquare*: 0.9647
- *dfe*: 62

- *adjrsquare*: 0.9551
- *rmse*: 0.0777

Just by looking at the sag-time graph it is clear that the complexity is caused by two different regimes. They appear to be two exponentials regimes, both natural responses of the system at the input of a switch. The switch in this case are represented by the turning-on and the turning-off of throttle at 30amps as seen previously in the report. In Figures 66a and 67a there are the two exponentials fitted with r-squared of, respectively, 0.9182 and 0.9068. These very high levels of r-squared suggest a very good model. Also, fitting with exponential is possible to extract the time constants of the transitions which are equal to  $1/-b$  and  $1/-d$  [Figures 66b and 67b].

```
General model Exp2:
val(x) = a*exp(b*x) + c*exp(d*x)
Coefficients (with 95% confidence bounds):
a =      1.351  (0.3162, 2.386)
b =     -0.1749 (-0.3359, -0.01386)
c =     11.24  (10.15, 12.33)
d =    0.0002347 (-0.004288, 0.004757)
```

(a) First exponential of the SAG

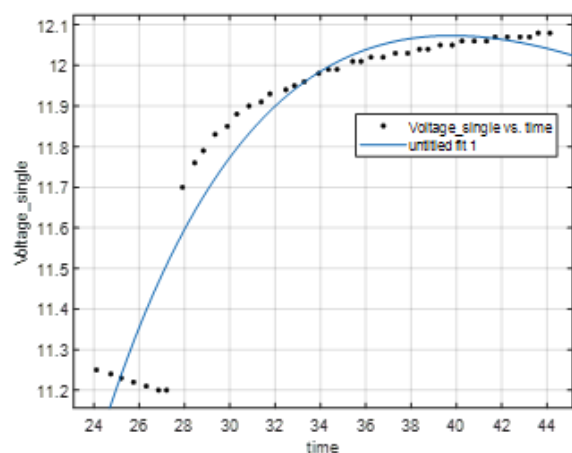


(b) Coefficients of the first exponential with 95% confidence bounds

Figure 66: First exponential

```
General model Exp2:
val(x) = a*exp(b*x) + c*exp(d*x)
Coefficients (with 95% confidence bounds):
a =     14.03  (8.192, 19.87)
b =   -0.003076 (-0.011, 0.004848)
c =   -29.42  (-86.98, 28.13)
d =   -0.1121 (-0.2521, 0.02796)
```

(a) Second exponential of the SAG



(b) Coefficients of the second exponential with 95% confidence bounds

Figure 67: Second exponential

Wind Tunnel Testing

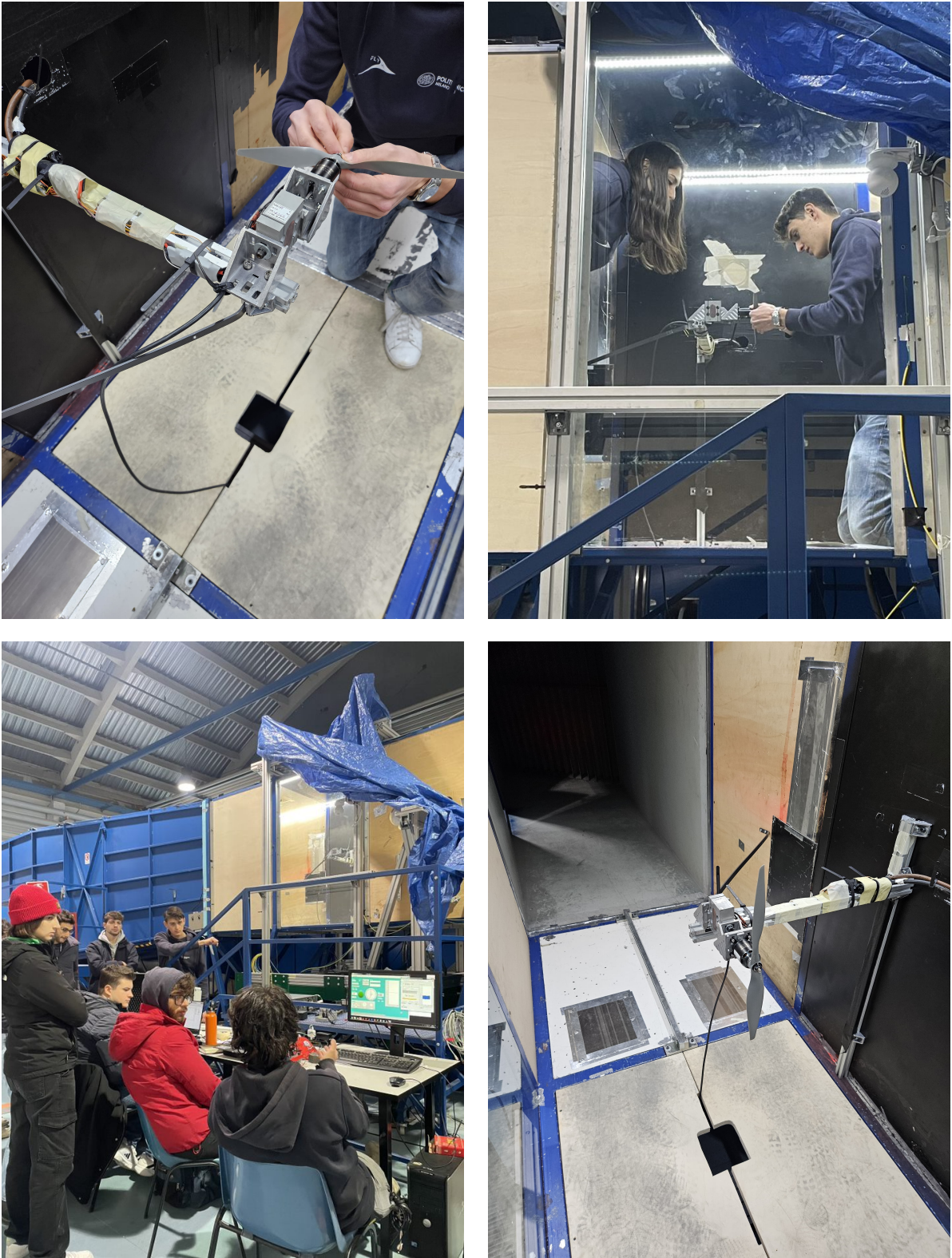
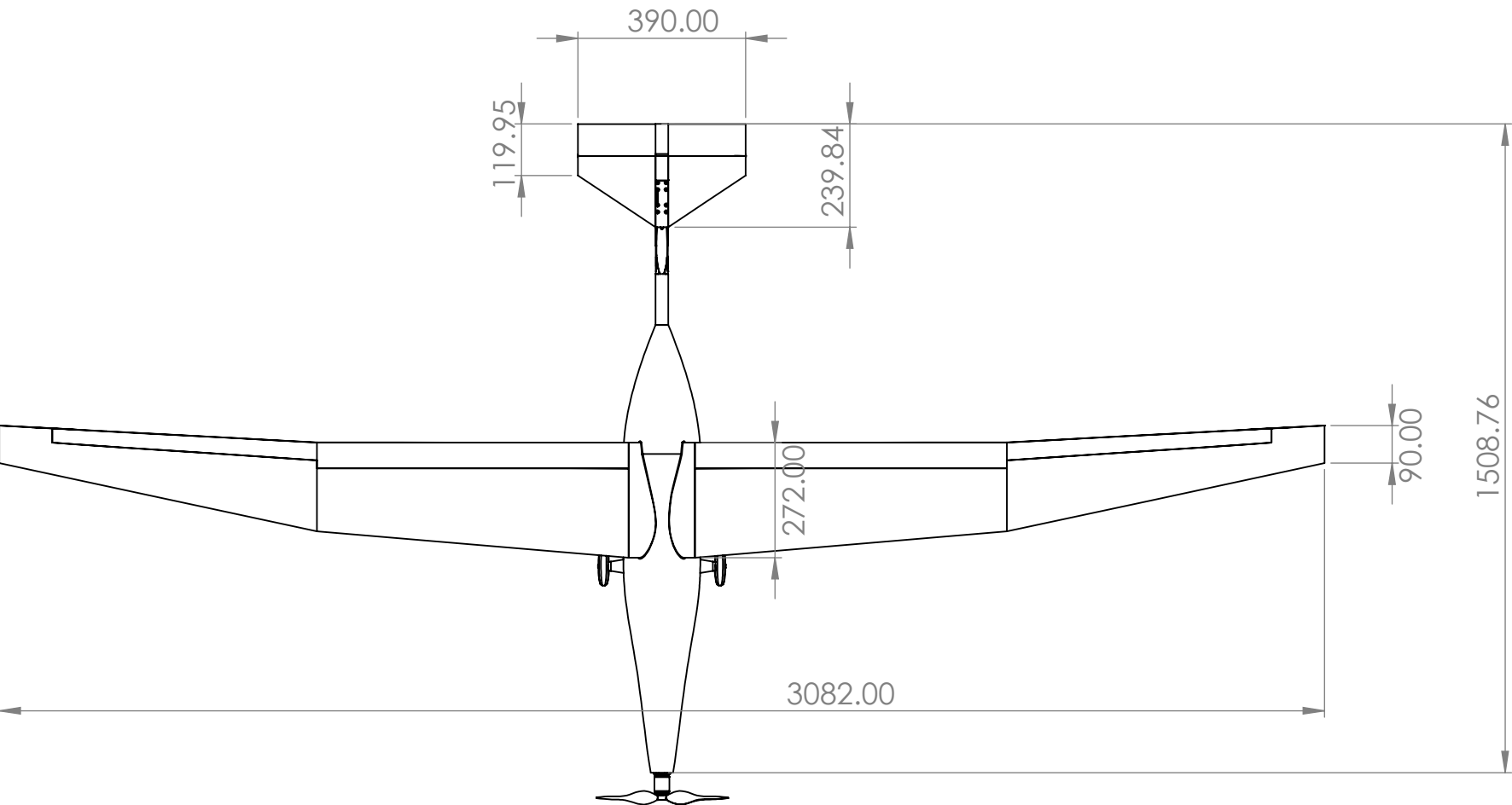
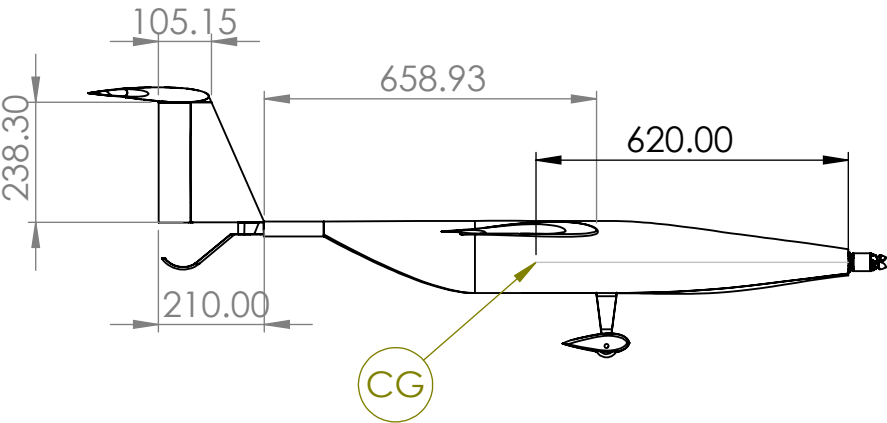
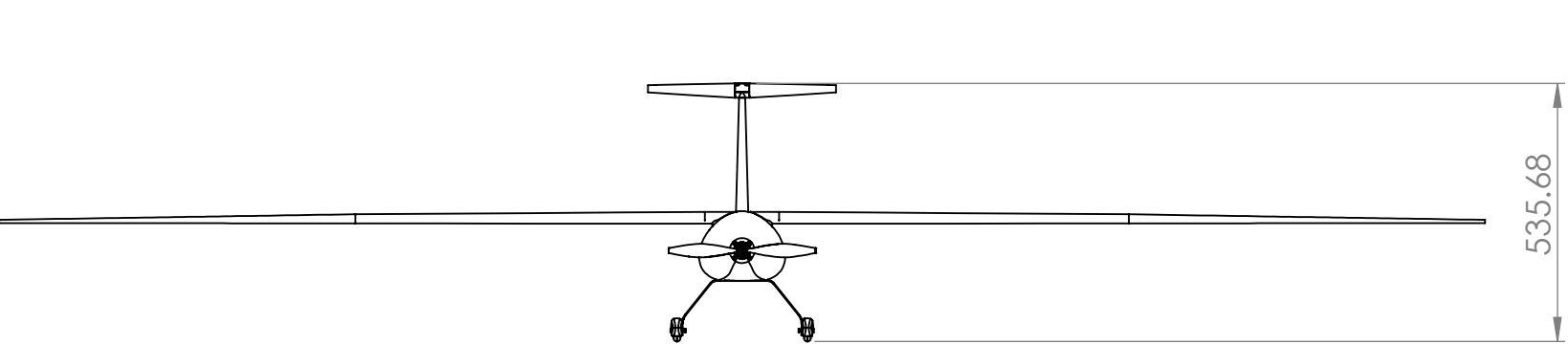
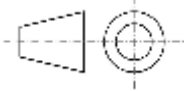


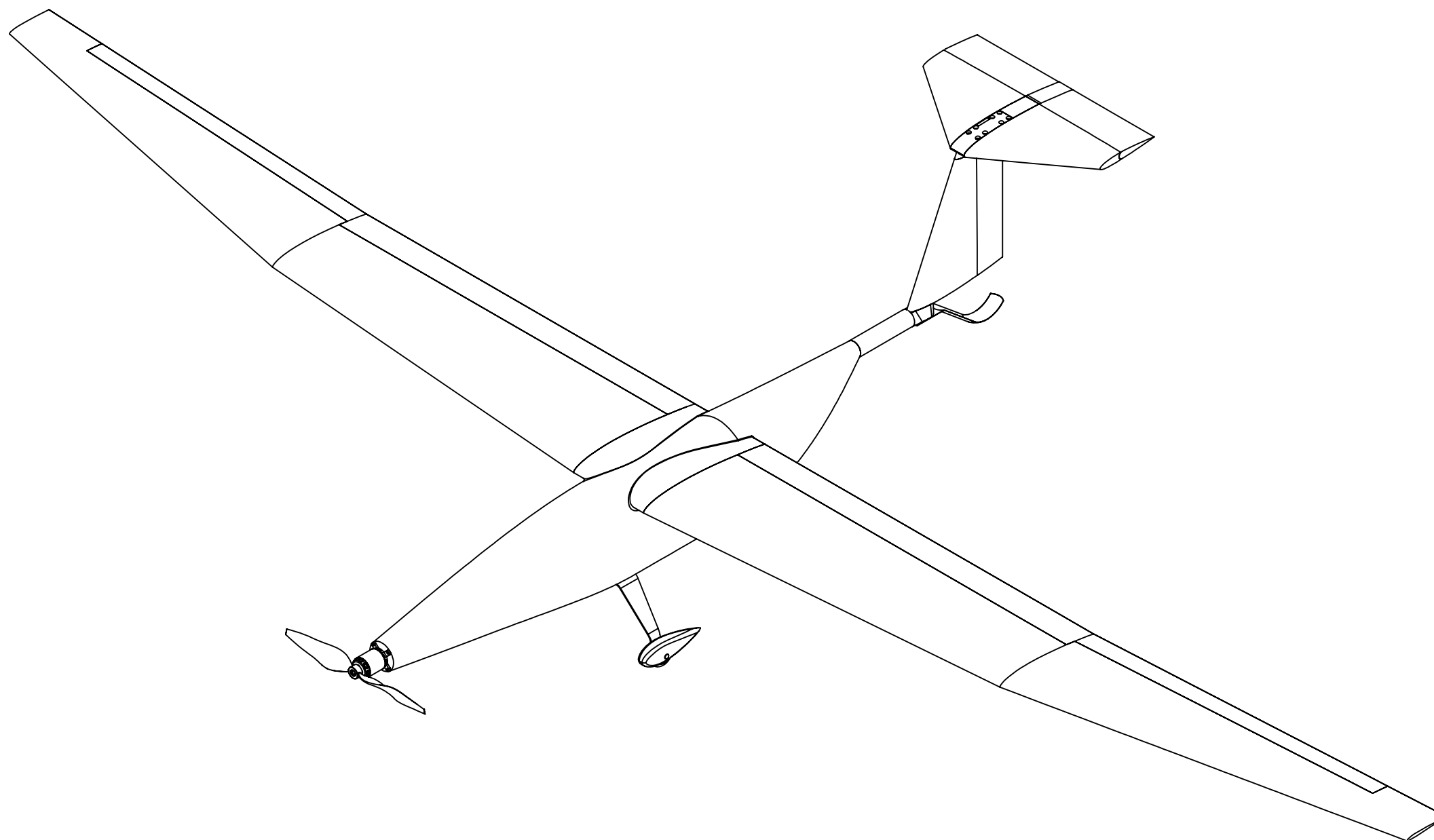
Figure 68: WT testing



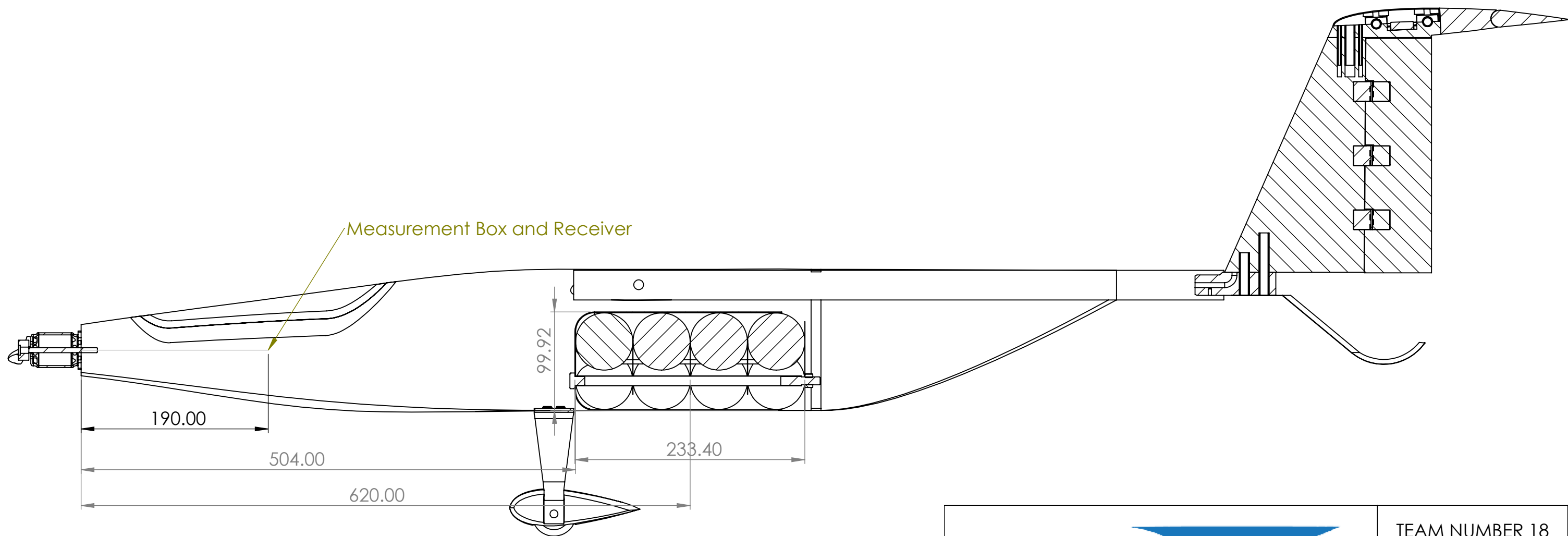
|                     |                        |                |
|---------------------|------------------------|----------------|
| Wing Area           | 0.617 m <sup>2</sup>   |                |
| MAC                 | 0.214 m                |                |
| Fuselage Volume     | 0.01413 m <sup>3</sup> |                |
| Rear Capote Surface | 0.11254 m <sup>2</sup> |                |
| Airfoils            | S9000                  | s9000optimized |

|                                                                                       |                                                                                       |                                                                                                                     |
|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
|  |                                                                                       | TEAM NUMBER 18                                                                                                      |
|                                                                                       |                                                                                       | <br>POLITECNICO<br>MILANO 1863 |
| A3                                                                                    |  | SCALE 1:15                                                                                                          |
| <b>3-View Drawing</b>                                                                 |                                                                                       |                                                                                                                     |

all measurements are in mm

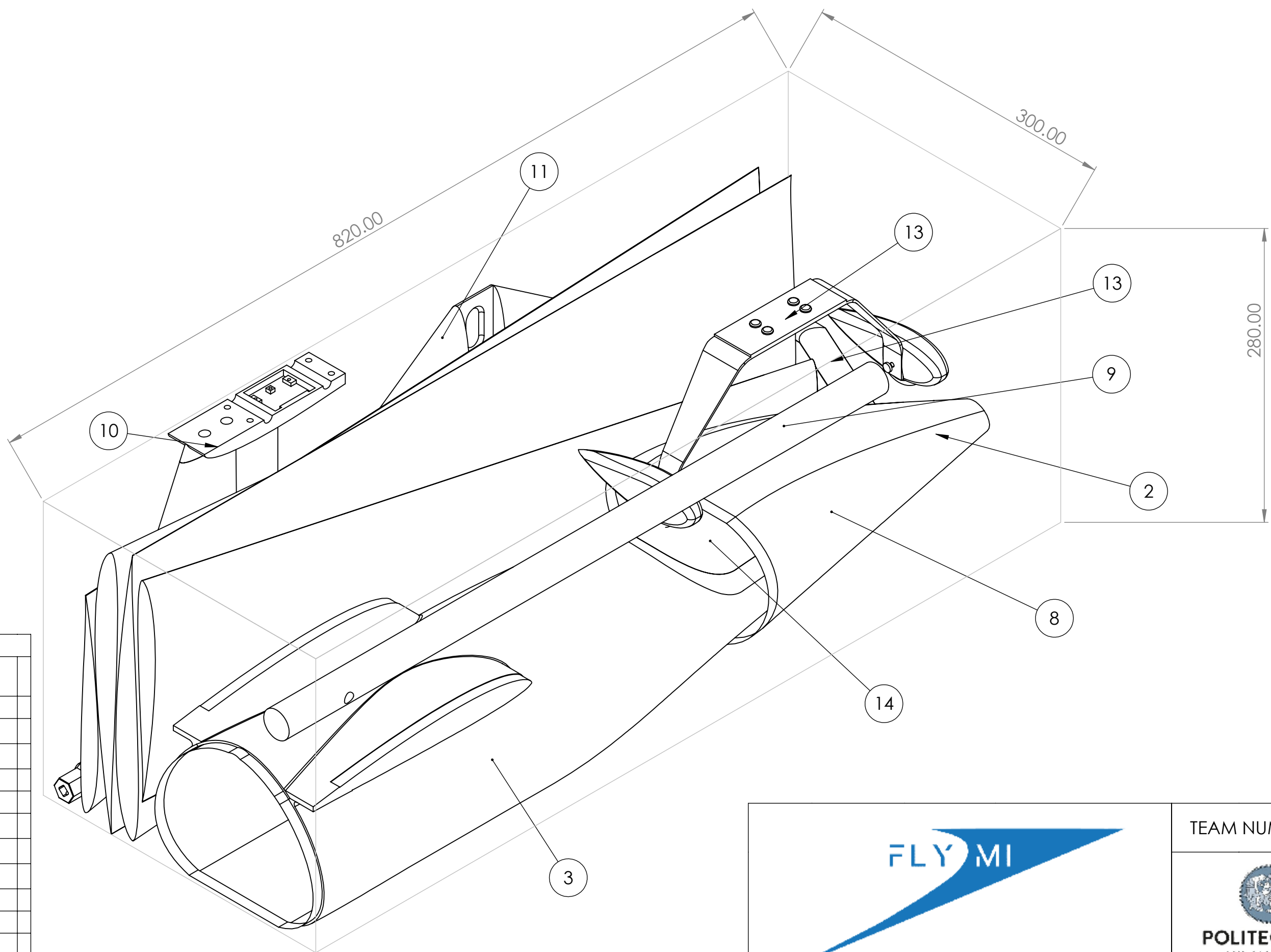


|                                                                                       |                                                                                       |                                                                                                                     |
|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
|  |                                                                                       | TEAM NUMBER 18                                                                                                      |
|                                                                                       |                                                                                       | <br>POLITECNICO<br>MILANO 1863 |
| A3                                                                                    |  | SCALE 1:8                                                                                                           |
| Isometric Perspective View                                                            |                                                                                       |                                                                                                                     |



|                                                                                       |                                                                                       |                                                                                                                            |
|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
|  |                                                                                       | TEAM NUMBER 18                                                                                                             |
|                                                                                       |                                                                                       | <br><b>POLITECNICO</b><br>MILANO 1863 |
| A3                                                                                    |  | SCALE 1:4                                                                                                                  |
| <b>Cargo Bay</b>                                                                      |                                                                                       |                                                                                                                            |

all measurements are in mm



| ITEM NO. | PART NUMBER        |  |
|----------|--------------------|--|
| 1        | Box                |  |
| 2        | Engine             |  |
| 3        | Fuselage           |  |
| 4        | Starboard_Inner    |  |
| 5        | Starboard_Outer    |  |
| 6        | Port_Outer         |  |
| 7        | Port_Inner         |  |
| 8        | Capote_Rear        |  |
| 9        | Tube               |  |
| 10       | VTail              |  |
| 11       | HTail              |  |
| 12       | Landing Gear_Front |  |
| 13       | Landing Gear_Rear  |  |
| 14       | Capote_Front       |  |

|                                                                                       |                                                                                       |                                                                                                                     |
|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
|  |                                                                                       | TEAM NUMBER 18                                                                                                      |
|                                                                                       |                                                                                       | <br>POLITECNICO<br>MILANO 1863 |
| A3                                                                                    |  | SCALE 1:3                                                                                                           |
| <b>Transportation Box</b>                                                             |                                                                                       |                                                                                                                     |

all measurements are in mm